

User Manual

for S32K1_S32M24X MCU Driver

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	07.03.2025	NXP RTD Team	S32K1_S32M24X Real-Time Drivers AUTOSAR R21-11 Version 3.0.0

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductors' AUTOSAR Mcu Driver for S32K1_S32M24x.

AUTOSAR Mcu Driver configuration parameters description can be found in the Tresos Configuration Plugin section. Deviations from the specification are described in the [Deviations from Requirements](#) section.

AUTOSAR Mcu driver requirements and APIs are described in the Mcu Driver Software Specification Document (version R21-11).

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k116_qfn32
- s32k116_lqfp48
- s32k118_lqfp48
- s32k118_lqfp64
- s32k142_lqfp48
- s32k142_lqfp64
- s32k142_lqfp100
- s32k142w_lqfp48
- s32k142w_lqfp64
- s32k144_lqfp48
- s32k144_lqfp64 / MWCT1014S_lqfp64
- s32k144_lqfp100 / MWCT1014S_lqfp100
- s32k144_mapbga100

- s32k144w_lqfp48
- s32k144w_lqfp64
- s32k146_lqfp64
- s32k146_lqfp100 / MWCT1015S_lqfp100
- s32k146_mapbga100 / MWCT1015S_mapbga100
- s32k146_lqfp144
- s32k148_lqfp100
- s32k148_mapbga100 / MWCT1016S_mapbga100
- s32k148_lqfp144
- s32k148_lqfp176
- s32m241_lqfp64
- s32m242_lqfp64
- s32m243_lqfp64
- s32m244_lqfp64

All of the above microcontroller devices are collectively named as S32K1_S32M24X. Note: MWCT part numbers contain NXP confidential IP for Qi Wireless Power

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
CAN	Controller Area Network
C/CPP	C and C++ Source Code
CS	Chip Select
CTU	Cross Trigger Unit
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
FIFO	First In First Out
LSB	Least Significant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MSB	Most Significant Bit
N/A	Not Applicable
RAM	Random Access Memory
SIU	Systems Integration Unit
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language

2.5 Reference List

#	Title	Version
1	Specification of Driver	AUTOSAR Release R21-11
2	S32K1xx Reference Manual	Rev.14.1, 01/2024
3	S32M24x Reference Manual	Rev. 5, 12/2024
4	S32K1xx Data Sheet	Rev. 14, 08/2021
5	S32M2xx Data Sheet	Rev. 7, 12/2024
6	S32K116 Errata	Mask Set Errata for Mask 0N96V Rev. 17/JAN/2024, 1/2024
7	S32K118 Errata	Mask Set Errata for Mask 0N97V Rev. 15/JAN/2024, 1/2024
8	S32K142 Errata	Mask Set Errata for Mask 0N33V Rev. 15/JAN/2024, 1/2024
9	S32K144 Errata	Mask Set Errata for Mask 0N57U Rev. 15/JAN/2024, 1/2024
10	S32K144W Errata	Mask Set Errata for Mask 0P64A Rev. 03/JUL/2024, 7/2024
11	S32K146 Errata	Mask Set Errata for Mask 0N73V Rev. 15/JAN/2024, 1/2024
12	S32K148 Errata	Mask Set Errata for Mask 0N20V Rev. 15/JAN/2024, 1/2024
13	S32M242 Errata	Mask Set Errata for Mask N33V+P69K Rev.1, 8/2024
14	S32M244 Errata	Mask Set Errata for Mask P64A+P69K Rev.1, 8/2024

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)
- [Usage in non AUTOSAR context](#)

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR R21-11 Mcu Driver Software Specification document (See Table [Reference List](#))

3.2 Driver Design Summary

The Mcu Driver controls the CLOCK, POWER and RAM modules of the S32K1_S32M24x device. It provides the following features:

- Configuration and initialization of the CLOCK.
- Configuration and initialization of the POWER.
- Configuration and initialization of the RAM.

3.3 Hardware Resources

The Mcu Driver consists of:

1. Clock IPs (CMU,PCC,PRAMC,SIM,SCG)
2. Power IPs (PMC,RCM,SIM,SMC,AE_PMC,AEC)
3. Ram IPs (PRAMC)

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR Mcu Driver Software Specification in some places.

The table [Mcu Requirements Deviations](#) identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out of scope for the Mcu Driver.

The table [Status Column Description](#) provides the "Status" column description.

3.4.1 Status Column Description

Table 3.1 Status Column Description Areas

Term	Definition
N/S	Not In Scope
N/I	Not Implemented
N/F	Not Fully Implemented

3.4.2 Mcu Requirements Deviations

Table 3.3 Mcu Requirements Deviations

Requirement	Status	Description	Notes
SWS_Mcu_00053	N/S	If clock failure notification is enabled in the configuration set and a clock source failure error occurs, the error code MCU_E_CLOCK_FAILURE shall be reported. (See also SWS_Mcu_00051).	DEMs cannot be reported in I↔SR contexts. For the clock failure case the error MCU_E_IS↔R_CLOCK_FAILURE was defined and if the error notification is configured by the application the CMU interrupt will report it.
SWS_Mcu_00056	N/S	The function Mcu_Distribute↔PllClock shall return without affecting the MCU hardware if the PLL clock has been automatically activated by the MCU hardware. (BSW12336)	The function Mcu_Distribute↔PllClock will change the Mcu hardware. The clock switching to PLL is not completed by Mcu_InitClock.
SWS_Mcu_00245	N/S	If the register can affect several hardware modules and if it is not an I/O register, it shall be initialised by this MCU driver. (BSW12125, BSW12461)	There is a separate plug-in that will cover shared ip's.

Requirement	Status	Description	Notes
SWS_Mcu_00257	N/S	Fail criteria for MCU_E_CLOCK_FAILURE: a clock source failure occurs	For the clock failure case the error MCU_E_ISR_CLOCK_FAILURE was defined and if the error notification is configured by the application the CMU interrupt will report it.
SWS_Mcu_00258	N/S	Pass criteria for MCU_E_CLOCK_FAILURE: no clock source failure occurs	For the clock failure case the error MCU_E_ISR_CLOCK_FAILURE was defined and if the error notification is configured by the application the CMU interrupt will report it.
SWS_Mcu_00259	N/S	DRAFT: The MCU Driver module shall reject configurations with partition mappings which are not supported by the implementation.	Based on ticket AAI-462, this requirement is not applicable.
SWS_Mcu_CONSTR_00001	N/S	DRAFT: The module will operate as an independent instance in each of the partitions, means the called API will only target the partition it is called in.	Based on ticket AAI-462, this requirement is not applicable.
CPR_RTD_00544.mcu	N/F	Driver shall support Autosar standard configuration format for the IP layer Note: EPD file for the IP shall be provided.	For S32K3XX,S32K1, CLOCK IP can't be supported because it will break compatibility.

3.5 Driver Limitations

- Can't disable FIRC clock.
- Clock initialization function (Mcu_InitClock/Clock_Ip_Init) must not be called while chip is operating in other than RUN mode.
- The notifications did not support in IPL.
- In IPL, the interrupt handler function for PMC and CMU are implemented separately: an interrupt handler function for only PMC and a similar one for CMU. Interrupt handler function that controls both CMU and PMC isn't supported.
- Errata e011063 (exist on S32K14x and S32M242) SMC: An asynchronous wakeup event during VLPS mode entry may result in possible system hang scenario.
- Errata e011114 and e050383 (exist on S32K116, S32K118, S32K14x and S32M24x) SMC: invalid data might be fetched while accessing Flash in VLP modes.
- Errata e052092, e052053 and e052094 (exist on S32K116, S32K118, S32K14x and S32M242) PMC: During the power up sequence, the microcontroller may stay in reset condition if the current demand is not satisfied.
- The power mode change notification function will not work without a previously initialized Clock driver. If done otherwise the system clock will not be updated correctly when power mode is changed.
- ECPD/EPC for Clock Ip is not supported. EPC importing to Clocks tool(S32DS) is not supported.

- Configuration of McuClockReferencePoint with "Mcu Clock Frequency Select" is "CUSTOM" needs to be updated manually in S32DS Mcu component after importing configuration from EPC.

3.6 Driver usage and configuration tips

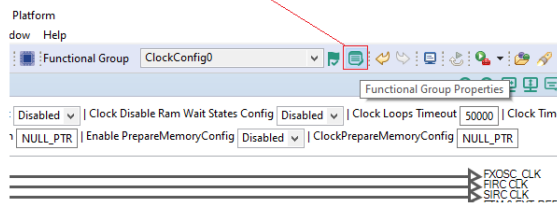
3.6.1 MCU Clock Management

- To generate a clock configuration in Clock Tool from Design Studio, a new project must be created. Project is created with an example clock configuration ClockConfig0. To create a new configuration from scratch (reset configuration), "add a new functional group" button must be pressed. To create a new configuration from an existing one, "copy functional group" button must be pressed. Clock configuration can be removed by removing functional group.

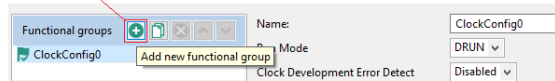
1) Go to clock tool



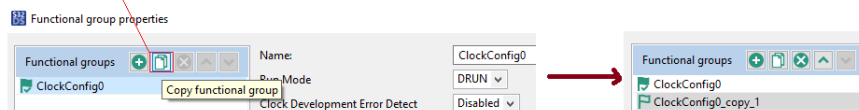
2) Functional group properties



- Add a new functional group



- Copy functional group



- Remove functional group

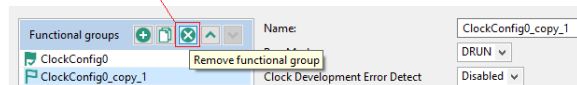


Figure 3.1 Clock tool snapshot for Functional Group Properties

- For reconfiguring the PLLs using Mcu_InitClock and Mcu_DistributePllClock the peripherals that are clocked using the PLL that needs to be reconfigured should be turned OFF using Mcu_SetMode to transition in a mode where that peripheral is OFF.
- For bypassing the configuration of a clock source during Mcu_InitClock, the "[source] under MCU control" checkbox should be unchecked. This will generate smaller configurations that will be updated faster and more efficiently. In addition, if the application is configuring some clocks in advance, the UnderMcuControl should be unchecked so the MCU driver does not overwrite the initial settings.
- When the clock tree is initialized before the MCU driver is used. e.g The bootloader or user code initializes the clock tree. After that the control is passed to AUTOSAR software, the MCU is used to configured the ECU and clock again. The following sequence is required to successfully and safely re-configure the clock tree.

- System clock frequency selected must adhere to the same clock divider ratios shown in Clocking use case examples of Reference Manual.
- If you want to use peripheral module, you must enable clock gate corresponding to the module in `McuPeripheralClockConfig` container.

1. `Mcu_Init`
2. `Mcu_InitClock` (This function is only called in Run mode.)
3. If the PLL is used as a clock source, call `Mcu_GetPllStatus` until it returns `MCU_PLL_LOCKED` and call `Mcu_DistributePllClock`.
4. `Mcu_SetMode`

3.6.2 Specific clock use case:

- The protocol clock of FlexCan has two clock source `SOSC` and `SYS_CLK`, which control by `CAN_CTRL1[CLKSRC]`. When `SOSC` is selected, `SYS_CLK` must be greater than 1.5x the protocol clock. When `SYS_CLK` is selected, the ratio of `SYS_CLK` and `FLEXCAN_CLK` can be 1:1.
- Maximum supported clock out frequency for this device is 20 MHz. There is no restrict corresponding to this range in EB Tresos due to frequency of `CLKOUT` is depended on working mode.

3.6.3 MCU Mode Management

- When power mode switching, the interrupt priority must be set to the highest priority of 0 in order to wake up normally, and the delay time must be sufficiently long for the process to finish (`McuTimeout > 20000`).

3.6.4 MCU RAM Configuration

- NA

3.7 Runtime errors

The driver generates the following DET errors at runtime.

Table 3.4 Default Errors (reported by DET)

Function	Error Code	Condition triggering the error
<code>Mcu_Init()</code>	<code>MCU_E_INIT_FAILED</code>	Invalid configuration pointer.
<code>Mcu_InitClock()</code>	<code>MCU_E_PARAM_CLOCK</code>	Invalid input parameter.
<code>Mcu_SetMode()</code>	<code>MCU_E_PARAM_MODE</code>	Invalid input parameter.
<code>Mcu_InitRamSection()</code>	<code>MCU_E_PARAM_RAMSECTION</code>	Invalid input parameter or invalid memory configuration.
<code>Mcu_DistributePllClock()</code>	<code>MCU_E_PLL_NOT_LOCKED</code>	One of the used PLL's failed to achieve lock
All functions, except <code>Mcu_Init()</code> and <code>Mcu_GetVersionInfo()</code>	<code>MCU_E_UNINIT</code>	The driver is in an uninitialized state.
<code>Mcu_GetVersionInfo()</code>	<code>MCU_E_PARAM_POINTER</code>	Invalid input parameter.
<code>Mcu_Init()</code>	<code>MCU_E_ALREADY_INITIALIZED</code>	The driver is already initialized.

Function	Error Code	Condition triggering the error
Mcu_DisableCmu()	MCU_E_CMU_INDEX_OUT↔ _OF_RANGE	Invalid input parameter.

The driver generates the following DEM errors at runtime.

Table 3.6 Default Errors (reported by DEM)

Function	Error Code	Condition triggering the error
Mcu_GetResetReason()	Mcu_E_TimeoutFailure	Reset flags could not be cleared.
Mcu_GetResetRawValue()	Mcu_E_TimeoutFailure	Reset flags could not be cleared.
Mcu_SetMode()	Mcu_E_TimeoutFailure	The MC_ME or LPU mode transition failed.
Mcu_Init()	Mcu_E_TimeoutFailure	The MC_ME mode transition failed.
Mcu_InitClock()	Mcu_E_TimeoutFailure	The MC_ME mode transition failed.
Mcu_DisableCmu()	Mcu_E_TimeoutFailure	Disable CMU failed.

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like: `#define <Mip>Conf_<Container_ShortName>_<Container_ID>`

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

3.9 Usage in non AUTOSAR context

For the non-ASR environments, both High-Level and IP layer interfaces are exposed and can be used by software units above. These may consist of either higher level NXP SW deliverables, like stacks and middleware, or the application code itself.

The goal of the product is to be versatile, easy to use and integrate with other SW environments, offering both standardized and optimized low-level interfaces.

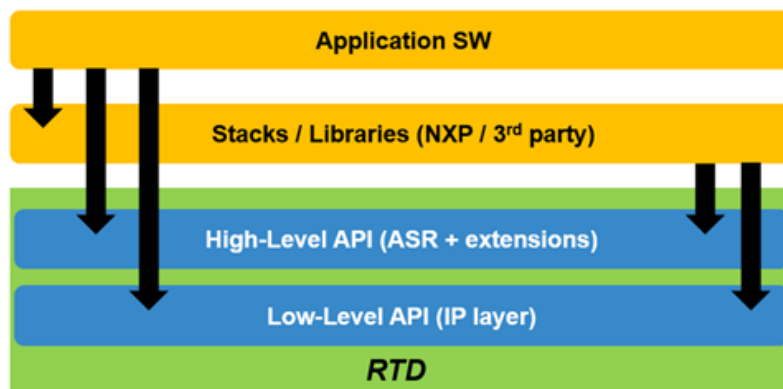


Figure 3.2 RTD Usage in non-ASR Context

In the non-AUTOSAR environment, there are no specific ECU states defined, but an external assumption for the application is that the correct sequence of initialization takes place prior to calling any runtime APIs.

The constraints considered for defining the APIs in the non-AUTOSAR environment are:

- Hardware resources used by the drivers must be properly initialized prior to runtime usage by calling the initialization function with configuration parameters generated by the configuration tools
- All dependencies must be initialized prior to the usage of a driver that depends on them; this translates to a general initialization sequence applicable for all applications, as follows:
 1. Initialization of the device clock tree so other periphery can be initialized
 2. Configuration of the pinout so peripherals with external connections can communicate with the outside world
 3. Configuration of the common resources (e.g. DMA channels, memory regions, access rights, etc) prior to the usage of any runtime functionality that has to benefit from this overall SoC configuration
 4. Initialization of specific drivers
 5. Runtime functionality
 6. Deinitialization (inverse initialization order)

High-level and IP layer functionalities cannot be used simultaneously over the same hardware resource. The granularity of the resource must be defined specifically for each driver, but as a general rule the default granularity considered is the peripheral instance (e.g. Gpt high level APIs cannot be called interchangeably with Pit_Ip low level APIs over the same PIT hardware instance).

The diagram below shows a generic, high-level sequence of calls to RTD interfaces in the non-AUTOSAR scenario:

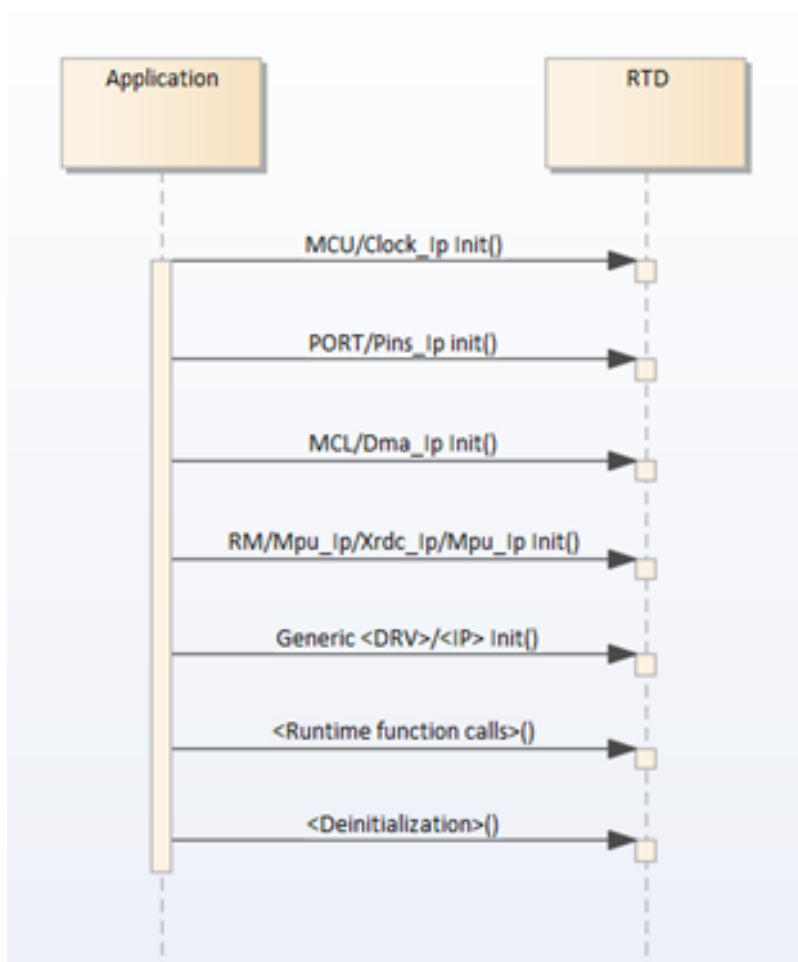


Figure 3.3 Non-AUTOSAR Sequence diagram

Chapter 4

Configuration Parameters

RTD provides configuration schema for the high level driver in EB Tresos proprietary format (**.xdm**), S32CT proprietary format (**.component**), but also in a standardized AUTOSAR format (**.epd**). For the low level driver configuration only the S32CT proprietary format (**.component**) is supported. The configuration data of the project can be saved in EB Tresos proprietary format (**.xdm**) and AUTOSAR standardized format (**.epc**) from EB Tresos and in S32CT proprietary format (**.mex**) and AUTOSAR standardized format (**.epc**) from S32 Design Studio. RTD provides templates for parsing the configuration data and generating configuration structures from it in XPath format, to be used with the EB Tresos generator and in JS format, to be used with the S32CT generator. Since the generation process is not standardized by AUTOSAR, only the two mentioned tools can be used with the provided templates. Using the same input configuration data, the same configuration structures will be obtained using any of the generators.

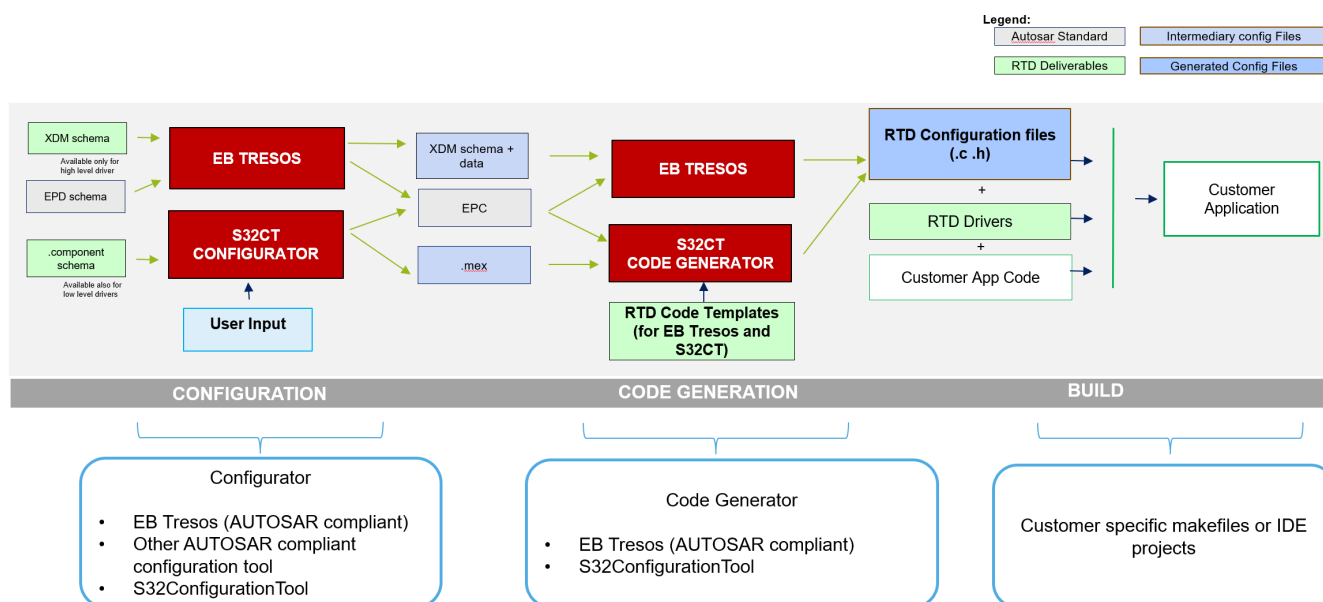


Figure 4.1 Configuration generation flow

- Mcu
- RAM
- POWER

4.1 Mcu

- Module [Mcu](#)
 - Container [McuGeneralConfiguration](#)
 - * Parameter [McuDevErrorDetect](#)
 - * Parameter [McuPowerNotifications](#)
 - * Parameter [McuVersionInfoApi](#)
 - * Parameter [McuGetRamStateApi](#)
 - * Parameter [McuInitClock](#)
 - * Parameter [McuNoPll](#)
 - * Parameter [McuEnterLowPowerMode](#)
 - * Parameter [McuTimeout](#)
 - * Parameter [McuEnableUserModeSupport](#)
 - * Parameter [McuMultiPartitionSupport](#)
 - * Parameter [McuPerformResetApi](#)
 - * Parameter [McuCalloutBeforePerformReset](#)
 - * Parameter [McuVeryLowPowerStopAbortNotification](#)
 - * Parameter [McuPerformResetCallout](#)
 - * Parameter [McuPmcAeNotification](#)
 - * Parameter [McuCmuNotification](#)
 - * Parameter [McuCmuErrorIsrUsed](#)
 - * Parameter [McuErrorIsrNotification](#)
 - * Parameter [McuDisableRcmInit](#)
 - * Parameter [McuDisablePmcInit](#)
 - * Parameter [McuDisableSmcInit](#)
 - * Parameter [McuTimeoutMethod](#)
 - * Parameter [McuRegisterValuesOptimization](#)
 - * Reference [McuInitEcucPartitionRef](#)
 - * Reference [McuEcucPartitionRef](#)
 - Container [McuDebugConfiguration](#)
 - * Parameter [McuDisableDemReportErrorStatus](#)
 - * Parameter [McuGetMidrStructureApi](#)
 - * Parameter [McuDisableCmuApi](#)
 - * Parameter [McuEnablePeripheralCMU](#)
 - * Parameter [McuSRAMRetentionConfigApi](#)
 - * Parameter [McuGetClockFrequencyApi](#)
 - * Parameter [McuGetPowerModeStatetApi](#)

- * Parameter [McuPmcAeConfigApi](#)
- * Parameter [McuAecResetConfigApi](#)
- Container [McuPublishedInformation](#)
 - * Container [McuResetReasonConf](#)
 - Parameter [McuResetReason](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ArReleaseMajorVersion](#)
 - * Parameter [ArReleaseMinorVersion](#)
 - * Parameter [ArReleaseRevisionVersion](#)
 - * Parameter [ModuleId](#)
 - * Parameter [SwMajorVersion](#)
 - * Parameter [SwMinorVersion](#)
 - * Parameter [SwPatchVersion](#)
 - * Parameter [VendorApiInfix](#)
 - * Parameter [VendorId](#)
- Container [McuModuleConfiguration](#)
 - * Parameter [McuNumberOfMcuModes](#)
 - * Parameter [McuRamSectors](#)
 - * Parameter [McuResetSetting](#)
 - * Parameter [McuRTC_CLKINFrequencyHz](#)
 - * Parameter [McuTCLK0_REF_CLKFrequencyHz](#)
 - * Parameter [McuTCLK1_REF_CLKFrequencyHz](#)
 - * Parameter [McuTCLK2_REF_CLKFrequencyHz](#)
 - * Parameter [McuClockSrcFailureNotification](#)
 - * Container [McuAllowedModes](#)
 - Parameter [McuAllowHighSpeedRunMode](#)
 - Parameter [McuAllowVeryLowPowerModes](#)
 - * Container [McuClockSettingConfig](#)
 - Parameter [McuClockSettingId](#)
 - Parameter [McuSysClockUnderMcuControl](#)
 - Parameter [McuScgClkOutSelect](#)
 - Reference [McuClockEcucPartitionRef](#)
 - Container [McuRunClockConfig](#)
 - Parameter [McuPreDivSystemClockFrequency](#)
 - Parameter [McuCoreClockFrequency](#)
 - Parameter [McuSystemClockFrequency](#)

- Parameter [McuBusClockFrequency](#)
- Parameter [McuFlashClockFrequency](#)
- Parameter [McuSystemClockSwitch](#)
- Parameter [McuCoreClockDivider](#)
- Parameter [McuBusClockDivider](#)
- Parameter [McuSlowClockDivider](#)
- Parameter [McuScgClkOutFrequency](#)
- Container [McuVlprClockConfig](#)
 - Parameter [McuPreDivSystemClockFrequency](#)
 - Parameter [McuCoreClockFrequency](#)
 - Parameter [McuSystemClockFrequency](#)
 - Parameter [McuBusClockFrequency](#)
 - Parameter [McuFlashClockFrequency](#)
 - Parameter [McuSystemClockSwitch](#)
 - Parameter [McuCoreClockDivider](#)
 - Parameter [McuBusClockDivider](#)
 - Parameter [McuSlowClockDivider](#)
 - Parameter [McuScgClkOutFrequency](#)
- Container [McuHsrClockConfig](#)
 - Parameter [McuPreDivSystemClockFrequency](#)
 - Parameter [McuCoreClockFrequency](#)
 - Parameter [McuSystemClockFrequency](#)
 - Parameter [McuBusClockFrequency](#)
 - Parameter [McuFlashClockFrequency](#)
 - Parameter [McuSystemClockSwitch](#)
 - Parameter [McuCoreClockDivider](#)
 - Parameter [McuBusClockDivider](#)
 - Parameter [McuSlowClockDivider](#)
 - Parameter [McuScgClkOutFrequency](#)
- Container [McuSystemOSCClockConfig](#)
 - Parameter [McuSOSCUnderMcuControl](#)
 - Parameter [McuSOSCFrequency](#)
 - Parameter [McuSOSCDiv2Frequency](#)
 - Parameter [McuSOSCDiv1Frequency](#)
 - Parameter [McuSOSCEnable](#)
 - Parameter [McuSOSCClockMonitorResetEnable](#)

- Parameter [McuSOSCClockMonitorEnable](#)
- Parameter [McuSOSCDiv2](#)
- Parameter [McuSOSCDiv1](#)
- Parameter [McuSOSCRangeSelect](#)
- Parameter [McuSOSCHighGainOscillatorSelect](#)
- Parameter [McuSOSCEXternalReferenceSelect](#)
- Container [McuSIRCClockConfig](#)
 - Parameter [McuSIRCUnderMcuControl](#)
 - Parameter [McuSIRCFrequency](#)
 - Parameter [McuSIRCDiv2Frequency](#)
 - Parameter [McuSIRCDiv1Frequency](#)
 - Parameter [McuSIRCEnable](#)
 - Parameter [McuSIRCLowPowerEnable](#)
 - Parameter [McuSIRCStopEnable](#)
 - Parameter [McuSIRCDiv2](#)
 - Parameter [McuSIRCDiv1](#)
 - Parameter [McuSIRCRangeSelect](#)
- Container [McuFIRCClockConfig](#)
 - Parameter [McuFIRCUnderMcuControl](#)
 - Parameter [McuFIRCFrequency](#)
 - Parameter [McuFIRCDiv2Frequency](#)
 - Parameter [McuFIRCDiv1Frequency](#)
 - Parameter [McuFIRCEnable](#)
 - Parameter [McuFIRCRegulatorEnable](#)
 - Parameter [McuFIRCDiv2](#)
 - Parameter [McuFIRCDiv1](#)
 - Parameter [McuFIRCRangeSelect](#)
- Container [McuSystemPll](#)
 - Parameter [McuSystemPllUnderMcuControl](#)
 - Parameter [McuSPLLFrequency](#)
 - Parameter [McuSPLLDiv2Frequency](#)
 - Parameter [McuSPLLDiv1Frequency](#)
 - Parameter [McuSPLLEnable](#)
 - Parameter [McuSPLLClockMonitorResetEnable](#)
 - Parameter [McuSPLLClockMonitorEnable](#)
 - Parameter [McuSPLLDiv2](#)

- Parameter [McuSPLLDiv1](#)
- Parameter [McuSPLLInputClkPreDivider](#)
- Parameter [McuSPLLReferenceFrequency](#)
- Parameter [McuSPLLInputFrequency](#)
- Parameter [McuSPLLMultiplier](#)
- Parameter [McuSPLLSelectSourceClock](#)
- Container [McuSIMClockConfig](#)
 - Parameter [McuSIMUnderMcuControl](#)
 - Parameter [McuEIMClockGatingEnable](#)
 - Parameter [McuERMClockGatingEnable](#)
 - Parameter [McuDMAClockGatingEnable](#)
 - Parameter [McuMPUClockGatingEnable](#)
 - Parameter [McuMSCMClockGatingEnable](#)
 - Parameter [McuGPIOClockGatingEnable](#)
- Container [McuSimChipConfiguration](#)
 - Parameter [McuDebugTraceDividerEnable](#)
 - Parameter [McuTRACECLKDivider](#)
 - Parameter [McuTRACECLKFraction](#)
 - Parameter [McuTRACECLKSelect](#)
 - Parameter [McuCLKOUTEnable](#)
 - Parameter [McuCLKOUTDivider](#)
 - Parameter [McuCLKOUTSelect](#)
- Container [McuSimLpoConfiguration](#)
 - Parameter [McuRTCClkSelect](#)
 - Parameter [McuLPOClkSelect](#)
 - Parameter [McuLPO_32KClockEnable](#)
 - Parameter [McuLPO_1KClockEnable](#)
- Container [McuSimFtmConfiguration](#)
 - Parameter [McuFTM3ExternalClockPinSelect](#)
 - Parameter [McuFTM2ExternalClockPinSelect](#)
 - Parameter [McuFTM1ExternalClockPinSelect](#)
 - Parameter [McuFTM0ExternalClockPinSelect](#)
 - Parameter [McuFTM7ExternalClockPinSelect](#)
 - Parameter [McuFTM6ExternalClockPinSelect](#)
 - Parameter [McuFTM5ExternalClockPinSelect](#)
 - Parameter [McuFTM4ExternalClockPinSelect](#)

- Container [McuClkMonitor](#)
 - Container [McuClkMonitor_0](#)
 - Parameter [McuClockMonitorUnderMcuControl](#)
 - Parameter [McuClkMonitorEn](#)
 - Parameter [McuCmuName](#)
 - Parameter [McuClockMonitorDeviation](#)
 - Parameter [McuClockMonitorLatency](#)
 - Parameter [McuAsyncFHHInterruptEn](#)
 - Parameter [McuAsyncFLLInterruptEn](#)
 - Container [McuClkMonitor_1](#)
 - Parameter [McuClockMonitorUnderMcuControl](#)
 - Parameter [McuClkMonitorEn](#)
 - Parameter [McuCmuName](#)
 - Parameter [McuClockMonitorDeviation](#)
 - Parameter [McuClockMonitorLatency](#)
 - Parameter [McuSyncFHHInterruptEn](#)
 - Parameter [McuSyncFLLInterruptEn](#)
- Container [McuPeripheralClockConfig](#)
 - Parameter [McuPeripheralClockUnderMcuControl](#)
 - Parameter [McuPerName](#)
 - Parameter [McuPeripheralClockEnable](#)
 - Parameter [McuPeripheralClockSelect](#)
 - Parameter [McuPeripheralClockDivider](#)
 - Parameter [McuPeripheralFractionalDivider](#)
 - Parameter [McuPeripheralClockFrequency](#)
- Container [McuClockReferencePoint](#)
 - Parameter [McuClockReferencePointFrequency](#)
 - Parameter [McuClockFrequencySelect](#)
- * Container [McuDemEventParameterRefs](#)
 - Reference [MCU_E_TIMEOUT_FAILURE](#)
 - Reference [MCU_E_CLOCK_FAILURE](#)
 - Reference [MCU_E_SWITCHMODE_FAILURE](#)
- * Container [McuModeSettingConf](#)
 - Parameter [McuMode](#)
 - Parameter [McuPowerMode](#)
 - Parameter [McuEnableSleepOnExit](#)

- Reference [McuPowerEcucPartitionRef](#)
- * Container [McuRamSectorSettingConf](#)
 - Parameter [McuRamSectorId](#)
 - Parameter [McuRamDefaultValue](#)
 - Parameter [McuRamSectionBaseAddress](#)
 - Parameter [McuRamSectionSize](#)
 - Parameter [McuRamSectionWriteSize](#)
 - Parameter [McuRamSectionBaseAddrLinkerSym](#)
 - Parameter [McuRamSectionSizeLinkerSym](#)
 - Reference [McuRamEcucPartitionRef](#)
- * Container [McuInterruptEvents](#)
 - Parameter [McuVoltageErrorEvent](#)
 - Parameter [McuAlternateResetEvent](#)
- * Container [McuResetConfig](#)
 - Parameter [McuResetPinFilterBusClockSelect](#)
 - Parameter [McuResetPinFilterInStopMode](#)
 - Parameter [McuResetPinFilterInRunAndWait](#)
 - Container [McuSystemInterruptEnable](#)
 - Parameter [McuResetDelayTime](#)
 - Parameter [McuStopAcknowledgeErrorInterrupt](#)
 - Parameter [McuMDMAPSystemResetInterrupt](#)
 - Parameter [McuSoftwareInterrupt](#)
 - Parameter [McuCoreLockupInterrupt](#)
 - Parameter [McuJTAGResetInterrupt](#)
 - Parameter [McuGlobalInterrupt](#)
 - Parameter [McuExternalResetPinInterrupt](#)
 - Parameter [McuWatchdogInterrupt](#)
 - Parameter [McuCMULossOfClockResetInterrupt](#)
 - Parameter [McuLossOfLockInterrupt](#)
 - Parameter [McuLossOfClockInterrupt](#)
 - Container [McuResetGeneratorConfiguration](#)
 - Parameter [McuRegsOtpReset](#)
 - Parameter [McuCanPhyReset](#)
 - Parameter [McuLinPhyHpReset](#)
 - Parameter [McuLinPhyLpReset](#)
 - Parameter [McuGduReset](#)

- Parameter [McuHviReset](#)
- Parameter [McuDpgaReset](#)
- Parameter [McuTempsensorReset](#)
- Parameter [McuCxpIReset](#)
- * Container [McuPowerControl](#)
 - Container [McuPMC_Config](#)
 - Parameter [McuLowVoltageDetectInterruptEnable](#)
 - Parameter [McuLowVoltageDetectResetEnable](#)
 - Parameter [McuLowVoltageWarningInterruptEnable](#)
 - Parameter [McuLPODisable](#)
 - Parameter [McuLPOTRIMSelect](#)
 - Parameter [McuClockBiasDisable](#)
 - Parameter [McuLowPowerBiasEnable](#)
 - Container [McuPMC_AE_Config](#)
 - Parameter [McuLowVoltageDetectInterruptsOnVLSEnable](#)
 - Parameter [McuLowVoltageDetectInterruptsOnVDDCEnable](#)
 - Parameter [McuHighVoltageDetectInterruptOnVDDEnable](#)
 - Parameter [McuHighVoltageDetectInterruptOnVDDINTAndVDD15Enable](#)
 - Parameter [McuLinphySupplyEnable](#)
 - Parameter [McuVDDCEnable](#)
 - Parameter [McuLvdVlsSelect](#)
 - Parameter [McuLinphySupplySelect](#)
 - Parameter [McuVddVoltageLevelSelect](#)

4.1.1 Module Mcu

Configuration of the MicroController Unit (MCU) module.

Included containers:

- [McuGeneralConfiguration](#)
- [McuDebugConfiguration](#)
- [McuPublishedInformation](#)
- [CommonPublishedInformation](#)
- [McuModuleConfiguration](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantSupport	true

Configuration Parameters

Property	Value
supportedConfigVariants	VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.1.2 Container McuGeneralConfiguration

This container contains the general configuration for the MCU driver.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.3 Parameter McuDevErrorDetect

Pre-processor switch for enabling the default error detection and reporting to the DET.

The switch McuDevErrorDetect shall switch the Default Error Tracer (Det) detection and notification ON or OFF.

The detection of default errors is configurable (ON/OFF) at precompile time.

#define MCU_DEV_ERROR_DETECT (STD_ON)/(STD_OFF) will be generated in Mcu_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.4 Parameter McuPowerNotifications

STD_ON to enable receive notifications from POWER driver.

#define MCU_POWER_NOTIFICATIONS (STD_ON)/(STD_OFF) will be generated in Mcu_Cfg.h file.

Note: 'Voltage trace interrupt' is enabled. This node must be enabled.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
default Value	true

4.1.5 Parameter McuVersionInfoApi

Pre-processor switch to enable/disable the API to read out the modules version information.

`#define MCU_VERSION_INFO_API (STD_ON)/(STD_OFF)` will be generated in `Mcu_Cfg.h` file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.6 Parameter McuGetRamStateApi

Pre-processor switch to enable/disable the API `Mcu_GetRamState`.

`#define MCU_GET_RAM_STATE_API (STD_ON)/(STD_OFF)` will be generated in `Mcu_Cfg.h` file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.7 Parameter McuInitClock

If this parameter is set to **FALSE**, the clock initialization has to be disabled from the MCU driver. This concept applies when there are some write once clock registers and a bootloader is present. If this parameter is set to **TRUE**, the MCU driver is responsible of the clock initialization

`#define MCU_INIT_CLOCK (STD_ON)/(STD_OFF)` will be generated in `Mcu_Cfg.h` file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.1.8 Parameter McuNoPll

This parameter shall be set **True**, if the H/W does not have a PLL or the PLL circuitry is enabled after the power on without S/W intervention. In this case `MCU_DistributePllClock` has to be disabled and `MCU_GetPllStatus` has to return `MCU_PLL_STATUS_UNDEFINED`. Otherwise this parameters has to be set **False**.

`#define MCU_NO_PLL (STD_ON)/(STD_OFF)` will be generated in `Mcu_Cfg.h` file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.1.9 Parameter McuEnterLowPowerMode

If this parameter has been configured to 'TRUE', the function 'Mcu_SetMode()' shall not be impacted and behave as specified.

If this parameter has been configured to 'FALSE', the function 'Mcu_SetMode()' shall not perform the transition to any low power modes as are 'STOP' or 'HALT' or any other mode, where the core stops execution.

#define MCU_ENTER_LOW_POWER_MODE (STD_ON)/(STD_OFF) will be generated in Mcu_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.1.10 Parameter McuTimeout

This parameter represents the maximum number of loops for blocking functionality.

The maximum time needed for a MC_ME transition from DRUN to DRUN with keeping PLL running is 3 ms.

Please take this into consideration when choosing the value for this parameter.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	50000
max	4294967295
min	0

4.1.11 Parameter McuEnableUserModeSupport

When this parameter is enabled, the MDL module will adapt to run from User Mode, with the following measures:

Configuration Parameters

a) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.

b) other module specific measures

for more information, please see chapter 5.7 User Mode Support in IM

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.12 Parameter McuMultiPartitionSupport

This parameter globally enables the possibility to support multi-partition.

If this parameter is enabled, at least one EcucPartition needs to be defined (in all variants).

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.13 Parameter McuPerformResetApi

Pre-processor switch to enable/disable the use the Mcu_PerformReset() API.

OFF - Mcu_PerformReset() API is not used.

ON - Mcu_PerformReset() API is used.

#define MCU_PERFORM_RESET_API (STD_ON)/(STD_OFF) will be generated in Mcu_Cfg.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.14 Parameter McuCalloutBeforePerformReset

Check this if you want a callout function, called by MCU right before `Mcu_PerformReset()`.

This parameter is available for configuration only if "McuPerformResetApi" is ON.

`#define MCU_RESET_CALLOUT_USED (STD_ON)/(STD_OFF)` will be generated in `Mcu_Cfg.h` file.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.15 Parameter McuVeryLowPowerStopAbortNotification

Function name of callout. This function will be called when entering VLPS is aborted.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	NULL_PTR

4.1.16 Parameter McuPerformResetCallout

Function name of callout.

The field is editable only if "McuCalloutBeforePerformReset" is ON.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	NULL_PTR

4.1.17 Parameter McuPmcAeNotification

Function pointer to callback function.

This notification will be called by the driver for PMC_AE

This node is supported on S32M24X derivatives only

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	NULL_PTR

4.1.18 Parameter McuCmuNotification

Function pointer to callback function.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	NULL_PTR

4.1.19 Parameter McuCmuErrorIsrUsed

Check this if clock source failure notifications are enabled (i.e. `McuModuleConfiguration/McuClockSrcFailureNotification = 'ENABLED'`).

`#define MCU_CMU_ERROR_ISR_USED (STD_ON)/(STD_OFF)` will be generated in `Mcu_Cfg.h` file.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.20 Parameter McuErrorIsrNotification

Function name of callout. This function will be called by the error ISR.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	NULL_PTR

4.1.21 Parameter McuDisableRcmInit

To allow integration with non-Autosar Application Initialization so that settings done by Applnit will not to be done again by MCU driver.

If this parameter is set to TRUE, the Reset Control Module (RMC) initialization has to be disabled in the MCU driver.

If this parameter is set to FALSE, the MCU driver is responsible for the Reset Control Module (RMC) initialization.

#define POWER_IP_DISABLE_RCM_INIT (STD_ON)/(STD_OFF) will be generated in Power_Ip_Cfg_Defines.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.22 Parameter McuDisablePmcInit

To allow integration with non-Autosar Application Initialization so that settings done by Applnit will not to be done again by MCU driver.

If this parameter is set to TRUE, the Power Management Controller (PMC) initialization has to be disabled in the MCU driver.

If this parameter is set to FALSE, the MCU driver is responsible for the Power Management Controller (PMC) initialization.

#define POWER_IP_DISABLE_PMC_INIT (STD_ON)/(STD_OFF) will be generated in Power_Ip_Cfg_Defines.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.23 Parameter McuDisableSmcInit

To allow integration with non-Autosar Application Initialization so that settings done by Applnit will not to be done again by MCU driver.

If this parameter is set to TRUE, the System Mode Controller (SMC) initialization has to be disabled in the MCU driver.

If this parameter is set to FALSE, the MCU driver is responsible for the System Mode Controller (SMC) initialization.

#define POWER_IP_DISABLE_SMC_INIT (STD_ON)/(STD_OFF) will be generated in Power_Ip_Cfg_Defines.h file.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.24 Parameter McuTimeoutMethod

McuTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: If OSIF_COUNTER_SYSTEM or OSIF_COUNTER_CUSTOM are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Configuration Parameters

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	OSIF_COUNTER_DUMMY
literals	['OSIF_COUNTER_DUMMY', 'OSIF_COUNTER_SYSTEM', 'OSIF_COUNTER_CUSTOM']

4.1.25 Parameter McuRegisterValuesOptimization

Check this if register values of CMUs, PCFS will be generated by configuration tool.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	DISABLED
literals	['ENABLED', 'DISABLED']

4.1.26 Reference McuInitEcucPartitionRef

Maps a config part of Mcu_Init to ECUC partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

4.1.27 Reference McuEcucPartitionRef

Maps the MCU driver to zero or multiple ECUC partitions to make the modules API available in this partition.

Tags: *atp.Status=draft*

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

4.1.28 Container McuDebugConfiguration

This container contains option for non-ASR APIs used for debug or extra-implementation.

Note: Implementation Specific Parameter.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.29 Parameter McuDisableDemReportErrorStatus

Enable/Disable the API for reporting the Dem Error.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.30 Parameter McuGetMidrStructureApi

Enable/Disable the API for `Mcu_GetMidrStructure()`.

Get information from SIUL2 MIDRn registers.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.31 Parameter McuDisableCmuApi

Enable/Disable the API for disabling the clock monitoring unit.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.32 Parameter McuEnablePeripheralCMU

Enable/Disable Peripheral CMU for S32K11X

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.33 Parameter McuSRAMRetentionConfigApi

Enable/Disable the API for SRAM retention configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.34 Parameter McuGetClockFrequencyApi

Enable/Disable the API for `Mcu_GetClockFrequency()`.

Return the frequency of a given clock.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.35 Parameter McuGetPowerModeStatetApi

Enable/Disable the API for `Get Power Mode state: Mcu_GetPowerMode_State()`.

Get information regarding current power mode, enabled clocks, etc (content of `SMC_PMSTART` register).

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.36 Parameter McuPmcAeConfigApi

Enable/Disable the API for `Mcu_PmcAeConfig()`.

Configure the Power Management Controller AE.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.37 Parameter McuAecResetConfigApi

Enable/Disable the API for `Mcu_AecResetConfig()`.

Configure the reset generator AEC.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.38 Container McuPublishedInformation

Container holding all MCU specific published information parameters.

Included subcontainers:

- [McuResetReasonConf](#)

Configuration Parameters

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.39 Container McuResetReasonConf

This container contains the configuration for the different type of reset reason that can be retrieved from `Mcu_GetResetReason` Api.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.1.40 Parameter McuResetReason

The parameter represents the different type of reset that a Micro supports. This parameter is referenced by the parameter `EcuMResetReason` in the ECU State manager module.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	5
max	255
min	0

4.1.41 Container CommonPublishedInformation

Common container, aggregated by all modules.

It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.42 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.1.43 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION

Configuration Parameters

Property	Value
defaultValue	7
max	7
min	7

4.1.44 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.45 Parameter ModuleId

Module ID of this module from Module List.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	101
max	101
min	101

4.1.46 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	3
max	3
min	3

4.1.47 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.48 Parameter SwPatchVersion

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.49 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the Implementation specific name is generated as follows:

`<ModuleName>_<VendorId>_<VendorApiInfix>`.

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity >

1. It shall not be used for modules with upper multiplicity =1.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	

4.1.50 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

4.1.51 Container McuModuleConfiguration

This container contains the configuration for the MCU driver.

Included subcontainers:

- [McuAllowedModes](#)
- [McuClockSettingConfig](#)
- [McuDemEventParameterRefs](#)
- [McuModeSettingConf](#)
- [McuRamSectorSettingConf](#)
- [McuInterruptEvents](#)
- [McuResetConfig](#)
- [McuPowerControl](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.52 Parameter McuNumberOfMcuModes

This parameter shall represent the number of Modes available for the MCU (from "McuModeSettingConf" list).

CalculationFormula = Number of configured "McuModeSettingConf".

This parameter is not used.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	255
min	1

4.1.53 Parameter McuRamSectors

This parameter shall represent the number of RAM sectors available for the MCU (from "McuRamSectorSettingConf" list).

CalculationFormula = Number of configured "McuRamSectorSettingConf".

This parameter is not used.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	4294967295
min	0

4.1.54 Parameter McuResetSetting

This parameters applies to the function `Mcu_PerformReset()`, which performs a microcontroller reset using the hardware feature of the microcontroller.

Note: This parameter is not used by the current Implementation.

Software Reset occurs when `Mcu_PerformReset()` function is called.

This parameter is not used.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	1
max	255
min	1

4.1.55 Parameter McuRTC_CLKINFrequencyHz

RTC_CLKIN Frequency [Hz].

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	32768.0
max	1000000.0
min	0.0

4.1.56 Parameter McuTCLK0_REF_CLKFrequencyHz

TCLK0_REF_CLK Frequency [Hz].

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	32000.0
max	2.0E7
min	0.0

4.1.57 Parameter McuTCLK1_REF_CLKFrequencyHz

TCLK1_REF_CLK Frequency [Hz].

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	32000.0
max	2.0E7
min	0.0

4.1.58 Parameter McuTCLK2_REF_CLKFrequencyHz

TCLK2_REF_CLK Frequency [Hz].

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	32000.0
max	2.0E7
min	0.0

4.1.59 Parameter McuClockSrcFailureNotification

Enables/Disables clock failure notification.

In case this feature is not supported by HW the setting should be disabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	DISABLED
literals	['ENABLED', 'DISABLED']

4.1.60 Container McuAllowedModes

Configures SMC_PMPROT register. The PMPROT register can be written only once after any system reset.

If the MCU is configured for a disallowed or reserved power mode, the MCU remains in its current power mode.

For example, if the MCU is in normal RUN mode and AVLP is 0, an attempt to enter VLPR mode using PMCTRL[RUNM] is blocked and PMCTRL[RUNM] remains 00b, indicating the MCU is still in Normal Run mode.

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.61 Parameter McuAllowHighSpeedRunMode

This is a write-once parameter

SMC_PMPROT[AHSRUN] - Allow High Speed Run mode

Provided the appropriate control bits are set up in PMCTRL, this write-once field allows the MCU to enter High Speed Run mode (HSRUN).

0 - HSRUN is not allowed

1 - HSRUN is allowed

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.62 Parameter McuAllowVeryLowPowerModes

This is a write-once parameter

Provided the appropriate control bits are set up in PMCTRL, this write-once field allows the MCU to enter any very-low-power mode (VLPR, and VLPS).

0 - VLPR and VLPS are not allowed.

1 - VLPR and VLPS are allowed.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.63 Container McuClockSettingConfig

This container contains the configuration for the Clock settings of the MCU.

Included subcontainers:

- [McuRunClockConfig](#)
- [McuVlprClockConfig](#)
- [McuHsrunClockConfig](#)
- [McuSystemOSCClockConfig](#)
- [McuSIRCClockConfig](#)
- [McuFIRCClockConfig](#)
- [McuSystemPII](#)
- [McuSIMClockConfig](#)
- [McuClkMonitor](#)
- [McuPeripheralClockConfig](#)
- [McuClockReferencePoint](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.1.64 Parameter McuClockSettingId

The Id of this McuClockSettingConfig to be used as argument for the API call `Mcu_InitClock()`.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	0
max	255
min	0

4.1.65 Parameter McuSysClockUnderMcuControl

0 - System clock tree is NOT under mcu control.

1 - System clock is under mcu control.

If this is set to false, the MCU code will not configure the SCG_xCCR register when Mcu_InitClock is called

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.66 Parameter McuScgClkOutSelect

SCG_CLKOUTCNFG[CLKOUTSEL] - SCG Clkout Select.

This register controls which SCG clock source is selected to be ported out to the CLKOUT pin.

0 - SCG SLOW Clock (FLASH_CLK Clock).

1 - System OSC

2 - Slow IRC

3 - Fast IRC

6 - System PLL

The selected clock must be enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	SLOW_CLK

Property	Value
literals	['SLOW_CLK', 'SOSC_CLK', 'SIRC_CLK', 'FIRC_CLK', 'SPLL_CLK']

4.1.67 Reference McuClockEcucPartitionRef

Maps a config part of Clock to ECUC partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.1.68 Container McuRunClockConfig

This container configures the system clock source and the system clock dividers for the core, platform, external and bus clock domains when in Run mode only.

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.69 Parameter McuPreDivSystemClockFrequency

Run Core clock - Pre Divide System Clock Frequency.

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: PREDIV_SYS_CLK is only available in S32K148.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	4.8E7
max	1.6E8
min	0.0

4.1.70 Parameter McuCoreClockFrequency

Run Core clock - Clocks the ARM core, divided by DIVCORE bits inside SCG

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	4.8E7
max	8.0E7
min	0.0

4.1.71 Parameter McuSystemClockFrequency

Run System clock - Clocks the Crossbar, NVIC, Flash controller, FTM and PDB, etc.

RUN_SYS_CLK can run up to CORE_CLK.

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	8.0E7
max	8.0E7
min	0.0

4.1.72 Parameter McuBusClockFrequency

Run Bus clock - BUS_CLK Clocks the Peripherals, divided by DIVBUS bits inside SCG

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	4.0E7
max	4.8E7
min	0.0

4.1.73 Parameter McuFlashClockFrequency

Run Flash clock - Clocks the flash module, divided by DIVSLOW bits inside SCG

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Configuring RUN_FLASH_CLK to lower frequencies than 24MHz (S32K11X) and 20Mhz (S32K14xW and S32M242) and 26.67MHz (S32K14X and S32M241) adds wait states and no power saving. It is recommended to configure it as close to 24MHz or 20Mhz or 26.67MHz.

Configuration Parameters

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	2.0E7
max	2.0E7
min	0.0

4.1.74 Parameter McuSystemClockSwitch

Run System Clock Select. Configure the SCG_RCCR[SCS] register field.

The system clock is either:

- System OSC (SCG_RCCR[SCS]=1)
- Slow IRC (SCG_RCCR[SCS]=2)
- Fast IRC (SCG_RCCR[SCS]=3)
- System PLL (SCG_RCCR[SCS]=6)

Value extracted from Resource: MCU.RunSystemClkSource.List

The selected clock must be enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FIRC_CLK
literals	['SOSC_CLK', 'SIRC_CLK', 'FIRC_CLK', 'SPLL_CLK']

4.1.75 Parameter McuCoreClockDivider

Configures the `SCG_RCCR[DIVCORE]` bitfield

This parameter represents the core clock divider.

Note: implementation specific parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	16
min	1

4.1.76 Parameter McuBusClockDivider

Configures the `SCG_RCCR[DIVBUS]` bitfield

This parameter represents the bus clock divider.

Note: implementation specific parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	16
min	1

4.1.77 Parameter McuSlowClockDivider

Configures the `SCG_RCCR[DIVSLOW]` bitfield.

Configuration Parameters

This parameter represents the flash clock divider.

Note: Configuring DIVSLOW to lower frequencies than supported flash frequencies mentioned in datasheet (fFLASH) adds wait states and no power saving. It is recommended to configure DIVSLOW as close to fFLASH.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	3
max	8
min	1

4.1.78 Parameter McuScgClkOutFrequency

This is frequency of SCG clockout. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	1000000.0
max	1.6E8
min	0.0

4.1.79 Container McuVlprClockConfig

Selects the clock source generating the system clock in VLPR mode.

The clock dividers cannot be changed while in VLPR mode.

They must be programmed prior to entering VLPR mode to guarantee

- the core/system and bus clocks are less than or equal to 4 MHz

- the flash memory clock is less than or equal to 1 MHz.

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.80 Parameter McuPreDivSystemClockFrequency

VLPR Core clock - Pre Divide System Clock Frequency.

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: PREDIV_SYS_CLK is only available in S32K148.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	8000000.0
max	8000000.0
min	0.0

4.1.81 Parameter McuCoreClockFrequency

VLPR Core clock - Clocks the ARM core, divided by DIVCORE bits inside SCG

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	1000000.0
max	1000000.0
min	0.0

4.1.82 Parameter McuSystemClockFrequency

VLPR System clock - Clocks the Crossbar, NVIC, Flash controller, FTM and PDB,etc.

VLPR_SYS_CLK can run up to CORE_CLK.

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	1000000.0
max	1000000.0
min	0.0

4.1.83 Parameter McuBusClockFrequency

VLPR Bus clock - BUS_CLK Clocks the Peripherals, divided by DIVBUS bits inside SCG

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	1000000.0
max	1000000.0
min	0.0

4.1.84 Parameter McuFlashClockFrequency

VLPR Flash clock - Clocks the flash module, divided by DIVSLOW bits inside SCG

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Configuring VLPR_FLASH_CLK to lower frequencies than 1MHz (S32K14X and S32K11X and S32M241) and 0.25MHz (S32K14xW and S32M242) adds wait states and no power saving. It is recommended to configure it as close to 1MHz or 0.25MHz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	250000.0
max	250000.0
min	0.0

4.1.85 Parameter McuSystemClockSwitch

VLPR System Clock Select. Configure the SCG_VCCR[SCS] register field.

The system clock is either:

- Slow IRC (SCG_VCCR[SCS]=2)

Configuration Parameters

Value extracted from Resource: *MCU.VlprSystemClkSource.List*

The selected clock must be enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	SIRC_CLK
literals	['SIRC_CLK']

4.1.86 Parameter McuCoreClockDivider

Configures the *SCG_VCCR[DIVCORE]* bitfield

This parameter represents the core clock divider.

Note: implementation specific parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	8
max	16
min	1

4.1.87 Parameter McuBusClockDivider

Configures the *SCG_VCCR[DIVBUS]* bitfield

This parameter represents the bus clock divider.

Note: implementation specific parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	1
max	16
min	1

4.1.88 Parameter McuSlowClockDivider

Configures the SCG_VCCR[DIVSLOW] bitfield.

This parameter represents the flash clock divider.

Note: Configuring DIVSLOW to lower frequencies than supported flash frequencies mentioned in datasheet (fFLASH) adds wait states and no power saving. It is recommended to configure DIVSLOW as close to fFLASH.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	4
max	8
min	1

4.1.89 Parameter McuScgClkOutFrequency

This is frequency of SCG clockout. It is given in Hz.

In VLPR mode, the FIRC, SOSC, SPLL clocks are disabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	1000000.0
max	1.6E8
min	0.0

4.1.90 Container McuHsruntimeClockConfig

This container configures the system clock source and the system clock dividers for the core, platform, external and bus clock domains when in HSRUN mode only.

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.91 Parameter McuPreDivSystemClockFrequency

HSRUN System clock - Pre Divide System Clock Frequency.

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: PREDIV_SYS_CLK is only available in S32K148.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	4.8E7
max	1.6E8
min	0.0

4.1.92 Parameter McuCoreClockFrequency

HSRUN Core clock - Clocks the ARM core, divided by DIVCORE bits inside SCG

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	4.8E7
max	1.12E8
min	0.0

4.1.93 Parameter McuSystemClockFrequency

HSRUN System clock - Clocks the Crossbar, NVIC, Flash controller, FTM and PDB, etc.

HSRUN_SYS_CLK can run up to CORE_CLK.

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	1.12E8
max	1.12E8
min	0.0

4.1.94 Parameter McuBusClockFrequency

HSRUN Bus clock - BUS_CLK Clocks the Peripherals, divided by DIVBUS bits inside SCG

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	5.6E7
max	5.6E7
min	0.0

4.1.95 Parameter McuFlashClockFrequency

HSRUN Flash clock - Clocks the flash module, divided by DIVSLOW bits inside SCG

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Configuring HSRUN_FLASH_CLK to lower frequencies than 28MHz adds wait states and no power saving. It is recommended to configure it as close to 28MHz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	2.8E7
max	2.8E7
min	0.0

4.1.96 Parameter McuSystemClockSwitch

HSRUN System Clock Select. Configure the SCG_HCCR[SCS] register field.

The system clock is either:

- Fast IRC (SCG_HCCR[SCS]=3)

- System PLL (SCG_HCCR[SCS]=6)

Value extracted from Resource: MCU.HsruntimeSystemClkSource.List

The selected clock must be enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	FIRC_CLK
literals	['FIRC_CLK', 'SPLL_CLK']

4.1.97 Parameter McuCoreClockDivider

Configures the SCG_HCCR[DIVCORE] bitfield

This parameter represents the core clock divider.

Configuration Parameters

Note: implementation specific parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	16
min	1

4.1.98 Parameter McuBusClockDivider

Configures the SCG_HCCR[DIVBUS] bitfield

This parameter represents the bus clock divider.

Note: implementation specific parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	16
min	1

4.1.99 Parameter McuSlowClockDivider

Configures the SCG_HCCR[DIVSLOW] bitfield.

This parameter represents the flash clock divider.

Note: Configuring DIVSLOW to lower frequencies than supported flash frequencies mentioned in datasheet (fFLASH) adds wait states and no power saving. It is recommended to configure DIVSLOW as close to fFLASH.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	2
max	8
min	1

4.1.100 Parameter McuScgClkOutFrequency

This is frequency of SCG clockout. It is given in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1000000.0
max	1.6E8
min	0.0

4.1.101 Container McuSystemOSCClockConfig

Configures System OSC registers.

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.102 Parameter McuSOSCUnderMcuControl

0 - System OSC is NOT under mcu control.

1 - System OSC is under mcu control.

If this is set to false, the MCU code will not configure the SOSC registers when *Mcu_InitClock* is called

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.1.103 Parameter McuSOSCFrequency

This is the SOSC frequency for the specific instance of the

McuClockReferencePoint container. It is expressed in Hz.

If PLL is used, then oscillator needs to be in high range only, *SCG_SOSCCFG[RANGE]* on 11 as used in reference clock.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	0.0

Property	Value
max	4.0E7
min	0.0

4.1.104 Parameter McuSOSCDiv2Frequency

This is the SOSC Divider 2 frequency for the specific instance of the McuClockReferencePoint container. It is expressed in Hz.

This should be configured to 40 MHz or less in RUN/HSRUN mode.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	8000000.0
max	4.0E7
min	0.0

4.1.105 Parameter McuSOSCDiv1Frequency

This is the SOSC Divider 1 frequency for the specific instance of the McuClockReferencePoint container. It is expressed in Hz.

This should be configured to 40 MHz or less in RUN/HSRUN mode.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

Configuration Parameters

Property	Value
defaultValue	8000000.0
max	4.0E7
min	0.0

4.1.106 Parameter McuSOSCEnable

SCG_SOSCCSR[SOSCEN] - System OSC Enable

0 - System OSC is disabled.

1 - System OSC is enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.107 Parameter McuSOSCClockMonitorResetEnable

SCG_SOSCCSR[SOSCCMRE] - System OSC Clock Monitor Reset Enable

0 - Clock Monitor generates interrupt when error detected.

1 - Clock Monitor generates reset when error detected.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.108 Parameter McuSOSCClockMonitorEnable

SCG_SOSCCSR[SOSCCM] - System OSC Clock Monitor

Enables the clock monitor, if the clock source is disabled in a low power mode then the clock monitor is also disabled in the low power mode. When the clock monitor is disabled in a low power mode, it remains disabled until the clock valid flag is set following exit from the low power mode.

0 - System OSC Clock Monitor is disabled.

1 - System OSC Clock Monitor is enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.109 Parameter McuSOSCDiv2

Configures SCG_SOSCDIV[SOSCDIV2].

System OSC Clock Divide 2.

Clock divider 2 for System OSC. Used by bus clock modules that need an asynchronous clock source.

0 - Output disabled.

Note: Implementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	64
min	0

4.1.110 Parameter McuSOSCDiv1

Configures SCG_SOSCDIV[SOSCDIV1]

System OSC Clock Divide 1.

Clock divider 1 for System OSC. Used to generate the system clock source and by platform clock modules that need an asynchronous clock source.

0 - Output disabled.

Note: Implementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	1
max	64
min	0

4.1.111 Parameter McuSOSCRangeSelect

SCG_SOSCCFG[RANGE] - System OSC Range Select

Selects the frequency range for the system crystal oscillator (OSC)

NOTE The following constraints are not checked by the xdm schema:

If PLL is used, then oscillator needs to be in high range only, SCG_SOSCCFG[RANGE] on 11 as used in reference clock.

- Medium frequency range selected for the crystal oscillator of 4 MHz to 8 MHz.

- High frequency range selected for the crystal oscillator of 8 MHz to 40 MHz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD

Property	Value
default Value	HIGH_FREQ_RANGE
literals	['MEDIUM_FREQ_RANGE', 'HIGH_FREQ_RANGE']

4.1.112 Parameter McuSOSCHighGainOscillatorSelect

SCG_SOSCCFG[HGO] - High Gain Oscillator Select

Controls the crystal oscillator power mode of operations.

unchecked - Configure crystal oscillator for low-power operation

checked - Configure crystal oscillator for high-gain operation

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.113 Parameter McuSOSCEXternalReferenceSelect

SCG_SOSCCFG[EREFS] - External Reference Select

Selects the source for the external reference clock.

unchecked - Internal oscillator of OSC requested.

checked - External reference clock from PAD pin selected

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Configuration Parameters

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
default Value	true

4.1.114 Container McuSIRCClockConfig

Configures Slow IRC (SIRC) registers.

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.115 Parameter McuSIRCUnderMcuControl

0 - Slow IRC is NOT under mcu control.

1 - Slow IRC is under mcu control.

If this is set to false, the MCU code will not configure the SIRC registers when Mcu_InitClock is called

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.116 Parameter McuSIRCFrequency

This is the SIRC frequency for the specific instance of the McuClockReferencePoint container. It is expressed in Hz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	0.0
max	8000000.0
min	0.0

4.1.117 Parameter McuSIRCDiv2Frequency

This is the SIRC Divider 2 frequency for the specific instance of the McuClockReferencePoint container. It is expressed in Hz.

This should be configured to 8 MHz or less.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	0.0
max	8000000.0
min	0.0

4.1.118 Parameter McuSIRCDiv1Frequency

This is the SIRC Divider 1 frequency for the specific instance of the McuClockReferencePoint container. It is expressed in Hz.

This should be configured to 8 MHz or less in RUN/HSRUN mode and to 4 MHz or less in VLPR mode.

Property	Value
type	ECUC-FLOAT-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	0.0
max	8000000.0
min	0.0

4.1.119 Parameter McuSIRCEnable

SCG_SIRCCSR[SIRCEN] - Slow IRC Enable

0 - Slow IRC is disabled.

1 - Slow IRC is enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	true

4.1.120 Parameter McuSIRCLowPowerEnable

SCG_SIRCCSR[SIRCLPEN] - Slow IRC Low Power Enable

0 - Slow IRC is disabled in VLP modes.

1 - Slow IRC is enabled in VLP modes.

SCG_SIRCCSR[SIRCLPEN] bit field is applicable for VLPS mode only.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.121 Parameter McuSIRCStopEnable

SCG_SIRCCSR[SIRCSTEN] - Slow IRC Enable

0 - Slow IRC is disabled.

1 - Slow IRC is enabled.

SCG_SIRCCSR[SIRCSTEN] bit field is not applicable and should be ignored.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.122 Parameter McuSIRCDiv2

Configures SCG_SIRCDIV[SIRCDIV2].

Slow IRC Clock Divider 2.

Clock divider 2 for Slow IRC. Used by bus clock modules that need an asynchronous clock source.

0 - Output disabled.

Note: Implementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Configuration Parameters

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	2
max	64
min	0

4.1.123 Parameter McuSIRCDiv1

Configures SCG_SIRCDIV[SIRCDIV1].

Slow IRC Clock Divider 1.

Clock divider 1 for Slow IRC. Used to generate the system clock source and by platform clock modules that need an asynchronous clock source.

0 - Output disabled.

Note: Implementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	64
min	0

4.1.124 Parameter McuSIRCRangeSelect

SCG_SIRCCFG[RANGE] - Selects the Frequency Range

Slow IRC low range clock (2 MHz)

Slow IRC high range clock (8 MHz)

Note: The SIRC clock is chosen as source clock that must be sacrificed to be ON at all times.

Add addition, Software should not configure the SCG_SIRCCFG[RANGE] to any value other than HIGH_RANGE_CLOCK.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	HIGH_RANGE_CLOCK
literals	['HIGH_RANGE_CLOCK']

4.1.125 Container McuFIRCClockConfig

Configures Fast IRC (FIRC) registers.

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.126 Parameter McuFIRCUnderMcuControl

0 - Fast IRC is NOT under mcu control.

1 - Fast IRC is under mcu control.

If this is set to false, the MCU code will not configure the FIRC registers when Mcu_InitClock is called

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.127 Parameter McuFIRCFrequency

This is the FIRC frequency for the specific instance of the *McuClockReferencePoint* container. It is expressed in Hz.

Note: Implementation specific Container.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	0.0
max	4.8E7
min	0.0

4.1.128 Parameter McuFIRCDiv2Frequency

This is the FIRC Divider 2 frequency for the specific instance of the *McuClockReferencePoint* container. It is expressed in Hz.

This should be configured to 48 MHz or less in RUN/HSRUN mode.

Note: Implementation specific Container.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	2.4E7
max	4.8E7
min	0.0

4.1.129 Parameter McuFIRCDiv1Frequency

This is the FIRC Divider 1 frequency for the specific instance of the *McuClockReferencePoint* container. It is expressed in Hz.

This should be configured to 48 MHz or less in RUN/HSRUN mode.

Note: Implementation specific Container.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	4.8E7
max	4.8E7
min	0.0

4.1.130 Parameter McuFIRCEnable

SCG_FIRCCSR[FIRCEN] - Fast IRC Enable

0 - Fast IRC is disabled.

1 - Fast IRC is enabled.

Note: Implementation specific Container.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.131 Parameter McuFIRCRregulatorEnable

Fast IRC Regulator Enable

0 - Fast IRC Regulator is disabled. (SCG_FIRCCSR[FIRCREGOFF] = 1)

1 - Fast IRC Regulator is enabled. (SCG_FIRCCSR[FIRCREGOFF] = 0)

When Fast IRC is used, FIRCREGOFF must be 0. Fast IRC cannot be operated with FIRCREGOFF=1.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.132 Parameter McuFIRCDiv2

Configures SCG_FIRCDIV[FIRCDIV2]

Fast IRC Clock Divider 2.

Clock divider 2 for the Fast IRC. Used by bus clock modules that need an asynchronous clock source.

0 - Output disabled.

Note: Implementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	2
max	64
min	0

4.1.133 Parameter McuFIRCDiv1

Configures SCG_FIRCDIV[FIRCDIV1]

Fast IRC Clock Divider 1.

Clock divider 1 for Fast IRC. Used to generate the system clock source and by platform clock modules that need an asynchronous clock source.

0 - Output disabled.

Note: Implementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	64
min	0

4.1.134 Parameter McuFIRCRangeSelect

SCG_FIRCCFG[RANGE] - Selects the Frequency Range.

00b - Fast IRC is trimmed to 48 MHz.

01b - Reserved.

10b - Reserved.

11b - Reserved.

Note: Software should not configure the SCG_FIRCCFG[RANGE] to any value other than Fast IRC is trimmed to 48 MHz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	TRIMMED_TO_48MHZ
literals	['TRIMMED_TO_48MHZ']

4.1.135 Container McuSystemPll

This container provides the specific configuration for the System PLL.

Note: Implementation Specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.136 Parameter McuSystemPllUnderMcuControl

Set this to TRUE if System PLL is under mcu control

If it is FALSE then the mcu driver will not write the corresponding registers.

The user must still set the values - they are used by the clock reference points

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.137 Parameter McuSPLLFrequency

This is the System PLL frequency for the specific instance of the *McuClockReferencePoint* container. It is expressed in Hz.

Note: $Pll_freq = (McuSPLLReferenceFrequency * McuSPLLMultiplier)/2$

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0.0
max	1.6E8
min	0.0

4.1.138 Parameter McuSPLLDiv2Frequency

This is the System PLL Divider 2 frequency for the specific instance of the *McuClockReferencePoint* container. It is expressed in Hz.

This should be configured to 40 MHz or less in RUN mode and 56 MHz or less in HSRUN mode.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	2.4E7
max	4.0E7
min	0.0

4.1.139 Parameter McuSPLLDiv1Frequency

This is the System PLL Divider 1 frequency for the specific instance of the McuClockReferencePoint container. It is expressed in Hz.

This should be configured to 80MHz or less in RUN mode and to 112 MHz or less in HSRUN mode.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	4.8E7
max	8.0E7
min	0.0

4.1.140 Parameter McuSPLLEnable

SCG_SPLLCSSR[SPLLEN] - System PLL Enable

0 - System PLL is disabled.

1 - System PLL is enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.141 Parameter McuSPLLClockMonitorResetEnable

SCG_SPLLCSR[SPLLCMRE] - System PLL Clock Monitor Reset Enable

0 - Clock Monitor generates interrupt when error detected.

1 - Clock Monitor generates reset when error detected.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.142 Parameter McuSPLLClockMonitorEnable

SCG_SPLLCSR[SPLLCM] - System PLL Clock Monitor

Enables the clock monitor, if the clock source is disabled in a low power mode then the clock monitor is also disabled in the low power mode. When the clock monitor is disabled in a low power mode, it remains disabled until the clock valid flag is set following exit from the low power mode.

0 - RTC OSC Clock Monitor is disabled.

1 - RTC OSC Clock Monitor is enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.143 Parameter McuSPLLDiv2

Configures SCG_SPLLDIV[SPLLDIV2].

System PLL Clock Divider 2.

Clock divider 2 for System PLL. Used by bus clock modules that need an asynchronous clock source.

0 - Output disabled.

Note: Implementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	4
max	64
min	0

4.1.144 Parameter McuSPLLDiv1

Configures SCG_SPLLDIV[SPLLDIV1]

System PLL Clock Divider 1.

Clock divider 1 for System PLL. Used to generate the system clock source and by platform clock modules that need an asynchronous clock source.

0 - Output disabled.

Note: Implementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	2
max	64
min	0

4.1.145 Parameter McuSPLLInputClkPreDivider

PLL Reference Clock Divider.

Set the SPLL: SCG_SPLLCFG[PREDIV] field register.

Selects the amount to divide down the reference clock for the System PLL. The resulting frequency must be in the range of 8 MHz to 50 MHz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	8
min	1

4.1.146 Parameter McuSPLLReferenceFrequency

FSPLL_REF is PLL reference frequency range after the PREDIV.

$F = \text{McuSPLLInputFrequency} / \text{McuSPLLInputClkPreDivider}$.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	8000000.0
max	1.6E7
min	0.0

4.1.147 Parameter McuSPLLInputFrequency

FSPLL_Input is PLL input frequency range before the PREDIV.

For S32K1XX, the valid range is [0 ... 40] MHz.

For S32R14xW, the valid range is [0 ... 48] MHz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	8000000.0
max	4.8E7
min	0.0

4.1.148 Parameter McuSPLLMultiplier

System PLL Multiplier.

Set the SCG_SPLLCFG[MULT] field register.

Valid range is in [16..47].

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	20
max	47
min	16

4.1.149 Parameter McuSPLLSelectSourceClock

SPLLCFG[SOURCE]: System PLL Clock Source.

Configures the input clock source for the System PLL.

The selected clock must be enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	SOSC_CLK
literals	['SOSC_CLK', 'FIRC_CLK']

4.1.150 Container McuSIMClockConfig

Configures SIM_CHIPCTL[TRACECLK_SEL], SIM_CHIPCTL[CLKOUTSEL] bits and SIM_PLATGC and SIM_CLKDIV4 registers.

Note: Implementation specific Container.

Included subcontainers:

- [McuSimChipConfiguration](#)
- [McuSimLpoConfiguration](#)
- [McuSimFtmConfiguration](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.151 Parameter McuSIMUnderMcuControl

0 - SIM is NOT under mcu control.

1 - SIM is under mcu control.

If this is set to false, the MCU code will not configure the SIM registers when *Mcu_InitClock* is called

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.152 Parameter McuEIMClockGatingEnable

SIM_PLATCGC[CGCEIM] - EIM Clock Gating Control

Controls the clock gating to the EIM.

0 - Clock disabled.

1 - Clock enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.153 Parameter McuERMClockGatingEnable

SIM_PLATCGC[CGCERM] - ERM Clock Gating Control

Controls the clock gating to the ERM.

0 - Clock disabled.

1 - Clock enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.154 Parameter McuDMAClockGatingEnable

SIM_PLATCGC[CGCDMA] - DMA Clock Gating Control

Controls the clock gating to the DMA module.

0 - Clock disabled.

1 - Clock enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.155 Parameter McuMPUClockGatingEnable

SIM_PLATCGC[CGCMPU] - MPU Clock Gating Control

Controls the clock gating to the MPU module.

0 - Clock disabled.

1 - Clock enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.156 Parameter McuMSCMClockGatingEnable

SIM_PLATCGC[CGCMSCM] - MSCM Clock Gating Control

Controls the clock gating to the MSCM module.

0 - Clock disabled.

1 - Clock enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.157 Parameter McuGPiOClockGatingEnable

SIM_PLATCGC[CGCGPIO] - GPIO Clock Gating Control

Controls the clock gating to the GPIO module.

0 - Clock disabled.

1 - Clock enabled.

Note: Implementation Specific Parameter. This bit is available in S32K11X variants only

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.158 Container McuSimChipConfiguration

This container contains the configuration for the SIM_CHIPCTL registers.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.159 Parameter McuDebugTraceDividerEnable

SIM_CLKDIV4[TRACEDIVEN] - Debug Trace Divider Control

0 - Debug trace divider disabled.

Configuration Parameters

1 - Debug trace divider enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	true

4.1.160 Parameter McuTRACECLKDivider

Configures the SIM_CLKDIV4[TRACEDIV] bitfield

Trace clock divider divisor - This field sets the divide value for the fractional clock divider used as a source for trace clock. The source clock for the trace clock is set by the SIM_CHIPCTRL[TRACECLK_SEL]. Divider output clock = Divider input clock * $\frac{1}{[(TRACEFRAC+1)/(TRACEDIV+1)]}$.

Note: implementation specific parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	1
max	8
min	1

4.1.161 Parameter McuTRACECLKFraction

Configures the SIM_CLKDIV4[TRACEFRAC] bitfield

This field value is TRACEFRAC+1.

Trace clock divider fraction - This field sets the divide value for the fractional clock divider used as a source for trace clock. The source clock for the trace clock is set by the SIM_CHIPCTRL[TRACECLK_SEL]. Divider output clock = Divider input clock * $\frac{1}{[(TRACEFRAC+1)/(TRACEDIV+1)]}$.

$[(TRACEFRAC+1)/(TRACEDIV+1)]$.

Note: implementation specific parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	2
min	1

4.1.162 Parameter McuTRACECLKSelect

SIM_CHIPCTL[TRACECLK_SEL] - Debug trace clock select

Selects core clock or platform clock as the trace clock source.

The selected clock must be enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	CORE_CLK
literals	['CORE_CLK']

4.1.163 Parameter McuCLKOUTEnable

SIM_CHIPCTL[CLKOUTEN] - CLKOUT enable

unchecked - Clockout disabled.

checked - Clockout enabled.

Configuration Parameters

Maximum supported clock out frequency for this device is 20 MHz.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.164 Parameter McuCLKOUTDivider

Configures the `SIM_CHIPCTL[CLKOUTDIV]` bitfield

CLKOUT Divide Ratio.

Note: implementation specific parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	8
min	1

4.1.165 Parameter McuCLKOUTSelect

`SIM_CHIPCTL[CLKOUTSEL]` - CLKOUT select

Selects the clock to output on the CLKOUT pin.

0 SCG_CLKOUT

2 SOSC_DIV2

4 SIRC_DIV2

5 For S32K148: QSPI_SFIF_CLK_HYP_PREMUX_CLK: Divide by 2 clock (configured through SCLKCONFIG[5]) for HyperRAM going to sfif clock to QSPI; For others: Reserved

6 FIRC_DIV2

7 HCLK

8 For S32K14x: SPLLDIV2_CLK For S32K11x: Reserved

9 BUS_CLK

A LPO_128K_CLK

B For S32K148: QSPI_CLK; For others: Reserved

C LPO_CLK as selected by SIM_LPOCLKS[LPOCLKSEL]

D For S32K148: QSPI_SFIF_CLK; For others: Reserved

E RTC_CLK as selected by SIM_CLK 32 KHz Select

F For S32K148: QSPI_2xSFIF_CLK; For others: Reserved

The selected clock must be enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	SCG_CLKOUT_CLK
literals	['SCG_CLKOUT_CLK', 'SOSCDIV2_CLK', 'SIRCDIV2_CLK', 'FIRCDIV2_CLK', 'HCLK', 'SPLLDIV2_CLK', 'BUS_CLK', 'LPO_128K_CLK', 'LPO_CLK', 'RTC_CLK']

4.1.166 Container McuSimLpoConfiguration

This container contains the configuration for the SIM_LPOCLKS registers.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.167 Parameter McuRTCClkSelect

This is a write-once parameter

SIM_LPOCLKS[RTCCLKSEL] - 32 kHz clock source select.

Selects 32 kHz clock source for peripherals.

0 - SOSCDIV1_CLK

1 - 32 kHz LPO clock

2 - 32 kHz RTC_CLKIN clock

3 - FIRCDIV1_CLK

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	SOSCDIV1_CLK
literals	['SOSCDIV1_CLK', 'LPO_32K_CLK', 'RTC_CLKIN', 'FIRCDIV1_CLK']

4.1.168 Parameter McuLPOClkSelect

This is a write-once parameter

SIM_LPOCLKS[LPOCLKSEL] - LPO clock source select

Selects LPO clock source for peripherals

0 - 128 kHz LPO clock

1 - No clock

2 - 32 kHz LPO clock which is divided by the 128 kHz LPO clock

3 - 1 kHz LPO clock which is divided by the 128 kHz LPO clock

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	LPO_128K_CLK
literals	['LPO_128K_CLK', 'NO_CLOCK', 'LPO_32K_CLK', 'LPO_1K_CLK']

4.1.169 Parameter McuLPO_32KClockEnable

This is a write-once parameter

SIM_LPOCLKS[LPO32KCLKEN] - 32 kHz LPO clock enable

0 - Disable 32 kHz LPO clock output

1 - Enable 32 kHz LPO clock output

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.170 Parameter McuLPO_1KClockEnable

This is a write-once parameter

SIM_LPOCLKS[LPO1KCLKEN] - 1 kHz LPO clock enable

0 - Disable 1 kHz LPO clock output

1 - Enable 1 kHz LPO clock output

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.171 Container McuSimFtmConfiguration

This container contains the configuration for the SIM_FTMOPT0 registers.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.172 Parameter McuFTM3ExternalClockPinSelect

SIM_FTMOPT0[FTM3CLKSEL] - FTM3 External Clock Pin Select

Selects the external pin used to drive the clock to the FTM3 module.

The selected pin must also be configured for the FTM external clock function through the appropriate Pin Control Register in the Port Control module.

0 - FTM3 external clock driven by TCLK0 pin.

1 - FTM3 external clock driven by TCLK1 pin.

2 - FTM3 external clock driven by TCLK2 pin.

3 - No clock input.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TCLK0_PIN
literals	['TCLK0_PIN', 'TCLK1_PIN', 'TCLK2_PIN', 'NO_CLOCK']

4.1.173 Parameter McuFTM2ExternalClockPinSelect

SIM_FTMOPT0[FTM2CLKSEL] - FTM2 External Clock Pin Select

Selects the external pin used to drive the clock to the FTM2 module.

The selected pin must also be configured for the FTM external clock function through the appropriate Pin Control Register in the Port Control module.

0 - FTM2 external clock driven by TCLK0 pin.

1 - FTM2 external clock driven by TCLK1 pin.

2 - FTM2 external clock driven by TCLK2 pin.

3 - No clock input.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TCLK0_PIN
literals	['TCLK0_PIN', 'TCLK1_PIN', 'TCLK2_PIN', 'NO_CLOCK']

4.1.174 Parameter McuFTM1ExternalClockPinSelect

SIM_FTMOPT0[FTM1CLKSEL] - FTM1 External Clock Pin Select

Selects the external pin used to drive the clock to the FTM1 module.

The selected pin must also be configured for the FTM external clock function through the appropriate Pin Control Register in the Port Control module.

0 - FTM1 external clock driven by TCLK0 pin.

1 - FTM1 external clock driven by TCLK1 pin.

2 - FTM1 external clock driven by TCLK2 pin.

3 - No clock input.

Note: Implementation Specific Parameter.

Configuration Parameters

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TCLK0_PIN
literals	['TCLK0_PIN', 'TCLK1_PIN', 'TCLK2_PIN', 'NO_CLOCK']

4.1.175 Parameter McuFTM0ExternalClockPinSelect

SIM_FTMOPT0[FTM0CLKSEL] - FTM0 External Clock Pin Select

Selects the external pin used to drive the clock to the FTM0 module.

The selected pin must also be configured for the FTM external clock function through the appropriate Pin Control Register in the Port Control module.

0 - FTM0 external clock driven by TCLK0 pin.

1 - FTM0 external clock driven by TCLK1 pin.

2 - FTM0 external clock driven by TCLK2 pin.

3 - No clock input.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TCLK0_PIN
literals	['TCLK0_PIN', 'TCLK1_PIN', 'TCLK2_PIN', 'NO_CLOCK']

4.1.176 Parameter McuFTM7ExternalClockPinSelect

SIM_FTMOPT0[FTM7CLKSEL] - FTM7 External Clock Pin Select

Selects the external pin used to drive the clock to the FTM7 module.

The selected pin must also be configured for the FTM external clock function through the appropriate Pin Control Register in the Port Control module.

0 - FTM7 external clock driven by TCLK0 pin.

1 - FTM7 external clock driven by TCLK1 pin.

2 - FTM7 external clock driven by TCLK2 pin.

3 - No clock input.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TCLK0_PIN
literals	['TCLK0_PIN', 'TCLK1_PIN', 'TCLK2_PIN', 'NO_CLOCK']

4.1.177 Parameter McuFTM6ExternalClockPinSelect

SIM_FTMOPT0[FTM6CLKSEL] - FTM6 External Clock Pin Select

Selects the external pin used to drive the clock to the FTM6 module.

The selected pin must also be configured for the FTM external clock function through the appropriate Pin Control Register in the Port Control module.

0 - FTM6 external clock driven by TCLK0 pin.

1 - FTM6 external clock driven by TCLK1 pin.

2 - FTM6 external clock driven by TCLK2 pin.

3 - No clock input.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TCLK0_PIN
literals	['TCLK0_PIN', 'TCLK1_PIN', 'TCLK2_PIN', 'NO_CLOCK']

4.1.178 Parameter McuFTM5ExternalClockPinSelect

SIM_FTMOPT0[FTM5CLKSEL] - FTM5 External Clock Pin Select

Selects the external pin used to drive the clock to the FTM5 module.

The selected pin must also be configured for the FTM external clock function through the appropriate Pin Control Register in the Port Control module.

0 - FTM5 external clock driven by TCLK0 pin.

1 - FTM5 external clock driven by TCLK1 pin.

2 - FTM5 external clock driven by TCLK2 pin.

3 - No clock input.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TCLK0_PIN
literals	['TCLK0_PIN', 'TCLK1_PIN', 'TCLK2_PIN', 'NO_CLOCK']

4.1.179 Parameter McuFTM4ExternalClockPinSelect

SIM_FTMOPT0[FTM7CLKSEL] - FTM4 External Clock Pin Select

Selects the external pin used to drive the clock to the FTM4 module.

The selected pin must also be configured for the FTM external clock function through the appropriate Pin Control Register in the Port Control module.

0 - FTM4 external clock driven by TCLK0 pin.

1 - FTM4 external clock driven by TCLK1 pin.

2 - FTM4 external clock driven by TCLK2 pin.

3 - No clock input.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TCLK0_PIN
literals	['TCLK0_PIN', 'TCLK1_PIN', 'TCLK2_PIN', 'NO_CLOCK']

4.1.180 Container McuClkMonitor

This container contains the specific configuration (parameters) of the Clock Monitor Unit.

Each CMU is independently programmed. SIRC is used as the clock monitor references.

Detailed information on the CMUs can be found in the Clock Monitor Unit chapter.

This parameter is enabled only if "McuClockSrcFailureNotification" is enabled.

This container only for S32K11x derivatives.

Note: Implementation Specific Parameter.

Included subcontainers:

- [McuClkMonitor_0](#)
- [McuClkMonitor_1](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.181 Container McuClkMonitor_0

This container contains the specific configuration (parameters) of the Monitor_0.

Clock Monitor Unit for Motor Clock.

This parameter is enabled only if "McuClockSrcFailureNotification" is enabled.

Note: Implementation Specific Parameter.

Included subcontainers:

Configuration Parameters

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.182 Parameter McuClockMonitorUnderMcuControl

Set this to *TRUE* if this clock monitor is under mcu control

If it is *FALSE* then the mcu driver will not write the corresponding registers.

The user must still set the values - they are used by the clock reference points

This container only for S32K11x derivatives.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.183 Parameter McuClkMonitorEn

Enables/Disables the clock monitor (CMU_FC_GCR[FCE]).

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.184 Parameter McuCmuName

This is the name of the CMU0.

With name convention: CMU_FC_[Number Of CMU Unit]_[Name of Monitored clock].

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	CMU_FC_0_FIRC_MON1_CLK
literals	['CMU_FC_0_FIRC_MON1_CLK']

4.1.185 Parameter McuClockMonitorDeviation

Configure Maximum allowed frequency deviation of monitored clock.

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Enumeration entry

Enumeration description

0

Maximum_5_percentage_deviation

Maximum allowed frequency deviation is 5%

1

Maximum_4_point_9_percentage_deviation

Maximum allowed frequency deviation is 4.9%

2

Maximum_4_point_8_percentage_deviation

Configuration Parameters

Maximum allowed frequency deviation is 4.8%

3

Maximum_4_point_7_percentage_deviation

Maximum allowed frequency deviation is 4.7%

4

Maximum_4_point_6_percentage_deviation

Maximum allowed frequency deviation is 4.6%

5

Maximum_4_point_5_percentage_deviation

Maximum allowed frequency deviation is 4.5%

6

Maximum_4_point_4_percentage_deviation

Maximum allowed frequency deviation is 4.4%

7

Maximum_4_point_3_percentage_deviation

Maximum allowed frequency deviation is 4.3%

8

Maximum_4_point_2_percentage_deviation

Maximum allowed frequency deviation is 4.2%

9

Maximum_4_point_1_percentage_deviation

Maximum allowed frequency deviation is 4.1%

10

Maximum_4_percentage_deviation

Maximum allowed frequency deviation is 4%

11

Maximum_3_point_9_percentage_deviation

Maximum allowed frequency deviation is 3.9%

12

Maximum_3_point_8_percentage_deviation

Maximum allowed frequency deviation is 3.8%

13

Maximum_3_point_7_percentage_deviation

Maximum allowed frequency deviation is 3.7%

14

Maximum_3_point_6_percentage_deviation

Maximum allowed frequency deviation is 3.6%

15

Maximum_3_point_5_percentage_deviation

Maximum allowed frequency deviation is 3.5%

16

Maximum_3_point_4_percentage_deviation

Maximum allowed frequency deviation is 3.4%

17

Maximum_3_point_3_percentage_deviation

Maximum allowed frequency deviation is 3.3%

18

Maximum_3_point_2_percentage_deviation

Maximum allowed frequency deviation is 3.2%

19

Maximum_3_point_1_percentage_deviation

Maximum allowed frequency deviation is 3.1%

20

Maximum_3_percentage_deviation

Maximum allowed frequency deviation is 3%

21

Maximum_2_point_9_percentage_deviation

Maximum allowed frequency deviation is 2.9%

22

Maximum_2_point_8_percentage_deviation

Maximum allowed frequency deviation is 2.8%

23

Maximum_2_point_7_percentage_deviation

Maximum allowed frequency deviation is 2.7%

24

Maximum_2_point_6_percentage_deviation

Maximum allowed frequency deviation is 2.6%

25

Maximum_2_point_5_percentage_deviation

Maximum allowed frequency deviation is 2.5%

26

Maximum_2_point_4_percentage_deviation

Maximum allowed frequency deviation is 2.4%

27

Maximum_2_point_3_percentage_deviation

Maximum allowed frequency deviation is 2.3%

Configuration Parameters

28

Maximum_2_point_2_percentage_deviation

Maximum allowed frequency deviation is 2.2%

29

Maximum_2_point_1_percentage_deviation

Maximum allowed frequency deviation is 2.1%

30

Maximum_2_percentage_deviation

Maximum allowed frequency deviation is 2%

31

Maximum_1_point_9_percentage_deviation

Maximum allowed frequency deviation is 1.9%

32

Maximum_1_point_8_percentage_deviation

Maximum allowed frequency deviation is 1.8%

33

Maximum_1_point_7_percentage_deviation

Maximum allowed frequency deviation is 1.7%

34

Maximum_1_point_6_percentage_deviation

Maximum allowed frequency deviation is 1.6%

35

Maximum_1_point_5_percentage_deviation

Maximum allowed frequency deviation is 1.5%

36

Maximum_1_point_4_percentage_deviation

Maximum allowed frequency deviation is 1.4%

37

Maximum_1_point_3_percentage_deviation

Maximum allowed frequency deviation is 1.3%

38

Maximum_1_point_2_percentage_deviation

Maximum allowed frequency deviation is 1.2%

39

Maximum_1_point_1_percentage_deviation

Maximum allowed frequency deviation is 1.1%

40

Maximum_1_percentage_deviation

Maximum allowed frequency deviation is 1%

41

Maximum_0_point_9_percentage_deviation

Maximum allowed frequency deviation is 0.9%

42

Maximum_0_point_8_percentage_deviation

Maximum allowed frequency deviation is 0.8%

43

Maximum_0_point_7_percentage_deviation

Maximum allowed frequency deviation is 0.7%

44

Maximum_0_point_6_percentage_deviation

Maximum allowed frequency deviation is 0.6%

45

Maximum_0_point_5_percentage_deviation

Maximum allowed frequency deviation is 0.5%

46

Maximum_0_point_4_percentage_deviation

Maximum allowed frequency deviation is 0.4%

47

Maximum_0_point_3_percentage_deviation

Maximum allowed frequency deviation is 0.3%

48

Maximum_0_point_2_percentage_deviation

Maximum allowed frequency deviation is 0.2%

49

Maximum_0_point_1_percentage_deviation

Maximum allowed frequency deviation is 0.1%

50

Maximum_0_point_09_percentage_deviation

Maximum allowed frequency deviation is 0.09%

51

Maximum_0_point_08_percentage_deviation

Maximum allowed frequency deviation is 0.08%

52

Maximum_0_point_07_percentage_deviation

Maximum allowed frequency deviation is 0.07%

53

Configuration Parameters

Maximum_0_point_06_percentage_deviation

Maximum allowed frequency deviation is 0.06%

54

Maximum_0_point_05_percentage_deviation

Maximum allowed frequency deviation is 0.05%

55

Maximum_0_point_04_percentage_deviation

Maximum allowed frequency deviation is 0.04%

56

Maximum_0_point_03_percentage_deviation

Maximum allowed frequency deviation is 0.03%

57

Maximum_0_point_02_percentage_deviation

Maximum allowed frequency deviation is 0.02%

58

Maximum_0_point_01_percentage_deviation

Maximum allowed frequency deviation is 0.01%

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Maximum_5_percentage_deviation

Property	Value
literals	['Maximum_5_percentage_deviation', 'Maximum_4_point_9_percentage← _deviation', 'Maximum_4_point_8_percentage_deviation', 'Maximum← _4_point_7_percentage_deviation', 'Maximum_4_point_6_percentage← _deviation', 'Maximum_4_point_5_percentage_deviation', 'Maximum← _4_point_4_percentage_deviation', 'Maximum_4_point_3_percentage← _deviation', 'Maximum_4_point_2_percentage_deviation', 'Maximum← _4_point_1_percentage_deviation', 'Maximum_4_percentage_deviation', 'Maximum_3_point_9_percentage_deviation', 'Maximum_3_point_8← _percentage_deviation', 'Maximum_3_point_7_percentage_deviation', 'Maximum_3_point_6_percentage_deviation', 'Maximum_3_point_5← _percentage_deviation', 'Maximum_3_point_4_percentage_deviation', 'Maximum_3_point_3_percentage_deviation', 'Maximum_3_point_2← _percentage_deviation', 'Maximum_3_point_1_percentage_deviation', 'Maximum_3_percentage_deviation', 'Maximum_2_point_9_percentage← _deviation', 'Maximum_2_point_8_percentage_deviation', 'Maximum← _2_point_7_percentage_deviation', 'Maximum_2_point_6_percentage← _deviation', 'Maximum_2_point_5_percentage_deviation', 'Maximum← _2_point_4_percentage_deviation', 'Maximum_2_point_3_percentage← _deviation', 'Maximum_2_point_2_percentage_deviation', 'Maximum← _2_point_1_percentage_deviation', 'Maximum_2_percentage_deviation', 'Maximum_1_point_9_percentage_deviation', 'Maximum_1_point_8← _percentage_deviation', 'Maximum_1_point_7_percentage_deviation', 'Maximum_1_point_6_percentage_deviation', 'Maximum_1_point_5← _percentage_deviation', 'Maximum_1_point_4_percentage_deviation', 'Maximum_1_point_3_percentage_deviation', 'Maximum_1_point_2← _percentage_deviation', 'Maximum_1_point_1_percentage_deviation', 'Maximum_1_percentage_deviation', 'Maximum_0_point_9_percentage← _deviation', 'Maximum_0_point_8_percentage_deviation', 'Maximum_← 0_point_7_percentage_deviation', 'Maximum_0_point_6_percentage_← deviation', 'Maximum_0_point_5_percentage_deviation', 'Maximum_← 0_point_4_percentage_deviation', 'Maximum_0_point_3_percentage_← deviation', 'Maximum_0_point_2_percentage_deviation', 'Maximum_0← _point_1_percentage_deviation', 'Maximum_0_point_09_percentage_← deviation', 'Maximum_0_point_08_percentage_deviation', 'Maximum_0← _point_07_percentage_deviation', 'Maximum_0_point_06_percentage_← deviation', 'Maximum_0_point_05_percentage_deviation', 'Maximum_0← _point_04_percentage_deviation', 'Maximum_0_point_03_percentage_← deviation', 'Maximum_0_point_02_percentage_deviation', 'Maximum_0_← point_01_percentage_deviation']

4.1.186 Parameter McuClockMonitorLatency

Configure Latency of clock monitor reaction.

```
table, th, td {
border:1px solid black;
}
```

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Enumeration entry

Enumeration description

Configuration Parameters

0

Lowest_Latency

Clock monitor is configured to react as fast as possible. A clock failure is detected by clock monitor unit in less than one microsecond.

1

Low_Latency_1us

Clock monitor reaction is taking 1us. A clock failure is detected by clock monitor unit in one microsecond.

2

Medium_Latency_10us

Clock monitor reaction is taking 10us. A clock failure is detected by clock monitor unit in 10 microseconds.

3

High_Latency_100us

Clock monitor reaction is taking 100us. A clock failure is detected by clock monitor unit in 100 microseconds.

4

Very_High_Latency_1ms

Clock monitor reaction is taking 1ms. A clock failure is detected by clock monitor unit in 1 millisecond.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Low_Latency_1us
literals	['Lowest_Latency', 'Low_Latency_1us', 'Medium_Latency_10us', 'High_Latency_100us', 'Very_High_Latency_1ms']

4.1.187 Parameter McuAsyncFHHInterruptEn

This field is used to enable/disable FHH asynchronous interrupt at the module boundary. (CMU_FC_IER[FHHAIE]).

0 - Asynchronous FHH Interrupt is Disabled

1 - Asynchronous FHH Interrupt is Enabled

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.188 Parameter McuAsyncFLLInterruptEn

This field is used to enable/disable FLL asynchronous interrupt at the module boundary. (CMU_FC_IER[FLLAIE]).

0 - Asynchronous FLL Interrupt is Disabled

1 - Asynchronous FLL Interrupt is Enabled

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.189 Container McuClkMonitor_1

This container contains the specific configuration (parameters) of the Monitor_1.

Clock Monitor Unit for Motor Clock.

This parameter is enabled only if "McuClockSrcFailureNotification" is enabled.

Note: Implementation Specific Parameter.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.190 Parameter McuClockMonitorUnderMcuControl

Set this to *TRUE* if this clock monitor is under mcu control

If it is *FALSE* then the mcu driver will not write the corresponding registers.

The user must still set the values - they are used by the clock reference points

This container only for S32K11x derivatives.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.191 Parameter McuClkMonitorEn

Enables/Disables the clock monitor (CMU_FC_GCR[FCE]).

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.192 Parameter McuCmuName

This is the name of the CMU1.

With name convention: CMU_FC_[Number Of CMU Unit]_[Name of Monitored clock].

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	CMU_FC_1_FIRC_MON2_CLK
literals	['CMU_FC_1_FIRC_MON2_CLK']

4.1.193 Parameter McuClockMonitorDeviation

Configure Maximum allowed frequency deviation of monitored clock.

table, th, td {

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}

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Enumeration entry

Enumeration description

0

Maximum_5_percentage_deviation

Maximum allowed frequency deviation is 5%

1

Maximum_4_point_9_percentage_deviation

Maximum allowed frequency deviation is 4.9%

2

Maximum_4_point_8_percentage_deviation

Maximum allowed frequency deviation is 4.8%

3

Maximum_4_point_7_percentage_deviation

Maximum allowed frequency deviation is 4.7%

4

Configuration Parameters

Maximum_4_point_6_percentage_deviation

Maximum allowed frequency deviation is 4.6%

5

Maximum_4_point_5_percentage_deviation

Maximum allowed frequency deviation is 4.5%

6

Maximum_4_point_4_percentage_deviation

Maximum allowed frequency deviation is 4.4%

7

Maximum_4_point_3_percentage_deviation

Maximum allowed frequency deviation is 4.3%

8

Maximum_4_point_2_percentage_deviation

Maximum allowed frequency deviation is 4.2%

9

Maximum_4_point_1_percentage_deviation

Maximum allowed frequency deviation is 4.1%

10

Maximum_4_percentage_deviation

Maximum allowed frequency deviation is 4%

11

Maximum_3_point_9_percentage_deviation

Maximum allowed frequency deviation is 3.9%

12

Maximum_3_point_8_percentage_deviation

Maximum allowed frequency deviation is 3.8%

13

Maximum_3_point_7_percentage_deviation

Maximum allowed frequency deviation is 3.7%

14

Maximum_3_point_6_percentage_deviation

Maximum allowed frequency deviation is 3.6%

15

Maximum_3_point_5_percentage_deviation

Maximum allowed frequency deviation is 3.5%

16

Maximum_3_point_4_percentage_deviation

Maximum allowed frequency deviation is 3.4%

17

Maximum_3_point_3_percentage_deviation

Maximum allowed frequency deviation is 3.3%

18

Maximum_3_point_2_percentage_deviation

Maximum allowed frequency deviation is 3.2%

19

Maximum_3_point_1_percentage_deviation

Maximum allowed frequency deviation is 3.1%

20

Maximum_3_percentage_deviation

Maximum allowed frequency deviation is 3%

21

Maximum_2_point_9_percentage_deviation

Maximum allowed frequency deviation is 2.9%

22

Maximum_2_point_8_percentage_deviation

Maximum allowed frequency deviation is 2.8%

23

Maximum_2_point_7_percentage_deviation

Maximum allowed frequency deviation is 2.7%

24

Maximum_2_point_6_percentage_deviation

Maximum allowed frequency deviation is 2.6%

25

Maximum_2_point_5_percentage_deviation

Maximum allowed frequency deviation is 2.5%

26

Maximum_2_point_4_percentage_deviation

Maximum allowed frequency deviation is 2.4%

27

Maximum_2_point_3_percentage_deviation

Maximum allowed frequency deviation is 2.3%

28

Maximum_2_point_2_percentage_deviation

Maximum allowed frequency deviation is 2.2%

29

Maximum_2_point_1_percentage_deviation

Configuration Parameters

Maximum allowed frequency deviation is 2.1%

30

Maximum_2_percentage_deviation

Maximum allowed frequency deviation is 2%

31

Maximum_1_point_9_percentage_deviation

Maximum allowed frequency deviation is 1.9%

32

Maximum_1_point_8_percentage_deviation

Maximum allowed frequency deviation is 1.8%

33

Maximum_1_point_7_percentage_deviation

Maximum allowed frequency deviation is 1.7%

34

Maximum_1_point_6_percentage_deviation

Maximum allowed frequency deviation is 1.6%

35

Maximum_1_point_5_percentage_deviation

Maximum allowed frequency deviation is 1.5%

36

Maximum_1_point_4_percentage_deviation

Maximum allowed frequency deviation is 1.4%

37

Maximum_1_point_3_percentage_deviation

Maximum allowed frequency deviation is 1.3%

38

Maximum_1_point_2_percentage_deviation

Maximum allowed frequency deviation is 1.2%

39

Maximum_1_point_1_percentage_deviation

Maximum allowed frequency deviation is 1.1%

40

Maximum_1_percentage_deviation

Maximum allowed frequency deviation is 1%

41

Maximum_0_point_9_percentage_deviation

Maximum allowed frequency deviation is 0.9%

42

Maximum_0_point_8_percentage_deviation

Maximum allowed frequency deviation is 0.8%

43

Maximum_0_point_7_percentage_deviation

Maximum allowed frequency deviation is 0.7%

44

Maximum_0_point_6_percentage_deviation

Maximum allowed frequency deviation is 0.6%

45

Maximum_0_point_5_percentage_deviation

Maximum allowed frequency deviation is 0.5%

46

Maximum_0_point_4_percentage_deviation

Maximum allowed frequency deviation is 0.4%

47

Maximum_0_point_3_percentage_deviation

Maximum allowed frequency deviation is 0.3%

48

Maximum_0_point_2_percentage_deviation

Maximum allowed frequency deviation is 0.2%

49

Maximum_0_point_1_percentage_deviation

Maximum allowed frequency deviation is 0.1%

50

Maximum_0_point_09_percentage_deviation

Maximum allowed frequency deviation is 0.09%

51

Maximum_0_point_08_percentage_deviation

Maximum allowed frequency deviation is 0.08%

52

Maximum_0_point_07_percentage_deviation

Maximum allowed frequency deviation is 0.07%

53

Maximum_0_point_06_percentage_deviation

Maximum allowed frequency deviation is 0.06%

54

Maximum_0_point_05_percentage_deviation

Maximum allowed frequency deviation is 0.05%

Configuration Parameters

55

Maximum_0_point_04_percentage_deviation

Maximum allowed frequency deviation is 0.04%

56

Maximum_0_point_03_percentage_deviation

Maximum allowed frequency deviation is 0.03%

57

Maximum_0_point_02_percentage_deviation

Maximum allowed frequency deviation is 0.02%

58

Maximum_0_point_01_percentage_deviation

Maximum allowed frequency deviation is 0.01%

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Maximum_5_percentage_deviation

Property	Value
literals	['Maximum_5_percentage_deviation', 'Maximum_4_point_9_percentage← _deviation', 'Maximum_4_point_8_percentage_deviation', 'Maximum← _4_point_7_percentage_deviation', 'Maximum_4_point_6_percentage← _deviation', 'Maximum_4_point_5_percentage_deviation', 'Maximum← _4_point_4_percentage_deviation', 'Maximum_4_point_3_percentage← _deviation', 'Maximum_4_point_2_percentage_deviation', 'Maximum← _4_point_1_percentage_deviation', 'Maximum_4_percentage_deviation', 'Maximum_3_point_9_percentage_deviation', 'Maximum_3_point_8← _percentage_deviation', 'Maximum_3_point_7_percentage_deviation', 'Maximum_3_point_6_percentage_deviation', 'Maximum_3_point_5← _percentage_deviation', 'Maximum_3_point_4_percentage_deviation', 'Maximum_3_point_3_percentage_deviation', 'Maximum_3_point_2← _percentage_deviation', 'Maximum_3_point_1_percentage_deviation', 'Maximum_3_percentage_deviation', 'Maximum_2_point_9_percentage← _deviation', 'Maximum_2_point_8_percentage_deviation', 'Maximum← _2_point_7_percentage_deviation', 'Maximum_2_point_6_percentage← _deviation', 'Maximum_2_point_5_percentage_deviation', 'Maximum← _2_point_4_percentage_deviation', 'Maximum_2_point_3_percentage← _deviation', 'Maximum_2_point_2_percentage_deviation', 'Maximum← _2_point_1_percentage_deviation', 'Maximum_2_percentage_deviation', 'Maximum_1_point_9_percentage_deviation', 'Maximum_1_point_8← _percentage_deviation', 'Maximum_1_point_7_percentage_deviation', 'Maximum_1_point_6_percentage_deviation', 'Maximum_1_point_5← _percentage_deviation', 'Maximum_1_point_4_percentage_deviation', 'Maximum_1_point_3_percentage_deviation', 'Maximum_1_point_2← _percentage_deviation', 'Maximum_1_point_1_percentage_deviation', 'Maximum_1_percentage_deviation', 'Maximum_0_point_9_percentage← _deviation', 'Maximum_0_point_8_percentage_deviation', 'Maximum_← 0_point_7_percentage_deviation', 'Maximum_0_point_6_percentage_← deviation', 'Maximum_0_point_5_percentage_deviation', 'Maximum_← 0_point_4_percentage_deviation', 'Maximum_0_point_3_percentage_← deviation', 'Maximum_0_point_2_percentage_deviation', 'Maximum_0← _point_1_percentage_deviation', 'Maximum_0_point_09_percentage_← deviation', 'Maximum_0_point_08_percentage_deviation', 'Maximum_0← _point_07_percentage_deviation', 'Maximum_0_point_06_percentage_← deviation', 'Maximum_0_point_05_percentage_deviation', 'Maximum_0← _point_04_percentage_deviation', 'Maximum_0_point_03_percentage_← deviation', 'Maximum_0_point_02_percentage_deviation', 'Maximum_0_← point_01_percentage_deviation']

4.1.194 Parameter McuClockMonitorLatency

Configure Latency of clock monitor reaction.

table, th, td {

border:1px solid black;

}

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Enumeration entry

Enumeration description

Configuration Parameters

0

Lowest_Latency

Clock monitor is configured to react as fast as possible. A clock failure is detected by clock monitor unit in less than one microsecond.

1

Low_Latency_1us

Clock monitor reaction is taking 1us. A clock failure is detected by clock monitor unit in one microsecond.

2

Medium_Latency_10us

Clock monitor reaction is taking 10us. A clock failure is detected by clock monitor unit in 10 microseconds.

3

High_Latency_100us

Clock monitor reaction is taking 100us. A clock failure is detected by clock monitor unit in 100 microseconds.

4

Very_High_Latency_1ms

Clock monitor reaction is taking 1ms. A clock failure is detected by clock monitor unit in 1 millisecond.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Low_Latency_1us
literals	['Lowest_Latency', 'Low_Latency_1us', 'Medium_Latency_10us', 'High_Latency_100us', 'Very_High_Latency_1ms']

4.1.195 Parameter McuSyncFHHInterruptEn

This field is used to enable/disable FHH synchronous interrupt at the module boundary. (CMU_FC_IER[FHHIE]).

0 - Synchronous FHH Interrupt is Disabled

1 - Synchronous FHH Interrupt is Enabled

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.196 Parameter McuSyncFLLInterruptEn

This field is used to enable/disable FLL synchronous interrupt at the module boundary. (CMU_FC_IER[FLLIE]).

0 - Synchronous FLL Interrupt is Disabled

1 - Synchronous FLL Interrupt is Enabled

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.197 Container McuPeripheralClockConfig

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation Specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	30
upperMultiplicity	30
postBuildVariantMultiplicity	N/A

Configuration Parameters

Property	Value
multiplicityConfigClasses	N/A

4.1.198 Parameter McuPeripheralClockUnderMcuControl

0 - Peripheral Clock is NOT under mcu control.

1 - Peripheral Clock is under mcu control.

If this is set to false, the MCU code will not configure the PCC registers when Mcu_InitClock is called

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.1.199 Parameter McuPerName

This is the name of the peripheral.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	FTFM
literals	['FTFM', 'DMAMUX0', 'FLEXCAN0', 'FLEXCAN1', 'FTM3', 'ADC1', 'LPSPI0', 'LPSPI1', 'LPSPI2', 'PDB1', 'CRC0', 'PDB0', 'LPIT0', 'FTM0', 'FTM1', 'FTM2', 'ADC0', 'RTC0', 'LPTMR0', 'PORTA', 'PORTB', 'PORTC', 'PORTD', 'PORTE', 'FlexIO', 'EWM0', 'LPI2C0', 'LPUART0', 'LPUART1', 'CMP0']

4.1.200 Parameter McuPeripheralClockEnable

Sets *PCC_[peripheral][CGC]* bit.

This read/write bit enables the clock for the peripheral.

0 - Clock disabled.

1 - Clock enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.201 Parameter McuPeripheralClockSelect

Configures *PCC_[peripheral][PCS]*.

This is used for peripherals that support various clock selections.

If the peripheral does not support various clock selections the field won't be editable.

This field can only be written when the CGC bit is 0 (clock disabled). Likewise, if the INUSE flag is set, this field is locked.

0 - Clock is off (or external clock as selected by FTMnCLKSEL for FTM modules).

1 - SOSCDIV2_CLK (SOCDIV1_CLK for FTM modules)

2 - SIRCDIV2_CLK (SIRCDIV1_CLK for FTM modules)

3 - FIRCDIV2_CLK (FIRCDIV1_CLK for FTM modules)

6 - SPLLDIV2_CLK (SPLLDIV1_CLK for FTM modules)

The selected clock must be enabled.

Note: Please make sure that the divider of clock source is not 0 .

If the field is not editable, it means the bit field is read only and the value will not be used.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	CLOCK_IS_OFF
literals	['CLOCK_IS_OFF', 'SOSC', 'SIRC', 'FIRC', 'SPLL', 'LPO_128K_CLK']

4.1.202 Parameter McuPeripheralClockDivider

Configures PCC_[peripheral][PCD].

This is used for peripherals that require a clock divider. At SOC integration, each peripheral is assigned either a divider or not.

This field can only be written when the CGC bit is 0 (clock disabled). Likewise, if the INUSE flag is set, this field is locked.

Allowed values are from 1 to 8.

Note: Implementation Specific Parameter.

If the field is not editable, it means the bit field is read only and the value will not be used.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	8
min	1

4.1.203 Parameter McuPeripheralFractionalDivider

Configures PCC_[peripheral][FRAC].

This field value is $FRAC+1$.

This sets the fraction multiply value for the fractional clock divider used as a clock source. Divider output clock = Divider input clock $\times [(FRAC+1)/(DIV+1)]$.

This field can only be written when the CGC bit is 0 (clock disabled). Likewise, if the INUSE flag is set, this field is locked.

1 - Fractional Multiply value is 1.

2 - Fractional Multiply value is 2.

Note: Implementation Specific Parameter.

If the field is not editable, it means the bit field is read only and the value will not be used.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	2
min	1

4.1.204 Parameter McuPeripheralClockFrequency

$\text{Divider output clock} = \text{Divider input clock} \times [(\text{FRAC}+1)/(\text{DIV}+1)]$.

This is only calculated if the clock source is selectable and if the peripheral is enabled.

This is the frequency for the specific instance of the "McuClockReferencePoint" container.

Value calculated for user info. It is given in Hz.

Note: The maximum frequency of LPUARTx, LPSPIx, LPI2Cx, FlexIO, LPTMR0 and LPIT0 are governed by BUS_CLK, FTMx are governed by SYS_CLK, ADCx are 50MHz but always less than BUS_CLK.

So please check configuration value is fit for SYS_CLK and BUS_CLK values correspond to MCU mode.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0.0
max	8.0E7
min	0.0

4.1.205 Container McuClockReferencePoint

This container defines a reference point in the Mcu Clock tree. It defines the frequency which then can be used by other modules as an input value. Lower multiplicity is 1, as even in the simplest case (only one frequency is used), there is one frequency to be defined.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.1.206 Parameter McuClockReferencePointFrequency

This is the frequency for the specific instance of the McuClockReferencePoint container.

It shall be given in Hz.

Calculated value.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	4.8E7
max	3.2E8
min	0.0

4.1.207 Parameter McuClockFrequencySelect

Select clock source for the specific instance of the McuClockReferencePoint container.

Note: The clock frequency configured in McuPeripheralClockConfig should be used to export the clock frequency through McuClockReferencePoint.

This reference point should be used in the configuration of the module that uses it (SPI, I2C, GPT, etc.).

If the configured module has also an internal clock selection (like FlexTimer for example),

the clock reference point should be configured taking the internal clock selection into account and the reference used should reflect the clock that finally enters the used peripheral.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	RUN_SYS_CLK
literals	['CUSTOM', 'FIRC_CLK', 'FIRCDIV2_CLK', 'FIRCDIV1_CLK', 'SIRC_CLK', 'SIRCDIV2_CLK', 'SIRCDIV1_CLK', 'SOSC_CLK', 'SOSCDIV2_CLK', 'SOSCDIV1_CLK', 'LPO_128K_CLK', 'LPO_32K_CLK', 'LPO_1K_CLK', 'SPLL_CLK', 'SPLLDIV2_CLK', 'SPLLDIV1_CLK', 'RUN_PRI_DIV_SYS_CLK', 'RUN_CORE_CLK', 'RUN_SYS_CLK', 'RUN_FLASH_CLK', 'RUN_BUS_CLK', 'VLPR_PRI_DIV_SYS_CLK', 'VLPR_CORE_CLK', 'VLPR_SYS_CLK', 'VLPR_FLASH_CLK', 'VLPR_BUS_CLK', 'FTM3_CLK', 'ADC1_CLK', 'LPSPi0_CLK', 'LPSPi1_CLK', 'LPSPi2_CLK', 'LPi0_CLK', 'FTM0_CLK', 'FTM1_CLK', 'FTM2_CLK', 'ADC0_CLK', 'LPTMR0_CLK', 'FlexIO_CLK', 'LPI2C0_CLK', 'LPUART0_CLK', 'LPUART1_CLK']

4.1.208 Container McuDemEventParameterRefs

Container for the references to *DemEventParameter* elements which shall be invoked using the API *Dem_ReportErrorStatus* API in case the corresponding error occurs.

The *EventId* is taken from the referenced *DemEventParameter*'s *DemEventId* value.

The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.1.209 Reference MCU_E_TIMEOUT_FAILURE

Reference to configured DEM event to report Timeout failure.

Property	Value
type	ECUC-REFERENCE-DEF

Configuration Parameters

Property	Value
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.1.210 Reference MCU_E_CLOCK_FAILURE

Reference to configured DEM event to report Clock source failure.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.1.211 Reference MCU_E_SWITCHMODE_FAILURE

Reference to configured DEM event to report Switch Mode failure.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.212 Container McuModeSettingConf

This container contains the configuration for the Mode setting of the MCU.

Note: Implementation Specific Parameter.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.1.213 Parameter McuMode

This parameter shall represent the ID of the MCU mode.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	255
min	0

4.1.214 Parameter McuPowerMode

This parameter selects the Power Mode to be used.

For valid Mode transitions refers to Power mode state diagram from Reference Manual.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	RUN
literals	['RUN', 'HSRUN', 'VLPR', 'VLPS', 'STOP1', 'STOP2']

4.1.215 Parameter McuEnableSleepOnExit

Indicates sleep-on-exit when returning from Handler mode to Thread mode:

0 - Do not sleep when returning to Thread mode.

1 - Enter sleep, or deep sleep, on return from an ISR.

Setting this bit to 1 enables an interrupt driven application to avoid returning to an empty main application.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.216 Reference McuPowerEcucPartitionRef

Maps a config part of Power Mode to ECUC partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

4.1.217 Container McuRamSectorSettingConf

This container contains the configuration for the RAM Sector setting.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.1.218 Parameter McuRamSectorId

This parameter shall represent the ID of the MCU RAM Sector configuration.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	0
max	4294967295
min	0

4.1.219 Parameter McuRamDefaultValue

This parameter shall represent the Data pre-setting to be initialized.

Default value is 0xbabababa.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	255
min	0

4.1.220 Parameter McuRamSectionBaseAddress

This parameter represents the RAM section base address.

The address must be aligned to 4 bytes.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	536838144
max	536899583
min	536838144

4.1.221 Parameter McuRamSectionSize

This parameter represents the RAM section size in bytes.

The size must be multiple of 4.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	1024
max	61440
min	0

4.1.222 Parameter McuRamSectionWriteSize

This parameter shall define the size in bytes of data which can be written into RAM at once.

The ram write size is currently restricted to {1, 2, 4, 8} bytes.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	8
max	4294967295
min	0

4.1.223 Parameter McuRamSectionBaseAddrLinkerSym

This parameter represents the RAM section base address.

The address must be aligned to 4 bytes.

If this parameter is empty, then the integer values from "McuRamSectionBaseAddress" will be used.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	

4.1.224 Parameter McuRamSectionSizeLinkerSym

This parameter represents the RAM section size in bytes.

The size must be multiple of 4.

If this parameter is empty, then the integer values from "McuRamSectionSize" will be used.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	

4.1.225 Reference McuRamEcucPartitionRef

Maps a config part of Ram to ECUC partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcucPartitionCollection/EcucPartition

4.1.226 Container McuInterruptEvents

Configuration for different interrupts handled by MCU.

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.227 Parameter McuVoltageErrorEvent

Power Management Unit Fault Monitoring Interrupts.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.228 Parameter McuAlternateResetEvent

Some events can generate an interrupt from RCM.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.229 Container McuResetConfig

The Reset Control Module (MC_RCM) centralizes the different reset sources and manages the reset sequence of the device.

Note: Implementation Specific Parameter.

Included subcontainers:

- [McuSystemInterruptEnable](#)
- [McuResetGeneratorConfiguration](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.230 Parameter McuResetPinFilterBusClockSelect

RCM_RPC[RSTFLTSEL] - Reset Pin Filter Bus Clock Select.

Selects the reset pin bus clock filter width. Transitions for less than (RSTFLTSEL+1) bus clock cycles are always filtered, transitions equal to (RSTFLTSEL+1) bus clock cycles may be filtered.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	31
min	0

4.1.231 Parameter McuResetPinFilterInStopMode

RCM_RPC[RSTFLTSS] - Reset Pin Filter Select in Stop Mode.

Selects how the reset pin filter is enabled in any stop mode.

0 - All filtering disabled.

1 - LPO clock filter enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ALL_FILTERING_DISABLE
literals	['ALL_FILTERING_DISABLE', 'LPO_CLOCK_FILTER_ENABLE']

4.1.232 Parameter McuResetPinFilterInRunAndWait

RCM_RPC[RSTFLTSRW] - Reset Pin Filter Select in Run and Wait Modes.

Selects how the reset pin filter is enabled in run and wait modes.

0 - All filtering disabled.

1 - Bus clock filter enabled for normal operation.

2 - LPO clock filter enabled for normal operation.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false

Configuration Parameters

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ALL_FILTERING_DISABLE
literals	['ALL_FILTERING_DISABLE', 'BUS_CLOCK_FILTER_ENABLE', 'LPO←_CLOCK_FILTER_ENABLE']

4.1.233 Container McuSystemInterruptEnable

Configures RCM_SRIE

This registers delays the assertion of a system reset for a period of time (DELAY field) while an interrupt is generated.

This allows software to perform a graceful shutdown.

A Chip POR source cannot be delayed by this feature, and entering Stop mode will terminate the delay.

The SRS will only update after the system reset occurs.

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.234 Parameter McuResetDelayTime

RCM_SRIE[DELAY] - Reset Delay Time.

Configures the maximum reset delay time from when the interrupt is asserted and the system reset occurs.

0 - 10 LPO cycles.

1 - 34 LPO cycles.

2 - 130 LPO cycles.

2 - 514 LPO cycles.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	DELAY_10_LPO_CYCLES
literals	['DELAY_10_LPO_CYCLES', 'DELAY_34_LPO_CYCLES', 'DELAY_130←_LPO_CYCLES', 'DELAY_514_LPO_CYCLES']

4.1.235 Parameter McuStopAcknowledgeErrorInterrupt

RCM_SRIE[SACKERR] - Stop Acknowledge Error Interrupt.

0 - Interrupt disabled.

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.236 Parameter McuMDMAPSystemResetInterrupt

RCM_SRIE[MDM_AP] - MDM-AP System Reset Request.

0 - Interrupt disabled.

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.237 Parameter McuSoftwareInterrupt

RCM_SRIE[SW] - Software Interrupt.

0 - Interrupt disabled.

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.238 Parameter McuCoreLockupInterrupt

RCM_SRIE[LOCKUP] - Core Lockup Interrupt.

0 - Interrupt disabled.

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.239 Parameter McuJTAGResetInterrupt

RCM_SRIE[JTAG] - JTAG generated reset.

0 - Interrupt disabled.

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.240 Parameter McuGlobalInterrupt

RCM_SRIE[GIE] - Global Interrupt Enable.

0 - All interrupt sources disabled.

1 - All interrupt sources enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.241 Parameter McuExternalResetPinInterrupt

RCM_SRIE[PIN] - External Reset Pin Interrupt.

0 - Interrupt disabled.

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.242 Parameter McuWatchdogInterrupt

RCM_SRIE[WDOG] - Watchdog Interrupt.

0 - Interrupt disabled.

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.243 Parameter McuCMULossOfClockResetInterrupt

RCM_SRIE[CMU_LOC] - CMU Loss-of-Clock Reset Interrupt.

0 - Interrupt disabled.

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.244 Parameter McuLossOfLockInterrupt

RCM_SRIE[LOL] - Loss of Lock Interrupt.

0 - Interrupt disabled.

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.245 Parameter McuLossOfClockInterrupt

RCM_SRIE[LOC] - Loss of Clock Interrupt.

0 - Interrupt disabled.

Configuration Parameters

1 - Interrupt enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.246 Container McuResetGeneratorConfiguration

Configures AEC_AE_RSTGEN_CFG[RSTGEN_CFG]- Configures reset generator.

Set this bit to activate IP (this deasserts hard reset input port of IP).

Individual control over:

0: regs_otp (resets OTP register interface, not mirror regs)

1: can_phy (do not activate CANPHY if VERID.VARIANT says CANPHY is disabled)

2: lin_phy HP (do not activate LINPHY if VERID.VARIANT says LINPHY is disabled)

3: lin_phy LP (do not activate LINPHY if VERID.VARIANT says LINPHY is disabled)

4: gdu

5: hvi

6: dpga

7: tempsensor

8: cxpi (do not activate CXPI if VERID.VARIANT says CXPI is disabled, or if using LINPHY in noncxpi mode)

Note: Implementation specific Container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.247 Parameter McuRegsOtpReset

Regs_otp (resets OTP register interface, not mirror regs)

0 - Regs Otp Reset disabled.

1 - Regs Otp Reset enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.248 Parameter McuCanPhyReset

Can_phy (do not activate CANPHY if VERID.VARIANT says CANPHY is disabled)

0 - CANPHY Reset disabled.

1 - CANPHY Reset enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.249 Parameter McuLinPhyHpReset

Lin_phy HP (do not activate LINPHY if VERID.VARIANT says LINPHY is disabled).

0 - LINPHY HP Reset disabled.

1 - LINPHY HP Reset enabled.

Configuration Parameters

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.250 Parameter McuLinPhyLpReset

Lin_phy LP (do not activate LINPHY if VERID.VARIANT says LINPHY is disabled).

0 - LINPHY LP Reset disabled.

1 - LINPHY LP Reset enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.251 Parameter McuGduReset

GDU resset.

0 - GDU Reset disabled.

1 - GDU Reset enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.252 Parameter McuHviReset

HVI resset.

0 - HVI Reset disabled.

1 - HVI Reset enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.253 Parameter McuDpgaReset

DPGA resset.

0 - DPGA Reset disabled.

1 - DPGA Reset enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.254 Parameter McuTempSensorReset

TempSensor reset.

0 - TempSensor Reset disabled.

1 - TempSensor Reset enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.255 Parameter McuCxpiReset

Cxpi (do not activate CXPI if VERID.VARIANT says CXPI is disabled, or if using LINPHY in noncxpi mode).

0 - Cxpi Reset disabled.

1 - Cxpi Reset enabled.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.256 Container McuPowerControl

Note: Implementation Specific Parameter.

Included subcontainers:

- [McuPMC_Config](#)
- [McuPMC_AE_Config](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.257 Container McuPMC_Config

This PMC Control Register contains the various control settings of the PMC block.

Note: Implementation Specific Parameter.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.258 Parameter McuLowVoltageDetectInterruptEnable

PMC_LVDSC1[LVDIE] - Low Voltage Detect Interrupt Enable.

This bit enables hardware interrupt requests for LVDF.

0 - Hardware interrupt disabled (use polling).

1 - Request a hardware interrupt when LVDF = 1.

Note:

- Implementation Specific Parameter.

Configuration Parameters

- Only support for S32K1XX series of devices.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.259 Parameter McuLowVoltageDetectResetEnable

PMC_LVDSC1[LVDRE] - Low Voltage Detect Reset Enable.

This bit enables the low voltage detect events to generate a system reset.

0 - No system resets on low voltage detect events.

1 - If the supply voltage falls below VLVD, a system reset will be generated.

Note:

- Implementation Specific Parameter.

- Only support for S32K1XX series of devices.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.260 Parameter McuLowVoltageWarningInterruptEnable

PMC_LVDSC2[LVWIE] - Low-Voltage Warning Interrupt Enable.

This bit enables hardware interrupt requests for LVWF.

0 - Hardware interrupt disabled (use polling).

1 - Request a hardware interrupt when LVWF = 1.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.261 Parameter McuLPODisable

PMC_REGSC[LPODIS] - This bit enables or disable the low power oscillator.

After disabling the LPO a time of 2 LPO clock cycles is required before it is allowed to enable it again. Violating this waiting time of 2 cycles can result in malfunction of the LPO.

unchecked - Low power oscillator enabled.

checked - Low power oscillator disabled.

Note: The reset delay feature requires the LPO clock to remain active.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.262 Parameter McuLPOTRIMSelect

LPOTRIM[LPOTRIM] - LPO trimming bits

Control the period trimming for the low power oscillator.

Configuration Parameters

If NO_CHANGED is selected, the register value will be the default value after any reset.

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}

LPOTRIM[4:0]

Decimal

Period of LPO clock

0b10000

-16

lowest

0b10001

-15

increasing

...

...

increasing

0b11110

-2

increasing

0b11111

-1

increasing

0b00000

0

typical 128 kHz

0b00001

+1

increasing

...

...

increasing

0b01110

+14

increasing

0b01111

+15

highest

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	NO_CHANGED
literals	['NO_CHANGED', 'DECREASE_16', 'DECREASE_15', 'DECREASE_14', 'DECREASE_13', 'DECREASE_12', 'DECREASE_11', 'DECREASE_10', 'DECREASE_09', 'DECREASE_08', 'DECREASE_07', 'DECREASE_06', 'DECREASE_05', 'DECREASE_04', 'DECREASE_03', 'DECREASE_02', 'DECREASE_01', 'TYPICAL', 'INCREASE_01', 'INCREASE_02', 'INCREASE_03', 'INCREASE_04', 'INCREASE_05', 'INCREASE_06', 'INCREASE_07', 'INCREASE_08', 'INCREASE_09', 'INCREASE_10', 'INCREASE_11', 'INCREASE_12', 'INCREASE_13', 'INCREASE_14', 'INCREASE_15']

4.1.263 Parameter McuClockBiasDisable

PMC_REGSC[CLKBIASDIS] - Clock Bias Disable Bit.

This bit disables the bias currents and reference voltages for some clock modules in order to further reduce power consumption in VLPS mode.

Note: While using this bit, it must be ensured that respective clock modules are disabled in VLPS mode.

Else, severe malfunction of clock modules will happen.

unchecked - No effect.

checked - In VLPS mode, the bias currents and reference voltages for the following clock modules are disabled: SIRC, FIRC, PLL.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.264 Parameter McuLowPowerBiasEnable

PMC_REGSC[BIASEN] - Bias Enable Bit.

This bit enables source and well biasing for the core logic in low power mode. In full performance mode this bit has no effect. This is useful to further reduce MCU power consumption in low power mode.

unchecked - Biasing disabled, core logic can run in full performance.

checked - Biasing enabled, core logic is slower and there are restrictions in allowed system clock speed.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.265 Container McuPMC_AE_Config

This PMC_AE Control Register contains the various control settings of the PMC_AE block.

Note: Implementation Specific Parameter.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.266 Parameter McuLowVoltageDetectInterruptsOnVLSEnable

PMC_MONITOR[LVDVLSIE] - LVD on VLS interrupt enable.

This is to enable interrupt for low voltage detect on VLS (GDU) power domain.

An interrupt will be requested in case of LVDVLSF=1.

0 - Low voltage detect interrupts on VLS disabled.

1 - Low voltage detect interrupts on VLS enabled.

Note:

- Implementation Specific Parameter.
- Only support for S32M2xx series of devices.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.267 Parameter McuLowVoltageDetectInterruptsOnVDDCEnable

PMC_MONITOR[LVDCIE] - LVD on VDDC interrupt enable.

This is to enable interrupt for low voltage detect on VDDC power domain.

An interrupt will be requested in case of LVDCF=1.

0 - Low voltage detect interrupts on VDDC disabled.

1 - Low voltage detect interrupts on VDDC enabled.

Note:

- Implementation Specific Parameter.
- Only support for S32M2xx series of devices.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.268 Parameter McuHighVoltageDetectInterruptOnVDDEnable

PMC_MONITOR[HVDVDDIE] - HVD on VDD interrupt enable.

This is to enable interrupt for high voltage detection on VDD supply.

An interrupt will be requested in case of HVDVDDF=1.

0 - High voltage detect interrupt disabled.

1 - High voltage detect interrupt enabled.

Note:

- Implementation Specific Parameter.
- Only support for S32M2xx series of devices.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.269 Parameter McuHighVoltageDetectInterruptOnVDDINTAndVDD15Enable

PMC_MONITOR[HVDINT15IE] - HVD on VDD interrupt enable.

This is to enable interrupt for high voltage detection on the A10 internal supplies VDDINT or VDD15.

An interrupt will be requested in case of HVD15F=1 or HVDINTF=1.

0 - High voltage detect interrupt disabled.

1 - High voltage detect interrupt enabled.

Note:

- Implementation Specific Parameter.
- Only support for S32M2xx series of devices.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.270 Parameter McuLinphySupplyEnable

PMC_CONFIG[LINSUPEN] - LINPHY supply enable bit.

Enables the LINPHY supply. If disabled LINPHY is turned high ohmic.

In case LINPHY is generally not used setting LINSUPEN=0 will save power.

0 - LINPHY supply is high ohmic (off).

1 - LINPHY supply is as selected by LINSUPSEL bit.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.271 Parameter McuVDDCEnable

PMC_CONFIG[VDDCEN] - VDDC enable bit.

This Bit enables the VDDC supply.

A PMC internal regulator generates the VDDC from VPRE supply.

VDDC is the 5V supply for the CAN physical interface IP.

0 - VDDC is disabled.

1 - VDDC is enabled and regulated to 5V.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.272 Parameter McuLvdVlsSelect

PMC_CONFIG[LVDVLSSEL] - LVD VLS select.

The LVD (Low-Voltage-Detection) threshold for generated VLS supply is selectable. There are two options.

5_5V - LVD threshold on VLS supply is 5.5V.

6_5V - LVD threshold on VLS supply is 6.5V.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	LVD_6_5V
literals	['LVD_5_5V', 'LVD_6_5V']

4.1.273 Parameter McuLinphySupplySelect

PMC_CONFIG[LINSUPSEL] - LINPHY supply select.

Selects LINPHY supply to be either connected to VSUP pin or HD pin (GDU).

In case using the boost option in GDU, the LINPHY supply must be on HD pin.

0b - LINPHY supply connects to VSUP pin.

1b - LINPHY supply connects to HD pin (of GDU).

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	VSUP_PIN
literals	['VSUP_PIN', 'HD_PIN']

4.1.274 Parameter McuVddVoltageLevelSelect

PMC_CONFIG[VDDSEL5V] - VDD voltage level select.

Selects VDD (supply for MCU) to be either 3.3V or 5V.

3_3V - VDD is regulated to 3.3V.

5V - VDD is regulated to 5V.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	VDD_3_3V
literals	['VDD_3_3V', 'VDD_5V']

4.2 RAM

The low level driver component can only be used with S32CT Configurator in S32 Design Studio.

- Module [RAM](#)
 - Container [RamUserCfgRef](#)
 - * Parameter [RamIpDevErrorDetect](#)
 - * Parameter [RamEnableUserModeSupport](#)
 - * Parameter [RamIpGetRamStateApi](#)

Configuration Parameters

- * Parameter [McuTimeout](#)
- * Parameter [McuTimeoutMethod](#)
- * Container [McuRamSectorSettingConf](#)
 - Parameter [McuRamSectorId](#)
 - Parameter [McuRamDefaultValue](#)
 - Parameter [McuRamSectionBaseAddress](#)
 - Parameter [McuRamSectionSize](#)
 - Parameter [McuRamSectionWriteSize](#)
 - Parameter [McuRamSectionBaseAddrLinkerSym](#)
 - Parameter [McuRamSectionSizeLinkerSym](#)
 - Reference [McuRamEcucPartitionRef](#)

4.2.1 Module RAM

N/A

Included containers:

- [RamUserCfgRef](#)

Property	Value
type	MODULE-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-POST-BUILD;VARIANT-PRE-COMPILE

4.2.2 Container RamUserCfgRef

N/A

Included subcontainers:

- [McuRamSectorSettingConf](#)

Property	Value
type	CONTAINER
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.2.3 Parameter RamIpDevErrorDetect

Pre-processor switch for enabling the default error detection and reporting to the DevAssert.

The detection of default errors is configurable (ON/OFF) at precompile time.

#define RAM_IP_DEV_ERROR_DETECT (STD_ON)/(STD_OFF) will be generated in Ram_Ip_Cfg.h file.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.2.4 Parameter RamEnableUserModeSupport

When this parameter is enabled, the MDL module will adapt to run from User Mode, with the following measures:

a) configuring REG_PROT for ABC1, ABC2 IPs so that the registers under protection can be accessed from user mode by setting UAA bit in REG_PROT_GCR to 1

b) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.

c) other module specific measures

for more information, please see chapter 5.7 User Mode Support in IM

Note: Implementation Specific Parameter.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.2.5 Parameter RamIpGetRamStateApi

Pre-processor switch to enable/disable the API `Mcu_GetRamState`.

`#define RAM_IP_GET_RAM_STATE_API (STD_ON)/(STD_OFF)` will be generated in `Ram_Ip_Cfg_Defines.h` file.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.3 POWER

The low level driver component can only be used with S32CT Configurator in S32 Design Studio.

- Module [POWER](#)
 - Container [PowerUserCfgRef](#)
 - * Container [McuGeneralConfiguration](#)
 - Parameter [PowerIpDevErrorDetect](#)
 - Parameter [McuEnterLowPowerMode](#)
 - Parameter [McuPerformResetApi](#)
 - Parameter [PowerEnableUserModeSupport](#)
 - Parameter [McuTimeout](#)
 - Parameter [McuTimeoutMethod](#)
 - Parameter [McuPmcAeNotification](#)
 - Parameter [McuErrorIsrNotification](#)
 - Parameter [McuNoPll](#)
 - Parameter [McuDisableRcmInit](#)
 - Parameter [McuDisablePmcInit](#)
 - Parameter [McuDisableSmcInit](#)
 - * Container [McuDebugConfiguration](#)
 - Parameter [McuEnablePeripheralCMU](#)
 - Parameter [McuSRAMRetentionConfigApi](#)

- Parameter [McuGetPowerModeStatetApi](#)
- Parameter [McuPmcAeConfigApi](#)
- Parameter [McuAecResetConfigApi](#)
- * Container [McuModuleConfiguration](#)
 - Container [McuResetConfig](#)
 - Parameter [McuResetPinFilterBusClockSelect](#)
 - Parameter [McuResetPinFilterInStopMode](#)
 - Parameter [McuResetPinFilterInRunAndWait](#)
 - Container [McuSystemInterruptEnable](#)
 - Container [McuResetGeneratorConfiguration](#)
- Container [McuResetReasonConf](#)
 - Parameter [McuResetReason](#)
- Container [McuPowerControl](#)
 - Container [McuPMC_Config](#)
 - Container [McuPMC_AE_Config](#)
- Container [McuModeSettingConf](#)
 - Parameter [McuMode](#)
 - Parameter [McuPowerMode](#)
 - Parameter [McuEnableSleepOnExit](#)
 - Reference [McuPowerEcucPartitionRef](#)
- Container [McuAllowedModes](#)
 - Parameter [McuAllowHighSpeedRunMode](#)
 - Parameter [McuAllowVeryLowPowerModes](#)
- Container [McuInterruptEvents](#)
 - Parameter [McuVoltageErrorEvent](#)
 - Parameter [McuAlternateResetEvent](#)
- Container [McuResetConfig](#)
 - Parameter [McuResetPinFilterBusClockSelect](#)
 - Parameter [McuResetPinFilterInStopMode](#)
 - Parameter [McuResetPinFilterInRunAndWait](#)
 - Container [McuSystemInterruptEnable](#)
 - Container [McuResetGeneratorConfiguration](#)
 - Container [McuSystemInterruptEnable](#)
 - Parameter [McuResetDelayTime](#)
 - Parameter [McuStopAcknowledgeErrorInterrupt](#)
 - Parameter [McuMDMAPSystemResetInterrupt](#)

- Parameter [McuSoftwareInterrupt](#)
- Parameter [McuCoreLockupInterrupt](#)
- Parameter [McuJTAGResetInterrupt](#)
- Parameter [McuGlobalInterrupt](#)
- Parameter [McuExternalResetPinInterrupt](#)
- Parameter [McuWatchdogInterrupt](#)
- Parameter [McuCMULossOfClockResetInterrupt](#)
- Parameter [McuLossOfLockInterrupt](#)
- Parameter [McuLossOfClockInterrupt](#)
- Container [McuResetGeneratorConfiguration](#)
 - Parameter [McuRegsOtpReset](#)
 - Parameter [McuCanPhyReset](#)
 - Parameter [McuLinPhyHpReset](#)
 - Parameter [McuLinPhyLpReset](#)
 - Parameter [McuGduReset](#)
 - Parameter [McuHviReset](#)
 - Parameter [McuDpgaReset](#)
 - Parameter [McuTempsensorReset](#)
 - Parameter [McuCxpiReset](#)
- Container [McuPowerControl](#)
 - Container [McuPMC_Config](#)
 - Container [McuPMC_AE_Config](#)
 - Container [McuPMC_Config](#)
 - Parameter [McuLowVoltageDetectInterruptEnable](#)
 - Parameter [McuLowVoltageDetectResetEnable](#)
 - Parameter [McuLowVoltageWarningInterruptEnable](#)
 - Parameter [McuLPODisable](#)
 - Parameter [McuLPOTRIMSelect](#)
 - Parameter [McuClockBiasDisable](#)
 - Parameter [McuLowPowerBiasEnable](#)
 - Container [McuPMC_AE_Config](#)
 - Parameter [McuLowVoltageDetectInterruptsOnVLSEnable](#)
 - Parameter [McuLowVoltageDetectInterruptsOnVDDCEnable](#)
 - Parameter [McuHighVoltageDetectInterruptOnVDDEnable](#)
 - Parameter [McuHighVoltageDetectInterruptOnVDDINTAndVDD15Enable](#)
 - Parameter [McuLinphySupplyEnable](#)

- Parameter [McuVDDCEnable](#)
- Parameter [McuLvdVlsSelect](#)
- Parameter [McuLinphySupplySelect](#)
- Parameter [McuVddVoltageLevelSelect](#)

4.3.1 Module POWER

N/A

Included containers:

- [PowerUserCfgRef](#)

Property	Value
type	MODULE-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-POST-BUILD;VARIANT-PRE-COMPILE

4.3.2 Container PowerUserCfgRef

N/A

Included subcontainers:

- [McuGeneralConfiguration](#)
- [McuDebugConfiguration](#)
- [McuModuleConfiguration](#)

Property	Value
type	CONTAINER
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3.3 Parameter PowerIpDevErrorDetect

Pre-processor switch for enabling the default error detection and reporting to the DevAssert.

The detection of default errors is configurable (ON/OFF) at precompile time.

#define POWER_IP_DEV_ERROR_DETECT (STD_ON)/(STD_OFF) will be generated in Power_Ip_Cfg.h file.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.3.4 Parameter PowerEnableUserModeSupport

When this parameter is enabled, the MDL module will adapt to run from User Mode, with the following measures:

- a) configuring REG_PROT for ABC1, ABC2 IPs so that the registers under protection can be accessed from user mode by setting UAA bit in REG_PROT_GCR to 1
- b) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.
- c) other module specific measures

for more information, please see chapter 5.7 User Mode Support in IM

Note: Implementation Specific Parameter.

Property	Value
type	BOOLEAN
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false



Chapter 5

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5.1 Software Specification

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Chapter 6

Module Documentation

6.1 Clock Ip Driver

6.1.1 Detailed Description

Data Structures

- struct [Clock_Ip_RegisterValueType](#)
Register value structure. Implements Clock_Ip_RegisterValueType_Class. [More...](#)
- struct [Clock_Ip_RegisterIndexType](#)
Register index structure. Implements Clock_Ip_RegisterIndexType_Class. [More...](#)
- struct [Clock_Ip_IrcoscConfigType](#)
Clock Source IRCOSC configuration structure. Implements Clock_Ip_IrcoscConfigType_Class. [More...](#)
- struct [Clock_Ip_XoscConfigType](#)
CGM Clock Source XOSC configuration structure. Implements Clock_Ip_XoscConfigType_Class. [More...](#)
- struct [Clock_Ip_PllConfigType](#)
CGM Clock Source PLLDIG configuration structure. Implements Clock_Ip_PllConfigType_Class. [More...](#)
- struct [Clock_Ip_SelectorConfigType](#)
Clock selector configuration structure. Implements Clock_Ip_SelectorConfigType_Class. [More...](#)
- struct [Clock_Ip_DividerConfigType](#)
Clock divider configuration structure. Implements Clock_Ip_DividerConfigType_Class. [More...](#)
- struct [Clock_Ip_DividerTriggerConfigType](#)
Clock divider trigger configuration structure. Implements Clock_Ip_DividerTriggerConfigType_Class. [More...](#)
- struct [Clock_Ip_FracDivConfigType](#)
Clock fractional divider configuration structure. Implements Clock_Ip_FracDivConfigType_Class. [More...](#)
- struct [Clock_Ip_ExtClkConfigType](#)
Clock external clock configuration structure. Implements Clock_Ip_ExtClkConfigType_Class. [More...](#)
- struct [Clock_Ip_PcfsConfigType](#)
Clock Source PCFS configuration structure. Implements Clock_Ip_PcfsConfigType_Class. [More...](#)
- struct [Clock_Ip_GateConfigType](#)
Clock gate clock configuration structure. Implements Clock_Ip_GateConfigType_Class. [More...](#)
- struct [Clock_Ip_CmuConfigType](#)

Clock cmu configuration structure. Implements Clock_Ip_CmuConfigType_Class. [More...](#)

- struct [Clock_Ip_ConfiguredFrequencyType](#)

Configured frequency structure. Implements Clock_Ip_ConfiguredFrequencyType_Class. [More...](#)

- struct [Clock_Ip_SpecificPeriphParamType](#)

Clock Specific peripheral configure. Implements Clock_Ip_SpecificPeriphParamType_Class. [More...](#)

- struct [Clock_IP_SpecificPeriphConfigType](#)

Clock Specific peripheral structure. Implements Clock_IP_SpecificPeriphConfigType_Class. [More...](#)

- struct [Clock_Ip_ClockConfigType](#)

Clock configuration structure. Implements Clock_Ip_ClockConfigType_Class. [More...](#)

Types Reference

- typedef void(* [Clock_Ip_NotificationsCallbackType](#)) ([Clock_Ip_NotificationType](#) Error, [Clock_Ip_NameType](#) ClockName)

Clock notifications callback type. Implements ClockNotificationsCallbackType_Class.

Enum Reference

- enum [Clock_Ip_ClockNameSourceType](#)

Clock ip source type.

- enum [Clock_Ip_PllStatusReturnType](#)

Clock pll status return codes.

- enum [Clock_Ip_DfsStatusType](#)

Clock dfs status return codes.

- enum [Clock_Ip_CommandType](#)

Clock ip specific commands.

- enum [Clock_Ip_PowerModesType](#)

Power modes.

- enum [Clock_Ip_PowerNotificationType](#)

Power mode notification.

- enum [Clock_Ip_NameType](#)

Clock names.

- enum [Clock_Ip_StatusType](#)

Clock ip status return codes.

- enum [Clock_Ip_PllStatusType](#)

Clock ip pll status return codes.

- enum [Clock_Ip_NotificationType](#)

Clock ip report error types.

- enum [Clock_Ip_TriggerDividerType](#)

Clock ip trigger divider type.

- enum [Clock_Ip_SpecificPeriphParamType](#)

specific peripheral.

- enum [Clock_Ip_MonitorFrequencyVariationType](#)

Clock ip monitor maximum allowed frequency variation.

- enum [Clock_Ip_MonitorLatencyType](#)

Clock ip latency of clock monitor reaction.

Function Reference

- [Clock_Ip_StatusType](#) [Clock_Ip_Init](#) ([Clock_Ip_ClockConfigType](#) const *Config)
Set clock configuration according to pre-defined structure.
- void [Clock_Ip_InitClock](#) ([Clock_Ip_ClockConfigType](#) const *Config)
Set the PLL and other MCU specific clock options.
- [Clock_Ip_PllStatusType](#) [Clock_Ip_GetPllStatus](#) (void)
Returns the lock status of the PLL.
- void [Clock_Ip_DistributePll](#) (void)
Activates the PLL in MCU clock distribution.
- void [Clock_Ip_InstallNotificationsCallback](#) ([Clock_Ip_NotificationsCallbackType](#) Callback)
Install a clock notifications callback.
- void [Clock_Ip_DisableClockMonitor](#) ([Clock_Ip_NameType](#) ClockName)
Disables a clock monitor.
- void [Clock_Ip_DisableModuleClock](#) ([Clock_Ip_NameType](#) ClockName)
Disables clock for a peripheral.
- void [Clock_Ip_EnableModuleClock](#) ([Clock_Ip_NameType](#) ClockName)
Enables clock for a peripheral.
- void [Clock_Ip_PowerNotifications](#) ([Clock_Ip_PowerModesType](#) PowerMode, [Clock_Ip_PowerNotificationType](#) Notification)
Sends notifications regarding power mode transition.
- uint64 [Clock_Ip_GetClockFrequency](#) ([Clock_Ip_NameType](#) ClockName)
Gets the clock frequency for a specific clock name.
- void [Clock_Ip_StartTimeout](#) (uint32 *StartTimeOut, uint32 *ElapsedTimeOut, uint32 *TimeoutTicksOut, uint32 TimeoutUs)
Initializes a starting reference point for timeout.
- boolean [Clock_Ip_TimeoutExpired](#) (uint32 *StartTimeInOut, uint32 *ElapsedTimeInOut, uint32 TimeoutTicks)
Checks for timeout condition.
- void [Clock_Ip_SetScgAsyncDiv1_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
Write Config to SCG_SIRCDIV1 register.
- void [Clock_Ip_SetScgAsyncDiv2_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
Write Config to SCG_SIRCDIV2 register.
- void [Clock_Ip_SetScgRunDivcore_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
Write Config to SCG_RCCR register.
- void [Clock_Ip_SetScgRunDivbus_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
Write Config to SCG_RCCR register.
- void [Clock_Ip_SetScgRunDivslow_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)

- Write Config to SCG_RCCR register.*
- void [Clock_Ip_SetScgVlprDivcore_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
- Write Config to SCG_VCCR register.*
- void [Clock_Ip_SetScgVlprDivbus_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
- Write Config to SCG_VCCR register.*
- void [Clock_Ip_SetScgVlprDivslow_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
- Write Config to SCG_VCCR register.*
- void [Clock_Ip_SetSimClkoutDiv_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
- Write Config to IP_SIM::CHIPCTL register.*
- void [Clock_Ip_SetPccPcdDivFrac_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
- Write Config to PCC register.*
- void [Clock_Ip_SetSimTraceDivMul_TrustedCall](#) ([Clock_Ip_DividerConfigType](#) const *Config)
- Write Config to IP_SIM::CLKDIV4 register.*
- void [Clock_Ip_ResetSOSC_TrustedCall](#) ([Clock_Ip_XoscConfigType](#) const *Config)
- Reset IP_SCG::SOSCCSR register.*
- void [Clock_Ip_SetSOSC_TrustedCall](#) ([Clock_Ip_XoscConfigType](#) const *Config)
- Write Config of SOSC to register.*
- void [Clock_Ip_EnableSOSC_TrustedCall](#) ([Clock_Ip_XoscConfigType](#) const *Config)
- Enable SOSC.*
- void [Clock_Ip_ClockSetSimLPO1KEnable_TrustedCall](#) ([Clock_Ip_GateConfigType](#) const *Config)
- Set gate of LPO1K_CLK.*
- void [Clock_Ip_ClockSetSimLPO32KEnable_TrustedCall](#) ([Clock_Ip_GateConfigType](#) const *Config)
- Set gate of LPO3K_CLK.*
- void [Clock_Ip_ClockSetSimClkoutEnable_TrustedCall](#) ([Clock_Ip_GateConfigType](#) const *Config)
- Set gate of CLKOUT_CLK.*
- void [Clock_Ip_ClockSetPccCgcEnable_TrustedCall](#) ([Clock_Ip_GateConfigType](#) const *Config)
- Set gate of PCC peripheral.*
- void [Clock_Ip_ClockSetSimGate_TrustedCall](#) ([Clock_Ip_GateConfigType](#) const *Config)
- Set gate of CLOCK_IP_SIM_PLATCGC_CGC clock.*
- void [Clock_Ip_ClockSetSimTraceEnable_TrustedCall](#) ([Clock_Ip_GateConfigType](#) const *Config)
- Set gate of TRACE_CLK.*
- void [Clock_Ip_SetSirc_TrustedCall](#) ([Clock_Ip_IrcoscConfigType](#) const *Config)
- Write Config to SIRC register.*
- void [Clock_Ip_EnableSirc_TrustedCall](#) ([Clock_Ip_IrcoscConfigType](#) const *Config)
- Enable SIRC.*
- void [Clock_Ip_DisableSirc_TrustedCall](#) ([Clock_Ip_NameType](#) Name)
- Disable SIRC.*
- void [Clock_Ip_SetSircVlp_TrustedCall](#) ([Clock_Ip_IrcoscConfigType](#) const *Config)
- Write Config for SIRC in Low power mode to register.*
- void [Clock_Ip_EnableSircVlp_TrustedCall](#) ([Clock_Ip_IrcoscConfigType](#) const *Config)
- Enable Sirc in Low power mode.*

- void [Clock_Ip_DisableSircVlp_TrustedCall](#) (Clock_Ip_NameType Name)
Disable Sirc in Low power mode.
- void [Clock_Ip_SetSircStop_TrustedCall](#) (Clock_Ip_IrcoscConfigType const *Config)
Write Config for SIRC in Stop mode to register.
- void [Clock_Ip_EnableSircStop_TrustedCall](#) (Clock_Ip_IrcoscConfigType const *Config)
Enable SIRC in Stop mode to register.
- void [Clock_Ip_DisableSircStop_TrustedCall](#) (Clock_Ip_NameType Name)
Disable SIRC in Stop mode to register.
- void [Clock_Ip_SetFirc_TrustedCall](#) (Clock_Ip_IrcoscConfigType const *Config)
Write Config for Firc to register.
- void [Clock_Ip_EnableFirc_TrustedCall](#) (Clock_Ip_IrcoscConfigType const *Config)
Enable Firc to register.
- void [Clock_Ip_DisableFirc_TrustedCall](#) (Clock_Ip_NameType Name)
Disable Firc to register.
- void [Clock_Ip_ResetSpll_TrustedCall](#) (Clock_Ip_PllConfigType const *Config)
Reset Spll register.
- void [Clock_Ip_SetSpll_TrustedCall](#) (Clock_Ip_PllConfigType const *Config)
Write Config to Spll register.
- void [Clock_Ip_DisableSpll_TrustedCall](#) (Clock_Ip_NameType PllName)
Disable Spll.
- void [Clock_Ip_EnableSpll_TrustedCall](#) (Clock_Ip_PllConfigType const *Config)
Enable Spll.
- void [Clock_Ip_ResetScgRunSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)
Reset SCG_RCCR selector register.
- void [Clock_Ip_SetScgRunSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)
Write Config to SCG_RCCR selector register.
- void [Clock_Ip_SetScgVlprSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)
Write Config to SCG_VCCR selector register.
- void [Clock_Ip_ResetSimRtcSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)
Reset SIM_LPOCLKS_RTCLKSEL register.
- void [Clock_Ip_SetSimRtcSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)
Write Config to RTC_CLK register.
- void [Clock_Ip_ResetSimLpoSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)
Reset LPO_CLK selector.
- void [Clock_Ip_SetSimLpoSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)
Write Config to LPO_CLK selector.
- void [Clock_Ip_ResetScgClkoutSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)
Reset CLKOUT_CLK selector.
- void [Clock_Ip_SetScgClkoutSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)
Write Config to LPO_CLK register.
- void [Clock_Ip_ResetSimFtmoptSel_TrustedCall](#) (Clock_Ip_SelectorConfigType const *Config)

Reset SIM->FTMOPT0 selector register.

- void [Clock_Ip_SetSimFtmoptSel_TrustedCall](#) ([Clock_Ip_SelectorConfigType](#) const *Config)

Write Config to SIM->FTMOPT0 register.

- void [Clock_Ip_ResetSimClkoutSel_TrustedCall](#) ([Clock_Ip_SelectorConfigType](#) const *Config)

Write Config to SIM_CHIPCTL_CLKOUTSEL register.

- void [Clock_Ip_SetSimClkoutSel_TrustedCall](#) ([Clock_Ip_SelectorConfigType](#) const *Config)

Write Config to SIM_CHIPCTL_CLKOUTSEL register.

- void [Clock_Ip_ResetPccPcsSelect_TrustedCall](#) ([Clock_Ip_SelectorConfigType](#) const *Config)

Reset register of PCC peripheral clock.

- void [Clock_Ip_SetPccPcsSelect_TrustedCall](#) ([Clock_Ip_SelectorConfigType](#) const *Config)

Write Config to PCC register.

- void [Clock_Ip_ResetSimTraceSel_TrustedCall](#) ([Clock_Ip_SelectorConfigType](#) const *Config)

Reset selector register of TRACE_CLK.

- void [Clock_Ip_SetSimTraceSel_TrustedCall](#) ([Clock_Ip_SelectorConfigType](#) const *Config)

Write Config to TRACE_CLK selector register.

- void [SetFircToResetValue_TrustedCall](#) (void)

Reset FIRC.

- void [SetSimLpoclksRegister_TrustedCall](#) ([Clock_Ip_ClockConfigType](#) const *Config)

Write Config whole SIM_LPOCLKS register.

- void [Power_Ip_CM4_DisableSleepOnExit](#) (void)

The function disables SLEEPONEXIT bit.

- void [Power_Ip_CM4_EnableSleepOnExit](#) (void)

The function enables SLEEPONEXIT bit.

- void [Power_Ip_CM4_EnableDeepSleep](#) (void)

The function enables DEEPSLEEP.

- void [Power_Ip_CM4_DisableDeepSleep](#) (void)

The function disables DEEPSLEEP.

- void [Power_Ip_CM4_SystemReset](#) (void)

The function initiates a system reset request to reset the SoC.

- void [Power_Ip_PMC_PowerInit](#) (const [Power_Ip_PMC_ConfigType](#) *ConfigPtr)

This function configure the Power Management Controller.

- void [Power_Ip_RCM_ResetInit](#) (const [Power_Ip_RCM_ConfigType](#) *ConfigPtr)

This function initializes the Reset parameters.

- uint32 [Power_Ip_RCM_GetResetReason](#) (void)

This function returns the Reset reason.

- [Power_Ip_RawResetType](#) [Power_Ip_RCM_GetResetRawValue](#) (void)

This function returns the Raw Reset value.

- void [Power_Ip_SIM_SRAMRetentionConfig](#) ([Power_Ip_SRAMRetenConfigType](#) SRAMRetenConfig)

Configuration for SRAM retention.

- void [Power_Ip_SMC_AllowedModesConfig](#) (const [Power_Ip_SMC_ConfigType](#) *ConfigPtr)

This function will configure the allowed modes.

- uint32 [Power_Ip_SMC_ModeConfig](#) (const [Power_Ip_ModeConfigType](#) *ModeConfigPtr)

This function switches the mode by writing SMC_PMCTRL and SMC_STOPCTRL.

6.1.2 Data Structure Documentation

6.1.2.1 struct Clock_Ip_RegisterValueType

Register value structure. Implements Clock_Ip_RegisterValueType_Class.

Definition at line 3255 of file Clock_Ip_Types.h.

Data Fields

uint32 *	RegisterAddr	Register address.
uint32	RegisterData	Register value.

6.1.2.2 struct Clock_Ip_RegisterIndexType

Register index structure. Implements Clock_Ip_RegisterIndexType_Class.

Definition at line 3266 of file Clock_Ip_Types.h.

Data Fields

uint16	StartIndex	Start index in register array.
uint16	EndIndex	End index in register array.

6.1.2.3 struct Clock_Ip_IrcoscConfigType

Clock Source IRCOSC configuration structure. Implements Clock_Ip_IrcoscConfigType_Class.

Definition at line 3279 of file Clock_Ip_Types.h.

Data Fields

Clock_Ip_NameType	Name	Clock name associated to ircosc
uint16	Enable	Enable ircosc .
uint8	Regulator	Enable regulator.
uint8	Range	Ircosc range.
uint8	LowPowerModeEnable	Ircosc enable in VLP mode
uint8	StopModeEnable	Ircosc enable in STOP mode

6.1.2.4 struct Clock_Ip_XoscConfigType

CGM Clock Source XOSC configuration structure. Implements Clock_Ip_XoscConfigType_Class.

Definition at line 3295 of file Clock_Ip_Types.h.

Data Fields

Clock_Ip_NameType	Name	Clock name associated to xosc
uint32	Freq	External oscillator frequency.

Data Fields

uint16	Enable	Enable xosc.
uint16	StartupDelay	Startup stabilization time.
uint8	BypassOption	XOSC bypass option
uint8	CompEn	Comparator enable
uint8	TransConductance	Crystal overdrive protection
uint8	Gain	Gain value
uint8	Monitor	Monitor type
uint8	AutoLevelController	Automatic level controller
uint8	LevelShifterCurrent	Level Shifter Current
uint8	ComparatorCurrent	Comparator Current

6.1.2.5 struct Clock_Ip_PllConfigType

CGM Clock Source PLLDIG configuration structure. Implements Clock_Ip_PllConfigType_Class.

Definition at line 3320 of file Clock_Ip_Types.h.

Data Fields

Clock_Ip_NameType	Name	Clock name associated to pll
uint16	Enable	Enable pll.
Clock_Ip_NameType	InputReference	Input reference.
uint8	Bypass	Bypass pll.
uint8	Predivider	Input clock predivider.
uint16	Multiplier	Clock multiplier.
uint8	Postdivider	Clock postdivider.
uint16	NumeratorFracLoopDiv	Numerator of fractional loop division factor (MFN)
uint8	MulFactorDiv	Multiplication factor divider (MFD)
uint8	FrequencyModulationBypass	Enable/disable modulation
uint8	ModulationType	Modulation type
uint16	ModulationPeriod	Stepsize - modulation period
uint16	IncrementStep	Stepno - step no
uint8	SigmaDelta	Sigma Delta Modulation Enable
uint8	DitherControl	Dither control enable
uint8	DitherControlValue	Dither control value
uint8	Monitor	Monitor type
uint16	Dividers[3U]	Dividers values
uint8	SoftwareDisable	Software Disable

6.1.2.6 struct Clock_Ip_SelectorConfigType

Clock selector configuration structure. Implements Clock_Ip_SelectorConfigType_Class.

Definition at line 3358 of file Clock_Ip_Types.h.

Data Fields

Clock_Ip_NameType	Name	Clock name associated to selector
Clock_Ip_NameType	Value	Name of the selected input source

6.1.2.7 struct Clock_Ip_DividerConfigType

Clock divider configuration structure. Implements Clock_Ip_DividerConfigType_Class.

Definition at line 3369 of file Clock_Ip_Types.h.

Data Fields

Clock_Ip_NameType	Name	Clock name associated to divider.
uint32	Value	Divider value - if value is zero then divider is disabled.
uint8	Options[1U]	Option divider value - this value depend hardware information.

6.1.2.8 struct Clock_Ip_DividerTriggerConfigType

Clock divider trigger configuration structure. Implements Clock_Ip_DividerTriggerConfigType_Class.

Definition at line 3380 of file Clock_Ip_Types.h.

Data Fields

Clock_Ip_NameType	Name	Clock name associated to divider for which trigger is configured.
Clock_Ip_TriggerDividerType	TriggerType	Trigger value - if value is zero then divider is updated immediately, divider is not triggered.
Clock_Ip_NameType	Source	Clock name of the common input source of all dividers from the same group that support a common update

6.1.2.9 struct Clock_Ip_FracDivConfigType

Clock fractional divider configuration structure. Implements Clock_Ip_FracDivConfigType_Class.

Definition at line 3394 of file Clock_Ip_Types.h.

Data Fields

Clock_Ip_NameType	Name	Clock name associated to fractional divider.
uint8	Enable	Enable control for port n
uint32	Value[2U]	Fractional dividers

6.1.2.10 struct Clock_Ip_ExtClkConfigType

Clock external clock configuration structure. Implements Clock_Ip_ExtClkConfigType_Class.

Definition at line 3406 of file Clock_Ip_Types.h.

Data Fields

Clock_Ip_NameType	Name	Clock name of the external clock.
uint32	Value	Enable value - if value is zero then clock is gated, otherwise is enabled in different modes.

6.1.2.11 struct Clock_Ip_PcfsConfigType

Clock Source PCFS configuration structure. Implements `Clock_Ip_PcfsConfigType_Class`.

Definition at line 3417 of file `Clock_Ip_Types.h`.

Data Fields

Clock_Ip_NameType	Name	Clock source from which ramp-down and to which ramp-up are processed.
uint32	MaxAllowableIDDchange	Maximum variation of current per time (mA/microsec) - max allowable IDD change is determined by the user's power supply design.
uint32	StepDuration	Step duration of each PCFS step
Clock_Ip_NameType	SelectorName	Name of the selector that supports PCFS and name is one the inputs that can be selected
uint32	ClockSourceFrequency	Frequency of the clock source from which ramp-down and to which ramp-up are processed.

6.1.2.12 struct Clock_Ip_GateConfigType

Clock gate clock configuration structure. Implements `Clock_Ip_GateConfigType_Class`.

Definition at line 3431 of file `Clock_Ip_Types.h`.

Data Fields

Clock_Ip_NameType	Name	Clock name associated to clock gate.
uint16	Enable	Enable or disable clock

6.1.2.13 struct Clock_Ip_CmuConfigType

Clock cmu configuration structure. Implements `Clock_Ip_CmuConfigType_Class`.

Definition at line 3442 of file `Clock_Ip_Types.h`.

Data Fields

Clock_Ip_NameType	Name	Clock name associated to clock monitor.
uint8	Enable	Enable/disable clock monitor
uint32	Interrupt	Enable/disable interrupt
Clock_Ip_MonitorFrequencyVariationType	Variation	Maximum allowed frequency variation.
Clock_Ip_MonitorLatencyType	Latency	Monitor latency range.

Data Fields

uint32	MonitoredClockFrequency	Frequency of the clock source from which ramp-down and to which ramp-up are processed.
Clock_Ip_RegisterIndexType	Indexes	Register index if register value optimization is enabled.

6.1.2.14 struct Clock_Ip_ConfiguredFrequencyType

Configured frequency structure. Implements [Clock_Ip_ConfiguredFrequencyType_Class](#).

Definition at line 3457 of file [Clock_Ip_Types.h](#).

Data Fields

Clock_Ip_NameType	Name	Clock name of the configured frequency value
uint32	ConfiguredFrequencyValue	Configured frequency value

6.1.2.15 struct Clock_Ip_SpecificPerpihParamType

Clock Specific peripheral configure. Implements [Clock_Ip_SpecificPerpihParamType_Class](#).

Definition at line 3467 of file [Clock_Ip_Types.h](#).

6.1.2.16 struct Clock_IP_SpecificPeriphConfigType

Clock Specific peripheral structure. Implements [Clock_IP_SpecificPeriphConfigType_Class](#).

Definition at line 3477 of file [Clock_Ip_Types.h](#).

6.1.2.17 struct Clock_Ip_ClockConfigType

Clock configuration structure. Implements [Clock_Ip_ClockConfigType_Class](#).

Definition at line 3487 of file [Clock_Ip_Types.h](#).

Data Fields

uint32	ClkConfigId	The ID for Clock configuration
const Clock_Ip_RegisterValueType (*	RegValues[]	Pointer to register values array
uint8	IrcoscsCount	IRCOSCs count
uint8	XoscsCount	XOSCs count
uint8	PllsCount	PLLs count
uint8	SelectorsCount	Selectors count
uint8	DividersCount	Dividers count
uint8	DividerTriggersCount	Divider triggers count
uint8	FracDivsCount	Fractional dividers count
uint8	ExtClksCount	External clocks count
uint8	GatesCount	Clock gates count
uint8	PcfsCount	Clock pcfs count
uint8	CmusCount	Clock cmus count

Data Fields

uint8	ConfigureFrequenciesCount	Configured frequencies count
Clock_Ip_IrcoscConfigType	Ircoscs[(2U)]	IRCOSCs
Clock_Ip_XoscConfigType	Xoscs[(1U)]	XOSCs
Clock_Ip_PllConfigType	Plls[(1U)]	PLLs
Clock_Ip_SelectorConfigType	Selectors[(26U)]	Selectors
Clock_Ip_DividerConfigType	Dividers[(17U)]	Dividers
Clock_Ip_DividerTriggerConfigType	DividerTriggers[1U]	Divider triggers
Clock_Ip_FracDivConfigType	FracDivs[1U]	Fractional dividers
Clock_Ip_ExtClkConfigType	ExtClks[(4U)]	External clocks
Clock_Ip_GateConfigType	Gates[(39U)]	Clock gates
Clock_Ip_PcfsConfigType	Pcfs[1U]	Progressive clock switching
Clock_Ip_CmuConfigType	Cmus[1U]	Clock cmus
Clock_IP_SpecificPeriphConfigType	SpecificPeriphConfiguration	Clock specific peripheral configuration
Clock_Ip_ConfiguredFrequencyType	ConfiguredFrequencies[(1U)]	Configured frequency values

6.1.3 Types Reference

6.1.3.1 Clock_Ip_NotificationsCallbackType

```
typedef void(* Clock_Ip_NotificationsCallbackType) (Clock\_Ip\_NotificationType Error, Clock\_Ip\_NameType
ClockName)
```

Clock notifications callback type. Implements ClockNotificationsCallbackType_Class.

Definition at line 3249 of file Clock_Ip_Types.h.

6.1.4 Enum Reference

6.1.4.1 Clock_Ip_ClockNameSourceType

```
enum Clock\_Ip\_ClockNameSourceType
```

Clock ip source type.

Enumerator

UNKNOWN_TYPE	Clock path from source to this clock name has at least one selector.
IRCOSC_TYPE	Source is an internal oscillator.
XOSC_TYPE	Source is an external oscillator.
PLL_TYPE	Source is a pll.
EXT_CLK_TYPE	Source is an external clock.
SERDES_TYPE	Source is a SERDES.

Definition at line 256 of file Clock_Ip_Private.h.

6.1.4.2 Clock_Ip_PllStatusReturnType

enum `Clock_Ip_PllStatusReturnType`

Clock pll status return codes.

Enumerator

STATUS_PLL_NOT_ENABLED	Not enabled
STATUS_PLL_UNLOCKED	Unlocked
STATUS_PLL_LOCKED	Locked

Definition at line 270 of file Clock_Ip_Private.h.

6.1.4.3 Clock_Ip_DfsStatusType

enum `Clock_Ip_DfsStatusType`

Clock dfs status return codes.

Enumerator

STATUS_DFS_NOT_ENABLED	Not enabled
STATUS_DFS_UNLOCKED	Unlocked
STATUS_DFS_LOCKED	Locked

Definition at line 280 of file Clock_Ip_Private.h.

6.1.4.4 Clock_Ip_CommandType

enum `Clock_Ip_CommandType`

Clock ip specific commands.

Enumerator

CLOCK_IP_RESERVED_COMMAND	Reserved command
CLOCK_IP_INITIALIZE_PLATFORM_COMMAND	Specific platform objects
CLOCK_IP_INITIALIZE_CLOCK_OBJECTS_COMMAND	Initialize clock objects
CLOCK_IP_SET_USER_ACCESS_ALLOWED_COMMAND	User access allowed
CLOCK_IP_DISABLE_SAFE_CLOCK_COMMAND	Disable safe clock

Definition at line 290 of file Clock_Ip_Private.h.

6.1.4.5 Clock_Ip_PowerModesTypeenum `Clock_Ip_PowerModesType`

Power modes.

Definition at line 194 of file `Clock_Ip_Types.h`.**6.1.4.6 Clock_Ip_PowerNotificationType**enum `Clock_Ip_PowerNotificationType`

Power mode notification.

Definition at line 208 of file `Clock_Ip_Types.h`.**6.1.4.7 Clock_Ip_NameType**enum `Clock_Ip_NameType`

Clock names.

Definition at line 219 of file `Clock_Ip_Types.h`.**6.1.4.8 Clock_Ip_StatusType**enum `Clock_Ip_StatusType`

Clock ip status return codes.

Enumerator

<code>CLOCK_IP_SUCCESS</code>	Clock tree was initialized successfully.
<code>CLOCK_IP_ERROR</code>	One of the elements timeout, clock tree couldn't be initialized.

Definition at line 3120 of file `Clock_Ip_Types.h`.**6.1.4.9 Clock_Ip_PllStatusType**enum `Clock_Ip_PllStatusType`

Clock ip pll status return codes.

Enumerator

<code>CLOCK_IP_PLL_LOCKED</code>	PLL is locked
<code>CLOCK_IP_PLL_UNLOCKED</code>	PLL is unlocked
<code>CLOCK_IP_PLL_STATUS_UNDEFINED</code>	PLL Status is unknown

Definition at line 3128 of file `Clock_Ip_Types.h`.

6.1.4.10 Clock_Ip_NotificationTypeenum `Clock_Ip_NotificationType`

Clock ip report error types.

Enumerator

<code>CLOCK_IP_CMU_ERROR</code>	Cmu Fccu notification.
<code>CLOCK_IP_REPORT_TIMEOUT_ERROR</code>	Report Timeout Error.
<code>CLOCK_IP_REPORT_FXOSC_CONFIGURATION_ERROR</code>	Report Fxosc Configuration Error.
<code>CLOCK_IP_REPORT_CLOCK_MUX_SWITCH_ERROR</code>	Report Clock Mux Switch Error.
<code>CLOCK_IP_RAM_MEMORY_CONFIG_ENTRY</code>	Ram config entry point.
<code>CLOCK_IP_RAM_MEMORY_CONFIG_EXIT</code>	Ram config exit point.
<code>CLOCK_IP_FLASH_MEMORY_CONFIG_ENTRY</code>	Flash config entry point.
<code>CLOCK_IP_FLASH_MEMORY_CONFIG_EXIT</code>	Flash config exit point.
<code>CLOCK_IP_ACTIVE</code>	Report Clock Active.
<code>CLOCK_IP_INACTIVE</code>	Report Clock Inactive.
<code>CLOCK_IP_REPORT_WRITE_PROTECTION_ERROR</code>	Report Write Protection Error.
<code>CLOCK_IP_SET_RAM_WAIT_STATES_ERROR</code>	Set Ram Wait States Error.

Definition at line 3137 of file `Clock_Ip_Types.h`.**6.1.4.11 Clock_Ip_TriggerDividerType**enum `Clock_Ip_TriggerDividerType`

Clock ip trigger divider type.

Enumerator

<code>IMMEDIATE_DIVIDER_UPDATE</code>	Immediate divider update.
<code>COMMON_TRIGGER_DIVIDER_UPDATE</code>	Common trigger divider update.

Definition at line 3154 of file `Clock_Ip_Types.h`.**6.1.4.12 Clock_Ip_SpecificPeriphParamType**enum `Clock_Ip_SpecificPeriphParamType`

specific peripheral.

Definition at line 3162 of file `Clock_Ip_Types.h`.**6.1.4.13 Clock_Ip_MonitorFrequencyVariationType**enum `Clock_Ip_MonitorFrequencyVariationType`

Clock ip monitor maximum allowed frequency variation.

Enumerator

CLOCK_IP_MAXIMUM_5_PERCENTAGE_DEVIATION	Maximum 5% variation.
CLOCK_IP_MAXIMUM_4_POINT_9_PERCENTAGE_DEVIATION	Maximum 4.9% variation.
CLOCK_IP_MAXIMUM_4_POINT_8_PERCENTAGE_DEVIATION	Maximum 4.8% variation.
CLOCK_IP_MAXIMUM_4_POINT_7_PERCENTAGE_DEVIATION	Maximum 4.7% variation.
CLOCK_IP_MAXIMUM_4_POINT_6_PERCENTAGE_DEVIATION	Maximum 4.6% variation.
CLOCK_IP_MAXIMUM_4_POINT_5_PERCENTAGE_DEVIATION	Maximum 4.5% variation.
CLOCK_IP_MAXIMUM_4_POINT_4_PERCENTAGE_DEVIATION	Maximum 4.4% variation.
CLOCK_IP_MAXIMUM_4_POINT_3_PERCENTAGE_DEVIATION	Maximum 4.3% variation.
CLOCK_IP_MAXIMUM_4_POINT_2_PERCENTAGE_DEVIATION	Maximum 4.2% variation.
CLOCK_IP_MAXIMUM_4_POINT_1_PERCENTAGE_DEVIATION	Maximum 4.1% variation.
CLOCK_IP_MAXIMUM_4_PERCENTAGE_DEVIATION	Maximum 4.0% variation.
CLOCK_IP_MAXIMUM_3_POINT_9_PERCENTAGE_DEVIATION	Maximum 3.9% variation.
CLOCK_IP_MAXIMUM_3_POINT_8_PERCENTAGE_DEVIATION	Maximum 3.8% variation.
CLOCK_IP_MAXIMUM_3_POINT_7_PERCENTAGE_DEVIATION	Maximum 3.7% variation.
CLOCK_IP_MAXIMUM_3_POINT_6_PERCENTAGE_DEVIATION	Maximum 3.6% variation.
CLOCK_IP_MAXIMUM_3_POINT_5_PERCENTAGE_DEVIATION	Maximum 3.5% variation.
CLOCK_IP_MAXIMUM_3_POINT_4_PERCENTAGE_DEVIATION	Maximum 3.4% variation.
CLOCK_IP_MAXIMUM_3_POINT_3_PERCENTAGE_DEVIATION	Maximum 3.3% variation.
CLOCK_IP_MAXIMUM_3_POINT_2_PERCENTAGE_DEVIATION	Maximum 3.2% variation.
CLOCK_IP_MAXIMUM_3_POINT_1_PERCENTAGE_DEVIATION	Maximum 3.1% variation.
CLOCK_IP_MAXIMUM_3_PERCENTAGE_DEVIATION	Maximum 3.0% variation.
CLOCK_IP_MAXIMUM_2_POINT_9_PERCENTAGE_DEVIATION	Maximum 2.9% variation.
CLOCK_IP_MAXIMUM_2_POINT_8_PERCENTAGE_DEVIATION	Maximum 2.8% variation.
CLOCK_IP_MAXIMUM_2_POINT_7_PERCENTAGE_DEVIATION	Maximum 2.7% variation.
CLOCK_IP_MAXIMUM_2_POINT_6_PERCENTAGE_DEVIATION	Maximum 2.6% variation.
CLOCK_IP_MAXIMUM_2_POINT_5_PERCENTAGE_DEVIATION	Maximum 2.5% variation.
CLOCK_IP_MAXIMUM_2_POINT_4_PERCENTAGE_DEVIATION	Maximum 2.4% variation.
CLOCK_IP_MAXIMUM_2_POINT_3_PERCENTAGE_DEVIATION	Maximum 2.3% variation.
CLOCK_IP_MAXIMUM_2_POINT_2_PERCENTAGE_DEVIATION	Maximum 2.2% variation.
CLOCK_IP_MAXIMUM_2_POINT_1_PERCENTAGE_DEVIATION	Maximum 2.1% variation.
CLOCK_IP_MAXIMUM_2_PERCENTAGE_DEVIATION	Maximum 2.0% variation.
CLOCK_IP_MAXIMUM_1_POINT_9_PERCENTAGE_DEVIATION	Maximum 1.9% variation.
CLOCK_IP_MAXIMUM_1_POINT_8_PERCENTAGE_DEVIATION	Maximum 1.8% variation.
CLOCK_IP_MAXIMUM_1_POINT_7_PERCENTAGE_DEVIATION	Maximum 1.7% variation.
CLOCK_IP_MAXIMUM_1_POINT_6_PERCENTAGE_DEVIATION	Maximum 1.6% variation.
CLOCK_IP_MAXIMUM_1_POINT_5_PERCENTAGE_DEVIATION	Maximum 1.5% variation.
CLOCK_IP_MAXIMUM_1_POINT_4_PERCENTAGE_DEVIATION	Maximum 1.4% variation.
CLOCK_IP_MAXIMUM_1_POINT_3_PERCENTAGE_DEVIATION	Maximum 1.3% variation.
CLOCK_IP_MAXIMUM_1_POINT_2_PERCENTAGE_DEVIATION	Maximum 1.2% variation.
CLOCK_IP_MAXIMUM_1_POINT_1_PERCENTAGE_DEVIATION	Maximum 1.1% variation.

Enumerator

CLOCK_IP_MAXIMUM_1_PERCENTAGE_DEVIATION	Maximum 1.0% variation.
CLOCK_IP_MAXIMUM_0_POINT_9_PERCENTAGE_DEVIATION	Maximum 0.9% variation.
CLOCK_IP_MAXIMUM_0_POINT_8_PERCENTAGE_DEVIATION	Maximum 0.8% variation.
CLOCK_IP_MAXIMUM_0_POINT_7_PERCENTAGE_DEVIATION	Maximum 0.7% variation.
CLOCK_IP_MAXIMUM_0_POINT_6_PERCENTAGE_DEVIATION	Maximum 0.6% variation.
CLOCK_IP_MAXIMUM_0_POINT_5_PERCENTAGE_DEVIATION	Maximum 0.5% variation.
CLOCK_IP_MAXIMUM_0_POINT_4_PERCENTAGE_DEVIATION	Maximum 0.4% variation.
CLOCK_IP_MAXIMUM_0_POINT_3_PERCENTAGE_DEVIATION	Maximum 0.3% variation.
CLOCK_IP_MAXIMUM_0_POINT_2_PERCENTAGE_DEVIATION	Maximum 0.2% variation.
CLOCK_IP_MAXIMUM_0_POINT_1_PERCENTAGE_DEVIATION	Maximum 0.1% variation.
CLOCK_IP_MAXIMUM_0_POINT_09_PERCENTAGE_DEVIATION	Maximum 0.09% variation.
CLOCK_IP_MAXIMUM_0_POINT_08_PERCENTAGE_DEVIATION	Maximum 0.08% variation.
CLOCK_IP_MAXIMUM_0_POINT_07_PERCENTAGE_DEVIATION	Maximum 0.07% variation.
CLOCK_IP_MAXIMUM_0_POINT_06_PERCENTAGE_DEVIATION	Maximum 0.06% variation.
CLOCK_IP_MAXIMUM_0_POINT_05_PERCENTAGE_DEVIATION	Maximum 0.05% variation.
CLOCK_IP_MAXIMUM_0_POINT_04_PERCENTAGE_DEVIATION	Maximum 0.04% variation.
CLOCK_IP_MAXIMUM_0_POINT_03_PERCENTAGE_DEVIATION	Maximum 0.03% variation.
CLOCK_IP_MAXIMUM_0_POINT_02_PERCENTAGE_DEVIATION	Maximum 0.02% variation.
CLOCK_IP_MAXIMUM_0_POINT_01_PERCENTAGE_DEVIATION	Maximum 0.01% variation.

Definition at line 3168 of file Clock_Ip_Types.h.

6.1.4.14 Clock_Ip_MonitorLatencyType

enum `Clock_Ip_MonitorLatencyType`

Clock ip latency of clock monitor reaction.

Enumerator

CLOCK_IP_LOWEST_LATENCY	Clock monitor is configured to react as fast as possible. A clock failure is detected by clock monitor unit in less than one microsecond.
CLOCK_IP_LOW_LATENCY_1US	Clock monitor reaction is taking 1us. A clock failure is detected by clock monitor unit in one microsecond.
CLOCK_IP_MEDIUM_LATENCY_10US	Clock monitor reaction is taking 10us. A clock failure is detected by clock monitor unit in 10 microseconds.
CLOCK_IP_HIGH_LATENCY_100US	Clock monitor reaction is taking 100us. A clock failure is detected by clock monitor unit in 100 microseconds.
CLOCK_IP_VERY_HIGH_LATENCY_1MS	Clock monitor reaction is taking 1ms. A clock failure is detected by clock monitor unit in 1 milisecond.

Definition at line 3233 of file Clock_Ip_Types.h.

6.1.5 Function Reference

6.1.5.1 Clock_Ip_Init()

```
Clock_Ip_StatusType Clock_Ip_Init (
    Clock_Ip_ClockConfigType const * Config )
```

Set clock configuration according to pre-defined structure.

This function sets system to target clock configuration; It sets the clock modules registers for clock mode change.

Parameters

in	<i>Config</i>	Pointer to configuration structure.
----	---------------	-------------------------------------

Returns

void

Note

If external clock is used in the target mode, please make sure it is enabled, for example, if the external oscillator is used, please setup correctly.

6.1.5.2 Clock_Ip_InitClock()

```
void Clock_Ip_InitClock (
    Clock_Ip_ClockConfigType const * Config )
```

Set the PLL and other MCU specific clock options.

This function initializes the PLL and other MCU specific clock options. The clock configuration parameters are provided via the configuration structure.

This function shall start the PLL lock procedure (if PLL shall be initialized) and shall return without waiting until the PLL is locked.

Parameters

in	<i>Config</i>	Pointer to configuration structure.
----	---------------	-------------------------------------

Returns

void

6.1.5.3 Clock_Ip_GetPllStatus()

```
Clock_Ip_PllStatusType Clock_Ip_GetPllStatus (
    void )
```

Returns the lock status of the PLL.

This function returns status of the PLL: undefined, unlocked or locked. This function returns undefined status if this function is called prior to calling of the function Clock_Ip_InitClock

Returns

Status. Pll lock status

6.1.5.4 Clock_Ip_DistributePll()

```
void Clock_Ip_DistributePll (
    void )
```

Activates the PLL in MCU clock distribution.

This function activates the PLL clock to the MCU clock distribution.

This function removes the current clock source (for example internal oscillator clock) from MCU clock distribution.

Application layer calls this function after the status of the PLL has been detected as locked by the function Clock_Ip_GetPllStatus.

The function Clock_Ip_DistributePll shall return without affecting the MCU hardware if the PLL clock has been automatically activated by the MCU hardware.

Returns

void

6.1.5.5 Clock_Ip_InstallNotificationsCallback()

```
void Clock_Ip_InstallNotificationsCallback (
    Clock_Ip_NotificationsCallbackType Callback )
```

Install a clock notifications callback.

This function installs a callback for reporting notifications from clock driver

Parameters

in	<i>Clock_Ip_NotificationsCallbackType</i>	notifications callback
----	---	------------------------

Returns

void

6.1.5.6 Clock_Ip_DisableClockMonitor()

```
void Clock_Ip_DisableClockMonitor (
    Clock_Ip_NameType ClockName )
```

Disables a clock monitor.

This function disables a clock monitor.

Parameters

in	<i>ClockName</i>	Clock Name.
----	------------------	-------------

Returns

void

6.1.5.7 Clock_Ip_DisableModuleClock()

```
void Clock_Ip_DisableModuleClock (
    Clock_Ip_NameType ClockName )
```

Disables clock for a peripheral.

This function disables clock for a peripheral.

Parameters

in	<i>ClockName</i>	Clock Name.
----	------------------	-------------

Returns

void

6.1.5.8 Clock_Ip_EnableModuleClock()

```
void Clock_Ip_EnableModuleClock (
    Clock_Ip_NameType ClockName )
```

Enables clock for a peripheral.

This function enables clock for a peripheral.

Parameters

in	<i>ClockName</i>	Clock Name.
----	------------------	-------------

Returns

void

6.1.5.9 Clock_Ip_PowerNotifications()

```
void Clock_Ip_PowerNotifications (
    Clock_Ip_PowerModesType PowerMode,
    Clock_Ip_PowerNotificationType Notification )
```

Sends notifications regarding power mode transition.

This function sends notifications regarding power mode transition. It is called by power driver each time power mode is changed.

Parameters

in	<i>PowerMode</i>	Power mode.
in	<i>PowerMode</i>	Power mode notification.

Returns

void

6.1.5.10 Clock_Ip_GetClockFrequency()

```
uint64 Clock_Ip_GetClockFrequency (
    Clock_Ip_NameType ClockName )
```

Gets the clock frequency for a specific clock name.

This function checks the current clock configurations and then calculates the clock frequency for a specific clock name defined in Clock_Ip_NameType. Clock modules must be properly configured before using this function. See features.h for supported clock names for different chip families. The returned value is in Hertz. If frequency is required for a peripheral and the module is not clocked, then 0 Hz frequency is returned.

Parameters

in	<i>ClockName</i>	Clock names defined in Clock_Ip_NameType
----	------------------	--

Returns

frequency Returned clock frequency value in Hertz

6.1.5.11 Clock_Ip_StartTimeout()

```
void Clock_Ip_StartTimeout (
    uint32 * StartTimeOut,
    uint32 * ElapsedTimeOut,
    uint32 * TimeoutTicksOut,
    uint32 TimeoutUs )
```

Initializes a starting reference point for timeout.

Parameters

out	<i>StartTimeOut</i>	The starting time from which elapsed time is measured
out	<i>ElapsedTimeOut</i>	The elapsed time to be passed to Clock_Ip_TimeoutExpired
out	<i>TimeoutTicksOut</i>	The timeout value (in ticks) to be passed to Clock_Ip_TimeoutExpired
in	<i>TimeoutUs</i>	The timeout value (in microseconds)

6.1.5.12 Clock_Ip_TimeoutExpired()

```
boolean Clock_Ip_TimeoutExpired (
    uint32 * StartTimeInOut,
    uint32 * ElapsedTimeInOut,
    uint32 TimeoutTicks )
```

Checks for timeout condition.

Parameters

in, out	<i>StartTimeInOut</i>	The starting time from which elapsed time is measured
in, out	<i>ElapsedTimeInOut</i>	The accumulated elapsed time from the starting time reference
in	<i>TimeoutTicks</i>	The timeout limit (in ticks)

6.1.5.13 Clock_Ip_SetScgAsyncDiv1_TrustedCall()

```
void Clock_Ip_SetScgAsyncDiv1_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to SCG_SIRCDIV1 register.

Parameters

<i>Config</i>	- pointer to configuration of SIRC divider 1
---------------	--

Returns

None

6.1.5.14 Clock_Ip_SetScgAsyncDiv2_TrustedCall()

```
void Clock_Ip_SetScgAsyncDiv2_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to SCG_SIRCDIV2 register.

Parameters

<i>Config</i>	- pointer to configuration of SIRC divider 2
---------------	--

Returns

None

6.1.5.15 Clock_Ip_SetScgRunDivcore_TrustedCall()

```
void Clock_Ip_SetScgRunDivcore_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to SCG_RCCR register.

Parameters

<i>Config</i>	- pointer to configuration of DIVCORE value in run mode
---------------	---

Returns

None

6.1.5.16 Clock_Ip_SetScgRunDivbus_TrustedCall()

```
void Clock_Ip_SetScgRunDivbus_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to SCG_RCCR register.

Parameters

<i>Config</i>	- pointer to configuration of DIVBUS value in run mode
---------------	--

Returns

None

6.1.5.17 Clock_Ip_SetScgRunDivslow_TrustedCall()

```
void Clock_Ip_SetScgRunDivslow_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to SCG_RCCR register.

Parameters

<i>Config</i>	- pointer to configuration of DIVSLOW value in run mode
---------------	---

Returns

None

6.1.5.18 Clock_Ip_SetScgVlprDivcore_TrustedCall()

```
void Clock_Ip_SetScgVlprDivcore_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to SCG_VCCR register.

Parameters

<i>Config</i>	- pointer to configuration of DIVCORE value in VLPR mode
---------------	--

Returns

None

6.1.5.19 Clock_Ip_SetScgVlprDivbus_TrustedCall()

```
void Clock_Ip_SetScgVlprDivbus_TrustedCall (
```



```
Clock_Ip_DividerConfigType const * Config )
```

Write Config to SCG_VCCR register.

Parameters

<i>Config</i>	- pointer to configuration of DIVBUS value in VLPR mode
---------------	---

Returns

None

6.1.5.20 Clock_Ip_SetScgVlprDivslow_TrustedCall()

```
void Clock_Ip_SetScgVlprDivslow_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to SCG_VCCR register.

Parameters

<i>Config</i>	- pointer to configuration of DIVSLOW value in VLPR mode
---------------	--

Returns

None

6.1.5.21 Clock_Ip_SetSimClkoutDiv_TrustedCall()

```
void Clock_Ip_SetSimClkoutDiv_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to IP_SIM::CHIPCTL register.

Parameters

<i>Config</i>	- pointer to configuration of divider value of CLKOUT0_CLK
---------------	--

Returns

None

6.1.5.22 Clock_Ip_SetPccPcdDivFrac_TrustedCall()

```
void Clock_Ip_SetPccPcdDivFrac_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to PCC register.

Parameters

<i>Config</i>	- pointer to configuration of divider value PCC peripheral clock
---------------	--

Returns

None

6.1.5.23 Clock_Ip_SetSimTraceDivMul_TrustedCall()

```
void Clock_Ip_SetSimTraceDivMul_TrustedCall (
    Clock_Ip_DividerConfigType const * Config )
```

Write Config to IP_SIM::CLKDIV4 register.

Parameters

<i>Config</i>	- pointer to configuration of divider value TRACE_CLK
---------------	---

Returns

None

6.1.5.24 Clock_Ip_ResetSOSC_TrustedCall()

```
void Clock_Ip_ResetSOSC_TrustedCall (
    Clock_Ip_XoscConfigType const * Config )
```

Reset IP_SCG::SOSCCSR register.

Parameters

<i>Config</i>	- can be NULL_PTR
---------------	-------------------

Returns

None

6.1.5.25 Clock_Ip_SetSOSC_TrustedCall()

```
void Clock_Ip_SetSOSC_TrustedCall (
    Clock_Ip_XoscConfigType const * Config )
```

Write Config of SOSC to register.

Parameters

<i>Config</i>	- pointer to configuration of SOSC
---------------	------------------------------------

Returns

None

6.1.5.26 Clock_Ip_EnableSOSC_TrustedCall()

```
void Clock_Ip_EnableSOSC_TrustedCall (
```

```
Clock_Ip_XoscConfigType const * Config )
```

Enable SOSOC.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.27 Clock_Ip_ClockSetSimLPO1KEnable_TrustedCall()

```
void Clock_Ip_ClockSetSimLPO1KEnable_TrustedCall (
    Clock_Ip_GateConfigType const * Config )
```

Set gate of LPO1K_CLK.

Parameters

<i>Config-</i>	pointer to configuration of LPO1K_CLK
----------------	---------------------------------------

Returns

None

6.1.5.28 Clock_Ip_ClockSetSimLPO32KEnable_TrustedCall()

```
void Clock_Ip_ClockSetSimLPO32KEnable_TrustedCall (
    Clock_Ip_GateConfigType const * Config )
```

Set gate of LPO3K_CLK.

Parameters

<i>Config-</i>	pointer to configuration of LPO3K_CLK
----------------	---------------------------------------

Returns

None

6.1.5.29 Clock_Ip_ClockSetSimClkoutEnable_TrustedCall()

```
void Clock_Ip_ClockSetSimClkoutEnable_TrustedCall (
    Clock_Ip_GateConfigType const * Config )
```

Set gate of CLKOUT_CLK.

Parameters

<i>Config-</i>	pointer to configuration of CLKOUT_CLK
----------------	--

Returns

None

6.1.5.30 Clock_Ip_ClockSetPccCgcEnable_TrustedCall()

```
void Clock_Ip_ClockSetPccCgcEnable_TrustedCall (
    Clock_Ip_GateConfigType const * Config )
```

Set gate of PCC peripheral.

Parameters

<i>Config-</i>	pointer to configuration of PCC peripheral clock
----------------	--

Returns

None

6.1.5.31 Clock_Ip_ClockSetSimGate_TrustedCall()

```
void Clock_Ip_ClockSetSimGate_TrustedCall (
    Clock_Ip_GateConfigType const * Config )
```

Set gate of CLOCK_IP_SIM_PLATCGC_CGC clock.

Parameters

<i>Config-</i>	pointer to configuration of CLOCK_IP_SIM_PLATCGC_CGC clock
----------------	--

Returns

None

6.1.5.32 Clock_Ip_ClockSetSimTraceEnable_TrustedCall()

```
void Clock_Ip_ClockSetSimTraceEnable_TrustedCall (
    Clock_Ip_GateConfigType const * Config )
```

Set gate of TRACE_CLK.

Parameters

<i>Config-</i>	pointer to configuration of TRACE_CLK
----------------	---------------------------------------

Returns

None

6.1.5.33 Clock_Ip_SetSirc_TrustedCall()

```
void Clock_Ip_SetSirc_TrustedCall (
```

```
Clock_Ip_IrcoscConfigType const * Config )
```

Write Config to SIRC register.

Parameters

<i>Config</i> -	pointer to configuration of SIRC
-----------------	----------------------------------

Returns

None

6.1.5.34 Clock_Ip_EnableSirc_TrustedCall()

```
void Clock_Ip_EnableSirc_TrustedCall (
    Clock_Ip_IrcoscConfigType const * Config )
```

Enable SIRC.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.35 Clock_Ip_DisableSirc_TrustedCall()

```
void Clock_Ip_DisableSirc_TrustedCall (
    Clock_Ip_NameType Name )
```

Disable SIRC.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.36 Clock_Ip_SetSircVlp_TrustedCall()

```
void Clock_Ip_SetSircVlp_TrustedCall (
    Clock_Ip_IrcoscConfigType const * Config )
```

Write Config for SIRC in Low power mode to register.

Parameters

<i>Config</i>	- pointer to configuration of SIRC low power mode
---------------	---

Returns

None

6.1.5.37 Clock_Ip_EnableSircVlp_TrustedCall()

```
void Clock_Ip_EnableSircVlp_TrustedCall (
    Clock_Ip_IrcoscConfigType const * Config )
```

Enable Sirc in Low power mode.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.38 Clock_Ip_DisableSircVlp_TrustedCall()

```
void Clock_Ip_DisableSircVlp_TrustedCall (
    Clock_Ip_NameType Name )
```

Disable Sirc in Low power mode.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.39 Clock_Ip_SetSircStop_TrustedCall()

```
void Clock_Ip_SetSircStop_TrustedCall (
    Clock_Ip_IrcoscConfigType const * Config )
```

Write Config for SIRC in Stop mode to register.

Parameters

<i>Config</i>	- pointer to configuration of SIRC in Stop mode
---------------	---

Returns

None

6.1.5.40 Clock_Ip_EnableSircStop_TrustedCall()

```
void Clock_Ip_EnableSircStop_TrustedCall (
```

```
Clock_Ip_IrcoscConfigType const * Config )
```

Enable SIRC in Stop mode to register.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.41 Clock_Ip_DisableSircStop_TrustedCall()

```
void Clock_Ip_DisableSircStop_TrustedCall (
    Clock_Ip_NameType Name )
```

Disable SIRC in Stop mode to register.

Parameters

<i>Config</i>	- pointer to configuration of SIRC in Stop mode
---------------	---

Returns

None

6.1.5.42 Clock_Ip_SetFirc_TrustedCall()

```
void Clock_Ip_SetFirc_TrustedCall (
    Clock_Ip_IrcoscConfigType const * Config )
```

Write Config for Firc to register.

Parameters

<i>Config</i>	- pointer to configuration of Firc
---------------	------------------------------------

Returns

None

6.1.5.43 Clock_Ip_EnableFirc_TrustedCall()

```
void Clock_Ip_EnableFirc_TrustedCall (
    Clock_Ip_IrcoscConfigType const * Config )
```

Enable Firc to register.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.44 Clock_Ip_DisableFirc_TrustedCall()

```
void Clock_Ip_DisableFirc_TrustedCall (
    Clock_Ip_NameType Name )
```

Disable Firc to register.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.45 Clock_Ip_ResetSppl_TrustedCall()

```
void Clock_Ip_ResetSppl_TrustedCall (
    Clock_Ip_PllConfigType const * Config )
```

Reset Sppl register.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.46 Clock_Ip_SetSppl_TrustedCall()

```
void Clock_Ip_SetSppl_TrustedCall (
    Clock_Ip_PllConfigType const * Config )
```

Write Config to Sppl register.

Parameters

<i>Config</i>	- pointer to configuration of SPLL
---------------	------------------------------------

Returns

None

6.1.5.47 Clock_Ip_DisableSppl_TrustedCall()

```
void Clock_Ip_DisableSppl_TrustedCall (
```



```
Clock_Ip_NameType PllName )
```

Disable Spll.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.48 Clock_Ip_EnableSppl_TrustedCall()

```
void Clock_Ip_EnableSppl_TrustedCall (
    Clock_Ip_PllConfigType const * Config )
```

Enable Sppl.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.49 Clock_Ip_ResetScgRunSel_TrustedCall()

```
void Clock_Ip_ResetScgRunSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Reset SCG_RCCR selector register.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.50 Clock_Ip_SetScgRunSel_TrustedCall()

```
void Clock_Ip_SetScgRunSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Write Config to SCG_RCCR selector register.

Parameters

<i>Config</i>	- pointer to configuration of system selector in run mode
---------------	---

Module Documentation

Returns

None

6.1.5.51 Clock_Ip_SetScgVlprSel_TrustedCall()

```
void Clock_Ip_SetScgVlprSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Write Config to SCG_VCCR selector register.

Parameters

<i>Config</i>	- pointer to configuration of system selector in low power mode
---------------	---

Returns

None

6.1.5.52 Clock_Ip_ResetSimRtcSel_TrustedCall()

```
void Clock_Ip_ResetSimRtcSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Reset SIM_LPOCLKS_RTCCLKSEL register.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.53 Clock_Ip_SetSimRtcSel_TrustedCall()

```
void Clock_Ip_SetSimRtcSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Write Config to RTC_CLK register.

Parameters

<i>Config</i>	- pointer to configuration of RTC_CLK selector
---------------	--

Returns

None

6.1.5.54 Clock_Ip_ResetSimLpoSel_TrustedCall()

```
void Clock_Ip_ResetSimLpoSel_TrustedCall (
```

```
    Clock_Ip_SelectorConfigType const * Config )
```

Reset LPO_CLK selector.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.55 Clock_Ip_SetSimLpoSel_TrustedCall()

```
void Clock_Ip_SetSimLpoSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Write Config to LPO_CLK selector.

Parameters

<i>Config</i>	- pointer to configuration of LPO_CLK selector
---------------	--

Returns

None

6.1.5.56 Clock_Ip_ResetScgClkoutSel_TrustedCall()

```
void Clock_Ip_ResetScgClkoutSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Reset CLKOUT_CLK selector.

Parameters

<i>Config</i>	- pointer to configuration of CLKOUT selector
---------------	---

Returns

None

6.1.5.57 Clock_Ip_SetScgClkoutSel_TrustedCall()

```
void Clock_Ip_SetScgClkoutSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Write Config to LPO_CLK register.

Parameters

<i>Config</i>	- pointer to configuration of LPO_CLK selector
---------------	--

Returns

None

6.1.5.58 Clock_Ip_ResetSimFtmoptSel_TrustedCall()

```
void Clock_Ip_ResetSimFtmoptSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Reset SIM->FTMOPT0 selector register.

Parameters

<i>Config</i>	- pointer to configuration of FTMOPT clock need to be reset
---------------	---

Returns

None

6.1.5.59 Clock_Ip_SetSimFtmoptSel_TrustedCall()

```
void Clock_Ip_SetSimFtmoptSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Write Config to SIM->FTMOPT0 register.

Parameters

<i>Config</i>	- pointer to configuration of FTMOPT clock need to be reset
---------------	---

Returns

None

6.1.5.60 Clock_Ip_ResetSimClkoutSel_TrustedCall()

```
void Clock_Ip_ResetSimClkoutSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Write Config to SIM_CHIPCTL_CLKOUTSEL register.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.61 Clock_Ip_SetSimClkoutSel_TrustedCall()

```
void Clock_Ip_SetSimClkoutSel_TrustedCall (
```

```
Clock_Ip_SelectorConfigType const * Config )
```

Write Config to SIM_CHIPCTL_CLKOUTSEL register.

Parameters

<i>Config</i>	- pointer to configuration of CLKOUT_CLK selector
---------------	---

Returns

None

6.1.5.62 Clock_Ip_ResetPccPcsSelect_TrustedCall()

```
void Clock_Ip_ResetPccPcsSelect_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Reset register of PCC peripheral clock.

Parameters

<i>Config</i>	- pointer to configuration clock need to be reset
---------------	---

Returns

None

6.1.5.63 Clock_Ip_SetPccPcsSelect_TrustedCall()

```
void Clock_Ip_SetPccPcsSelect_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Write Config to PCC register.

Parameters

<i>Config</i>	- pointer to configuration of PCC peripheral clock
---------------	--

Returns

None

6.1.5.64 Clock_Ip_ResetSimTraceSel_TrustedCall()

```
void Clock_Ip_ResetSimTraceSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Reset selector register of TRACE_CLK.

Parameters

<i>Config</i>	
---------------	--

Returns

None

6.1.5.65 Clock_Ip_SetSimTraceSel_TrustedCall()

```
void Clock_Ip_SetSimTraceSel_TrustedCall (
    Clock_Ip_SelectorConfigType const * Config )
```

Write Config to TRACE_CLK selector register.

Parameters

<i>Config</i>	- pointer to configuration of TRACE_CLK selector
---------------	--

Returns

None

6.1.5.66 SetFircToResetValue_TrustedCall()

```
void SetFircToResetValue_TrustedCall (
    void )
```

Reset FIRC.

Parameters

<i>None</i>	
-------------	--

Returns

None

6.1.5.67 SetSimLpoclksRegister_TrustedCall()

```
void SetSimLpoclksRegister_TrustedCall (
    Clock_Ip_ClockConfigType const * Config )
```

Write Config whole SIM_LPOCLKS register.

Parameters

<i>Config</i>	- pointer to configuration of all SIM_LPOCLKS clock
---------------	---

Returns

None

6.1.5.68 Power_Ip_CM4_DisableSleepOnExit()

```
void Power_Ip_CM4_DisableSleepOnExit (
```

```
void )
```

The function disables SLEEPONEXIT bit.

Parameters

<i>None</i>	
-------------	--

Returns

None

6.1.5.69 Power_Ip_CM4_EnableSleepOnExit()

```
void Power_Ip_CM4_EnableSleepOnExit (
    void )
```

The function enables SLEEPONEXIT bit.

Parameters

<i>None</i>	
-------------	--

Returns

None

6.1.5.70 Power_Ip_CM4_EnableDeepSleep()

```
void Power_Ip_CM4_EnableDeepSleep (
    void )
```

The function enables DEEPSLEEP.

Parameters

<i>None</i>	
-------------	--

Returns

None

6.1.5.71 Power_Ip_CM4_DisableDeepSleep()

```
void Power_Ip_CM4_DisableDeepSleep (
    void )
```

The function disables DEEPSLEEP.

Parameters

<i>None</i>	
-------------	--

Returns

None

6.1.5.72 Power_Ip_CM4_SystemReset()

```
void Power_Ip_CM4_SystemReset (  
    void )
```

The function initiates a system reset request to reset the SoC.

Parameters

<i>None</i>	
-------------	--

Returns

None

6.1.5.73 Power_Ip_PMC_PowerInit()

```
void Power_Ip_PMC_PowerInit (  
    const Power_Ip_PMC_ConfigType * ConfigPtr )
```

This function configure the Power Management Controller.

Parameters

<i>ConfigPtr</i>	Pointer to PMC configuration structure.
------------------	---

Returns

None

6.1.5.74 Power_Ip_RCM_ResetInit()

```
void Power_Ip_RCM_ResetInit (  
    const Power_Ip_RCM_ConfigType * ConfigPtr )
```

This function initializes the Reset parameters.

Parameters

<i>ConfigPtr</i>	Pointer to the RCM configuration structure.
------------------	---

Returns

None

6.1.5.75 Power_Ip_RCM_GetResetReason()

```
uint32 Power_Ip_RCM_GetResetReason (  
    void )
```



```
void )
```

This function returns the Reset reason.

Parameters

<i>None</i>	
-------------	--

Returns

None

6.1.5.76 Power_Ip_RCM_GetResetRawValue()

```
Power_Ip_RawResetType Power_Ip_RCM_GetResetRawValue (
    void )
```

This function returns the Raw Reset value.

Parameters

<i>None</i>	
-------------	--

Returns

Register value with the Reset status.

6.1.5.77 Power_Ip_SIM_SRAMRetentionConfig()

```
void Power_Ip_SIM_SRAMRetentionConfig (
    Power_Ip_SRAMRetenConfigType SRAMRetenConfig )
```

Configuration for SRAM retention.

Parameters

<i>None</i>	
-------------	--

Returns

None

6.1.5.78 Power_Ip_SMC_AllowedModesConfig()

```
void Power_Ip_SMC_AllowedModesConfig (
    const Power_Ip_SMC_ConfigType * ConfigPtr )
```

This function will configure the allowed modes.

Parameters

<i>ConfigPtr</i>	
------------------	--

Returns

None

6.1.5.79 Power_Ip_SMC_ModeConfig()

```
uint32 Power_Ip_SMC_ModeConfig (
    const Power_Ip_ModeConfigType * ModeConfigPtr )
```

This function switches the mode by writing SMC_PMCTRL and SMC_STOPCTRL.

Parameters

<i>ModeConfigPtr</i>	Pointer to mode configuration structure.
----------------------	--

Returns

uint32

6.2 Mcu Driver

6.2.1 Detailed Description

Data Structures

- struct [Mcu_ConfigType](#)
Initialization data for the MCU driver. [More...](#)

Macros

- `#define` [MCU_VENDOR_ID](#)
Import all data types from lower layers that should be exported. [Mcu.h](#) shall include [Mcu_Cfg.h](#) for the API pre-compiler switches.

Types Reference

- typedef [Power_Ip_HwIPsConfigType](#) [Mcu_HwIPsConfigType](#)
Mcu driver configuration structure.
- typedef [Clock_Ip_ClockConfigType](#) [Mcu_ClockConfigType](#)
Definition of a Clock configuration.
- typedef [Ram_Ip_RamConfigType](#) [Mcu_RamConfigType](#)
Definition of a Clock configuration.
- typedef [Power_Ip_ModeConfigType](#) [Mcu_ModeConfigType](#)
Definition of a Mode configuration.

Enum Reference

- enum [Mcu_ClockNotificationType](#)
Mcu_ClockNotificationType.

Function Reference

- void [Mcu_Init](#) (const [Mcu_ConfigType](#) *ConfigPtr)
MCU driver initialization function.
- Std_ReturnType [Mcu_InitRamSection](#) ([Mcu_RamSectionType](#) RamSection)
MCU driver initialization of Ram sections.
- Std_ReturnType [Mcu_InitClock](#) ([Mcu_ClockType](#) ClockSetting)
MCU driver clock initialization function.
- void [Mcu_SetMode](#) ([Mcu_ModeType](#) McuMode)
This function sets the MCU power mode.
- Std_ReturnType [Mcu_DistributePllClock](#) (void)
This function activates the PLL clock to the MCU clock distribution.
- [Mcu_PllStatusType](#) [Mcu_GetPllStatus](#) (void)
This function returns the lock status of the PLL.
- [Mcu_ResetType](#) [Mcu_GetResetReason](#) (void)
This function returns the Reset reason.

- `Mcu_RawResetType` [Mcu_GetResetRawValue](#) (void)
This function returns the Raw Reset value.
- void [Mcu_PerformReset](#) (void)
This function performs a microcontroller reset.
- void [Mcu_GetVersionInfo](#) (Std_VersionInfoType *versioninfo)
This function returns the Version Information for the MCU module.
- `Mcu_PowerModeStateType` [Mcu_GetPowerModeState](#) (void)
This function returns the System Status (power mode, clock settings) from MC_ME hw IP. The return value is the content register read from hardware (ME_GS). The return value shall evidentiate the current running mode, clock settings and activation, etc.
- void [Mcu_DisableCmu](#) ([Clock_Ip_NameType](#) ClockName)
Disable clock monitoring unit.
- `uint64` [Mcu_GetClockFrequency](#) ([Clock_Ip_NameType](#) ClockName)
Return the frequency of a given clock.
- void [Mcu_SleepOnExit](#) (`Mcu_SleepOnExitType` SleepOnExit)
This function disable/enable SleepOnExit.
- void [Mcu_ReportDemTimeoutError](#) (void)
Reports timeout error to DEM.
- void [Mcu_ReportDemSwitchModeError](#) (void)
Reports failed switch mode to DEM.
- void [Mcu_Ipw_SRAMRetentionConfig](#) (`Mcu_SRAMRetenConfigType` SRAMRetenConfig)
Configuration for SRAM retention.

6.2.2 Data Structure Documentation

6.2.2.1 struct `Mcu_ConfigType`

Initialization data for the MCU driver.

A pointer to such a structure is provided to the MCU initialization routines for configuration.

Definition at line 169 of file `Mcu.h`.

Data Fields

- `Mcu_ClockNotificationType` [ClkSrcFailureNotification](#)
Clock source failure notification enable configuration.
- `const` `Mcu_DemConfigType` * [DemConfigPtr](#)
Total number of RAM sections.
- `Mcu_RamSectionType` [NoRamConfigs](#)
Total number of MCU modes.
- `Mcu_ModeType` [NoModeConfigs](#)
Total number of MCU clock configurations.
- `const` `Mcu_RamConfigType`(* [RamConfigArrayPtr](#))(((uint32) 1U))
RAM data configuration.
- `const` `Mcu_ModeConfigType`(* [ModeConfigArrayPtr](#))(((uint32) 6U))

Clock data configuration.

- const `Mcu_HwIPsConfigType` * `HwIPsConfigPtr`

IPs data generic configuration.

Field Documentation

ClkSrcFailureNotification

`Mcu_ClockNotificationType` ClkSrcFailureNotification

Clock source failure notification enable configuration.

<

DEM error reporting configuration.

Definition at line 173 of file Mcu.h.

DemConfigPtr

`const Mcu_DemConfigType*` DemConfigPtr

Total number of RAM sections.

Definition at line 183 of file Mcu.h.

NoRamConfigs

`Mcu_RamSectionType` NoRamConfigs

Total number of MCU modes.

Definition at line 186 of file Mcu.h.

NoModeConfigs

`Mcu_ModeType` NoModeConfigs

Total number of MCU clock configurations.

Definition at line 189 of file Mcu.h.

RamConfigArrayPtr

`const Mcu_RamConfigType` (* RamConfigArrayPtr) [((uint32) 1U)]

RAM data configuration.

<

Power Modes data configuration.

Definition at line 197 of file Mcu.h.

ModeConfigArrayPtr

```
const Mcu_ModeConfigType (* ModeConfigArrayPtr) [ ((uint32) 6U) ]
```

Clock data configuration.

Definition at line 201 of file Mcu.h.

HwIPsConfigPtr

```
const Mcu_HwIPsConfigType* HwIPsConfigPtr
```

IPs data generic configuration.

<

Definition at line 208 of file Mcu.h.

6.2.3 Macro Definition Documentation

6.2.3.1 MCU_VENDOR_ID

```
#define MCU_VENDOR_ID
```

Import all data types from lower layers that should be exported. [Mcu.h](#) shall include Mcu_Cfg.h for the API pre-compiler switches.

Definition at line 63 of file Mcu.h.

6.2.4 Types Reference

6.2.4.1 Mcu_HwIPsConfigType

```
typedef Power_Ip_HwIPsConfigType Mcu_HwIPsConfigType
```

Mcu driver configuration structure.

Configuration for SIU reset configuration module. Configuration for power management and SSCM. Configuration for FLASH controller. Used by "Mcu_ConfigType" structure.

Definition at line 193 of file Mcu_Ipw__Types.h.

6.2.4.2 Mcu_ClockConfigType

```
typedef Clock_Ip_ClockConfigType Mcu_ClockConfigType
```

Definition of a Clock configuration.

This configuration is transmitted as parameter to Mcu_Ipw_InitClock() API. Used by "Mcu_ConfigType" structure.

Note

The structure `Mcu_ConfigType` shall provide a configurable (enable/ disable) clock failure notification if the MCU provides an interrupt for such detection.

Definition at line 206 of file `Mcu_Ipw_Types.h`.

6.2.4.3 Mcu_RamConfigType

```
typedef Ram_Ip_RamConfigType Mcu_RamConfigType
```

Definition of a Clock configuration.

This configuration is transmitted as parameter to `Mcu_Ipw_InitClock()` API. Used by "Mcu_ConfigType" structure.

Note

The structure `Mcu_ConfigType` shall provide a configurable (enable/ disable) clock failure notification if the MCU provides an interrupt for such detection.

Definition at line 220 of file `Mcu_Ipw_Types.h`.

6.2.4.4 Mcu_ModeConfigType

```
typedef Power_Ip_ModeConfigType Mcu_ModeConfigType
```

Definition of a Mode configuration.

This configuration is transmitted as parameter to `Mcu_Ipw_SetMode()` API. Used by "Mcu_ConfigType" structure.

Definition at line 230 of file `Mcu_Ipw_Types.h`.

6.2.5 Enum Reference

6.2.5.1 Mcu_ClockNotificationType

```
enum Mcu_ClockNotificationType
```

`Mcu_ClockNotificationType`.

Clock failure notification. Enable/disable clock failure interrupt generated by the MCU.

Enumerator

<code>MCU_CLK_NOTIF_DIS</code>	Disable clock notification.
<code>MCU_CLK_NOTIF_EN</code>	Enable clock notification.

Definition at line 172 of file `Mcu_Ipw_Types.h`.

6.2.6 Function Reference

6.2.6.1 Mcu_Init()

```
void Mcu_Init (
    const Mcu_ConfigType * ConfigPtr )
```

MCU driver initialization function.

This routine initializes the MCU Driver. The intention of this function is to make the configuration setting for power down, clock and Ram sections visible within the MCU Driver.

Parameters

in	<i>ConfigPtr</i>	Pointer to configuration structure.
----	------------------	-------------------------------------

Returns

void

6.2.6.2 Mcu_InitRamSection()

```
Std_ReturnType Mcu_InitRamSection (
    Mcu_RamSectionType RamSection )
```

MCU driver initialization of Ram sections.

Function initializes the ram section selected by RamSection parameter. The section base address, size and value to be written are provided from the configuration structure. The function will write the value specified in the configuration structure indexed by RamSection. After the write it will read back the RAM to verify that the requested value was written.

Parameters

in	<i>RamSection</i>	Index of ram section from configuration structure to be initialized.
----	-------------------	--

Returns

Command has or has not been accepted.

Return values

<i>E_OK</i>	Valid parameter, the driver state allowed execution and the RAM check was successful
<i>E_NOT_OK</i>	Invalid parameter, the driver state did not allowed execution or the RAM check was not successful

6.2.6.3 Mcu_InitClock()

```
Std_ReturnType Mcu_InitClock (
    Mcu_ClockType ClockSetting )
```

MCU driver clock initialization function.

This function intializes the PLL and MCU specific clock options. The clock setting is provided from the configuration structure.

Parameters

in	<i>ClockSetting</i>	Clock setting ID from config structure to be used.
----	---------------------	--

Returns

Command has or has not been accepted.

Return values

<i>E_OK</i>	The driver state allowed the execution of the function and the provided parameter was in range
<i>E_NOT_OK</i>	The driver state did not allowed execution or the parameter was invalid

6.2.6.4 Mcu_SetMode()

```
void Mcu_SetMode (
    Mcu_ModeType McuMode )
```

This function sets the MCU power mode.

This function activates MCU power mode from config structure selected by McuMode parameter. If the driver state is invalid or McuMode is not in range the function will skip changing the mcu mode.

Parameters

in	<i>McuMode</i>	MCU mode setting ID from config structure to be set.
----	----------------	--

Returns

void

6.2.6.5 Mcu_DistributePllClock()

```
Std_ReturnType Mcu_DistributePllClock (
    void )
```

This function activates the PLL clock to the MCU clock distribution.

Function completes the PLL configuration and then activates the PLL clock to MCU. If the MCU_NO_PLL is TRUE the Mcu_DistributePllClock has to be disabled. The function will not distribute the PLL clock if the driver state does not allow it, or the PLL is not stable.

Returns

Std_ReturnType

Return values

<i>E_OK</i>	Command has been accepted.
<i>E_NOT_OK</i>	Command has not been accepted.

6.2.6.6 Mcu_GetPllStatus()

```
Mcu_PllStatusType Mcu_GetPllStatus (
    void )
```

This function returns the lock status of the PLL.

The user takes care that the PLL is locked by executing Mcu_GetPllStatus. If the MCU_NO_PLL is TRUE the Mcu_GetPllStatus has to return MCU_PLL_STATUS_UNDEFINED. It will also return MCU_PLL_STATUS_UNDEFINED if the driver state was invalid

Returns

Mcu_PllStatusType Provides the lock status of the PLL.

Return values

<i>MCU_PLL_STATUS_UNDEFINED</i>	PLL Status is unknown.
<i>MCU_PLL_LOCKED</i>	PLL is locked.
<i>MCU_PLL_UNLOCKED</i>	PLL is unlocked.

6.2.6.7 Mcu_GetResetReason()

```
Mcu_ResetType Mcu_GetResetReason (
    void )
```

This function returns the Reset reason.

This routine returns the Reset reason that is read from the hardware.

Returns

Mcu_ResetType Reason of the Reset event.

6.2.6.8 Mcu_GetResetRawValue()

```
Mcu_RawResetType Mcu_GetResetRawValue (
    void )
```

This function returns the Raw Reset value.

This routine returns the Raw Reset value that is read from the hardware.

Returns

Mcu_RawResetType Description of the returned value.

Return values

<i>uint32</i>	Code of the Raw reset value.
---------------	------------------------------

6.2.6.9 Mcu_PerformReset()

```
void Mcu_PerformReset (
    void )
```

This function performs a microcontroller reset.

This function performs a microcontroller reset by using the hardware feature of the microcontroller. In case the function returns, the user must reset the platform using an alternate reset mechanism

Returns

void

6.2.6.10 Mcu_GetVersionInfo()

```
void Mcu_GetVersionInfo (
    Std_VersionInfoType * versioninfo )
```

This function returns the Version Information for the MCU module.

This function returns the vendor id, module id, major, minor and patch version.

Parameters

in, out	<i>versioninfo</i>	A pointer to a variable to store version info.
---------	--------------------	--

Returns

void

6.2.6.11 Mcu_GetPowerModeState()

```
Mcu_PowerModeStateType Mcu_GetPowerModeState (
    void )
```

This function returns the System Status (power mode, clock settings) from MC_ME hw IP. The return value is the content register read from hardware (ME_GS). The return value shall evidentiate the current running mode, clock settings and activation, etc.

Returns

Mcu_PowerModeStateType Get the state of the power mode.

Return values

<i>uint32</i>	Content of ME_GS register.
---------------	----------------------------

6.2.6.12 Mcu_DisableCmu()

```
void Mcu_DisableCmu (
    Clock_Ip_NameType ClockName )
```

Disable clock monitoring unit.

This function disables the selected clock monitoring unit.

Module Documentation

Precondition

Function requires an execution of [Mcu_Init\(\)](#) before it can be used.

Parameters

in	<i>ClockName</i>	Name of the monitor clock for which CMU must be disabled.
----	------------------	---

Returns

void

6.2.6.13 Mcu_GetClockFrequency()

```
uint64 Mcu_GetClockFrequency (
    Clock_Ip_NameType ClockName )
```

Return the frequency of a given clock.

This function returns the frequency of a given clock which is request by user.

Precondition

Function requires an execution of [Mcu_Init\(\)](#) before it can be used,

Parameters

in	<i>ClockName</i>	Name of the monitor clock for which CMU must be disabled.
----	------------------	---

Returns

uint32

6.2.6.14 Mcu_SleepOnExit()

```
void Mcu_SleepOnExit (
    Mcu_SleepOnExitType SleepOnExit )
```

This function disable/enable SleepOnExit.

Disable/enable Sleep on exit when returning from Handler mode to Thread mode.

Parameters

in	<i>Mcu_SleepOnExitType</i>	The value will be configured to SLEEPONEXIT bits. MCU_SLEEP_ON_EXIT_DISABLED - Disable SLEEPONEXIT bit. MCU_SLEEP_ON_EXIT_ENABLED - Enable SLEEPONEXIT bit.
----	----------------------------	---

Returns

void

6.2.6.15 Mcu_ReportDemTimeoutError()

```
void Mcu_ReportDemTimeoutError (
```

```
void )
```

Reports timeout error to DEM.

Checks if the timeout expired and reports the timeout error to DEM if that is the case.

Parameters

in	<i>u32Timeout</i>	Maximum timeout to be waited.
----	-------------------	-------------------------------

Returns

void.

6.2.6.16 Mcu_ReportDemSwitchModeError()

```
void Mcu_ReportDemSwitchModeError (
    void )
```

Reports failed switch mode to DEM.

Directly reports the clock multiplexer switch error to DEM.

Parameters

in	<i>void.</i>	
----	--------------	--

Returns

void.

6.2.6.17 Mcu_Ipw_SRAMRetentionConfig()

```
void Mcu_Ipw_SRAMRetentionConfig (
    Mcu_SRAMRetenConfigType SRAMRetenConfig )
```

Configuration for SRAM retention.

This function configure for both SRAML_RETEN and SRAMU_RETEN bits.

Returns

void

6.3 Power Ip Driver

6.3.1 Detailed Description

Data Structures

- struct [Power_Ip_AEC_ConfigType](#)
Configuration for AEC. [More...](#)
- struct [Power_Ip_PMC_ConfigType](#)
Configuration for PMC. [More...](#)
- struct [Power_Ip_RCM_ConfigType](#)
RCM IP configuration. [More...](#)
- struct [Power_Ip_SMC_ConfigType](#)
SMC IP configuration. [More...](#)
- struct [Power_Ip_ModeConfigType](#)
Definition of a MCU mode section in the configuration structure. [More...](#)

Types Reference

- typedef uint32 [Power_Ip_ModeType](#)
The [Power_Ip_ModeType](#) specifies the identification (ID) for a MCU mode, configured via configuration structure.
- typedef uint32 [Power_Ip_RawResetType](#)
The type [Mcu_RawResetType](#) specifies the reset reason in raw register format, read from a reset status register.
- typedef void(* [Power_Ip_ReportErrorsCallbackType](#)) ([Power_Ip_ReportErrorType](#) Error, uint8 ErrorCode)
Power report error callback structure. Implements [PowerReportErrorCallbackType_Class](#).

Enum Reference

- enum [Power_Ip_PMC_StatusType](#)
- enum [Power_Ip_LowVoltageType](#)
- enum [Power_Ip_PowerModeType](#)
Power Modes encoding.
- enum [Power_Ip_ReportErrorType](#)
Power ip report error types.

Function Reference

- void [Power_Ip_Init](#) (const [Power_Ip_HwIPsConfigType](#) *HwIPsConfigPtr)
Power initialization.
- void [Power_Ip_SetMode](#) (const [Power_Ip_ModeConfigType](#) *ModeConfigPtr)
Sets mode.
- void [Power_Ip_PerformReset](#) (const [Power_Ip_HwIPsConfigType](#) *HwIPsConfigPtr)
Performs reset.
- [Power_Ip_ResetType](#) [Power_Ip_GetResetReason](#) (void)
Returns reset type.
- [Power_Ip_RawResetType](#) [Power_Ip_GetResetRawValue](#) (void)

Returns raw reset type.

- void [Power_Ip_InstallNotificationsCallback](#) ([Power_Ip_ReportErrorsCallbackType](#) ReportErrorsCallback)

Install report error callback.

- void [Power_Ip_CM4_SystemReset](#) (void)

The function initiates a system reset request to reset the SoC.

- void [Power_Ip_StartTimeout](#) (uint32 *StartTimeOut, uint32 *ElapsedTimeOut, uint32 *TimeoutTicksOut, uint32 TimeoutUs)

Initializes a starting reference point for timeout.

- boolean [Power_Ip_TimeoutExpired](#) (uint32 *StartTimeInOut, uint32 *ElapsedTimeInOut, uint32 TimeoutTicks)

Checks for timeout condition.

6.3.2 Data Structure Documentation

6.3.2.1 struct [Power_Ip_AEC_ConfigType](#)

Configuration for AEC.

The Reset generator configuration ([AEC_RSTGEN_CFG](#))

Definition at line 125 of file [Power_Ip_AEC_Types.h](#).

Data Fields

uint16	Rstgencfg	Reset generator configuration (AEC_RSTGEN_CFG).
--------	-----------	---

6.3.2.2 struct [Power_Ip_PMC_ConfigType](#)

Configuration for PMC.

The power control unit (PMC) acts as a bridge for mapping the PMC peripheral to the PMC address space.

Definition at line 252 of file [Power_Ip_PMC_Types.h](#).

Data Fields

uint8	Lvdsc2	Low Voltage Detect Status and Control 2 Register (PMC_LVDSC2).
uint8	Regsc	Regulator Status and Control Register (PMC_REGSC).
uint8	Lpotrim	Low Power Oscillator Trim Register (PMC_LPOTRIM).
uint32	PmcAeConfig	PMC_AE Configuration Register (PMC_AE_CONFIG).
uint32	PmcAeMonitor	PMC_AE Monitor Register (PMC_AE_MONITOR).

6.3.2.3 struct [Power_Ip_RCM_ConfigType](#)

RCM IP configuration.

This structure contains information for peripheral configuration

Definition at line 206 of file [Power_Ip_RCM_Types.h](#).

6.3.2.4 struct Power_Ip_SMC_ConfigType

SMC IP configuration.

This structure contains information for allowed modes

Definition at line 157 of file Power_Ip_SMC_Types.h.

6.3.2.5 struct Power_Ip_ModeConfigType

Definition of a MCU mode section in the configuration structure.

Specifies the system behaviour during the selected target mode. Data set and configured by Mcu_SetMode call.

Definition at line 251 of file Power_Ip_Types.h.

Data Fields

Power_Ip_ModeType	ModeConfigId	The ID for Power Mode configuration. < Power modes control configuration
Power_Ip_PowerModeType	PowerMode	Indicates sleep-on-exit configuration.
uint8	SleepOnExit	

6.3.3 Types Reference

6.3.3.1 Power_Ip_ModeType

```
typedef uint32 Power\_Ip\_ModeType
```

The Power_Ip_ModeType specifies the identification (ID) for a MCU mode, configured via configuration structure.

The type shall be uint8, uint16 or uint32.

Definition at line 211 of file Power_Ip_Types.h.

6.3.3.2 Power_Ip_RawResetType

```
typedef uint32 Power\_Ip\_RawResetType
```

The type Mcu_RawResetType specifies the reset reason in raw register format, read from a reset status register.

The type shall be uint8, uint16 or uint32 based on best performance.

Destructive and Functional Reset Events Log.

Definition at line 274 of file Power_Ip_Types.h.

6.3.3.3 Power_Ip_ReportErrorsCallbackType

```
typedef void(* Power\_Ip\_ReportErrorsCallbackType) (Power\_Ip\_ReportErrorType Error, uint8 ErrorCode)
```

Power report error callback structure. Implements PowerReportErrorCallbackType_Class.

Definition at line 295 of file Power_Ip_Types.h.

6.3.4 Enum Reference

6.3.4.1 Power_Ip_PMC_StatusType

enum `Power_Ip_PMC_StatusType`

Enumerator

PMC_UNINIT	The PMC driver is uninitialized.
PMC_INIT	The PMC driver is initialized.

Definition at line 237 of file `Power_Ip_PMC_Types.h`.

6.3.4.2 Power_Ip_LowVoltageType

enum `Power_Ip_LowVoltageType`

Enumerator

POWER_IP_PMC_LOW_VOLTAGE_WARNING	Low Voltage Warning Flag (LVWF).
POWER_IP_PMC_NO_LOW_VOLTAGE	No Low Voltage Detect.

Definition at line 266 of file `Power_Ip_PMC_Types.h`.

6.3.4.3 Power_Ip_PowerModeType

enum `Power_Ip_PowerModeType`

Power Modes encoding.

Supported power modes for SMC hw IP.

Enumerator

POWER_IP_RUN_MODE	Run Mode.
POWER_IP_HSRUN_MODE	High Speed Mode.
POWER_IP_VLPR_MODE	Very Low Power Run Mode.
POWER_IP_VLPS_MODE	Very Low Power Stop Mode.
POWER_IP_STOP1_MODE	Stop 1 Mode.
POWER_IP_STOP2_MODE	Stop 2 Mode.

Definition at line 236 of file `Power_Ip_Types.h`.

6.3.4.4 Power_Ip_ReportErrorType

enum `Power_Ip_ReportErrorType`

Module Documentation

Power ip report error types.

Enumerator

POWER_IP_REPORT_TIMEOUT_ERROR	Report Timeout Error.
POWER_IP_ISR_ERROR	Notification Error.
POWER_IP_PMC_ERROR	Notification PMC.
POWER_IP_REPORT_SWITCH_MODE_ERROR	Report switch mode Error.
POWER_IP_REPORT_VLPSA_NOTIFICATION	Report the VLPS transition was aborted.
POWER_IP_ISR_VOLTAGE_HVD_VDDINT_DETECT	Report the Voltage on VDDINT is high-voltage detected.
POWER_IP_ISR_VOLTAGE_HVD_15_DETECT	Report the Voltage on VDD15 is high-voltage detected.
POWER_IP_ISR_VOLTAGE_HVD_VDD_DETECT	Report the Voltage on VDD is high-voltage detected.
POWER_IP_ISR_VOLTAGE_LVD_VDDC_DETECT	Report the Voltage on VDDC is low-voltage detected.
POWER_IP_ISR_VOLTAGE_LVD_VLS_DETECT	Report the Voltage on VLS is low-voltage detected.

Definition at line 277 of file Power_Ip_Types.h.

6.3.5 Function Reference

6.3.5.1 Power_Ip_Init()

```
void Power_Ip_Init (
    const Power_Ip_HwIPsConfigType * HwIPsConfigPtr )
```

Power initialization.

This function power initialization

Parameters

in	<i>HwIPsConfigPtr</i>	power initialization configuration.
----	-----------------------	-------------------------------------

Returns

void

6.3.5.2 Power_Ip_SetMode()

```
void Power_Ip_SetMode (
    const Power_Ip_ModeConfigType * ModeConfigPtr )
```

Sets mode.

This function sets mode.

Parameters

in	<i>ModeConfigPtr</i>	power set mote configuration.
----	----------------------	-------------------------------

Returns

void

6.3.5.3 Power_Ip_PerformReset()

```
void Power_Ip_PerformReset (
    const Power_Ip_HwIPsConfigType * HwIPsConfigPtr )
```

Performs reset.

This function performs reset.

Parameters

in	<i>HwIPsConfigPtr</i>	reset initialization configuration.
----	-----------------------	-------------------------------------

Returns

void

6.3.5.4 Power_Ip_GetResetReason()

```
Power_Ip_ResetType Power_Ip_GetResetReason (
    void )
```

Returns reset type.

This function returns reset type.

Returns

Power_Ip_ResetType Reset type

6.3.5.5 Power_Ip_GetResetRawValue()

```
Power_Ip_RawResetType Power_Ip_GetResetRawValue (
    void )
```

Returns raw reset type.

This function returns raw reset type.

Returns

Power_Ip_RawResetType Raw reset type

6.3.5.6 Power_Ip_InstallNotificationsCallback()

```
void Power_Ip_InstallNotificationsCallback (
    Power_Ip_ReportErrorsCallbackType ReportErrorsCallback )
```

Module Documentation

Install report error callback.

This function installs a callback for reporting errors from power driver

Parameters

in	<i>ReportErrorsCallback</i>	Callback to be installed.
----	-----------------------------	---------------------------

Returns

void

6.3.5.7 Power_Ip_CM4_SystemReset()

```
void Power_Ip_CM4_SystemReset (  
    void )
```

The function initiates a system reset request to reset the SoC.

The function initiates a system reset request to reset the SoC

Parameters

in	<i>none</i>	
----	-------------	--

Returns

void

6.3.5.8 Power_Ip_StartTimeout()

```
void Power_Ip_StartTimeout (  
    uint32 * StartTimeOut,  
    uint32 * ElapsedTimeOut,  
    uint32 * TimeoutTicksOut,  
    uint32 TimeoutUs )
```

Initializes a starting reference point for timeout.

Parameters

out	<i>StartTimeOut</i>	The starting time from which elapsed time is measured
out	<i>ElapsedTimeOut</i>	The elapsed time to be passed to Power_Ip_TimeoutExpired
out	<i>TimeoutTicksOut</i>	The timeout value (in ticks) to be passed to Power_Ip_TimeoutExpired
in	<i>TimeoutUs</i>	The timeout value (in microseconds)

6.3.5.9 Power_Ip_TimeoutExpired()

```
boolean Power_Ip_TimeoutExpired (  
    uint32 * StartTimeInOut,
```

```
uint32 * ElapsedTimeInOut,
uint32 TimeoutTicks )
```

Checks for timeout condition.

Parameters

in, out	<i>StartTimeInOut</i>	The starting time from which elapsed time is measured
in, out	<i>ElapsedTimeInOut</i>	The accumulated elapsed time from the starting time reference
in	<i>TimeoutTicks</i>	The timeout limit (in ticks)

6.4 Ram Ip Driver

6.4.1 Detailed Description

Data Structures

- struct [Ram_Ip_RamConfigType](#)

Definition of a RAM section within the configuration structure. The definitions for each RAM section within the structure `Ram_Ip_ConfigType` shall contain: [More...](#)

Types Reference

- typedef uint32 [Ram_Ip_RamSectionType](#)

The `Ram_Ip_RamSectionType` specifies the identification (ID) for a RAM section, configured via the configuration structure. The type shall be `uint8`, `uint16` or `uint32`, based on best performance.

- typedef uint32 [Ram_Ip_RamIndexType](#)

The `Ram_Ip_RamIndexType` specifies the variable for indexing RAM sections. The type shall be `uint8`, `uint16` or `uint32`, based on best performance.

- typedef `Ram_Ip_uintPtrType` [Ram_Ip_RamSizeType](#)

The `Ram_Ip_RamSizeType` specifies the RAM section size. The type shall be `uint8`, `uint16` or `uint32`, based on best performance.

- typedef uint32 [Ram_Ip_RamWriteSizeType](#)

The `Ram_Ip_RamWriteSizeType` specifies the RAM section write size. The type shall be `uint8`, `uint16` or `uint32`, based on best performance.

Enum Reference

- enum [Ram_Ip_StatusType](#)

Ram ip status return codes.

Function Reference

- [Ram_Ip_StatusType](#) [Ram_Ip_InitRamSection](#) (const [Ram_Ip_RamConfigType](#) *RamConfigPtr)

Initializes RAM section.

6.4.2 Data Structure Documentation

6.4.2.1 struct `Ram_Ip_RamConfigType`

Definition of a RAM section within the configuration structure. The definitions for each RAM section within the structure `Ram_Ip_ConfigType` shall contain:

- RAM section base address
- Section size
- Data pre-setting to be initialized
- RAM write size

Definition at line 172 of file `Ram_Ip_Types.h`.

Data Fields

Ram_Ip_RamSectionType	<code>RamSectorId</code>	The ID for Ram Sector configuration.
---------------------------------------	--------------------------	--------------------------------------

Data Fields

uint8(*)	RamBaseAddrPtr)[1U]	RAM section base address.
Ram_Ip_RamSizeType *	RamSize	RAM section size.
uint8	RamDefaultValue	RAM default value for initialization.
Ram_Ip_RamWriteSizeType	RamWriteSize	RAM section write size.

6.4.3 Types Reference

6.4.3.1 Ram_Ip_RamSectionType

```
typedef uint32 Ram_Ip_RamSectionType
```

The `Ram_Ip_RamSectionType` specifies the identification (ID) for a RAM section, configured via the configuration structure. The type shall be `uint8`, `uint16` or `uint32`, based on best performance.

Definition at line 140 of file `Ram_Ip_Types.h`.

6.4.3.2 Ram_Ip_RamIndexType

```
typedef uint32 Ram_Ip_RamIndexType
```

The `Ram_Ip_RamIndexType` specifies the variable for indexing RAM sections. The type shall be `uint8`, `uint16` or `uint32`, based on best performance.

Definition at line 147 of file `Ram_Ip_Types.h`.

6.4.3.3 Ram_Ip_RamSizeType

```
typedef Ram_Ip_uintPtrType Ram_Ip_RamSizeType
```

The `Ram_Ip_RamSizeType` specifies the RAM section size. The type shall be `uint8`, `uint16` or `uint32`, based on best performance.

Definition at line 154 of file `Ram_Ip_Types.h`.

6.4.3.4 Ram_Ip_RamWriteSizeType

```
typedef uint32 Ram_Ip_RamWriteSizeType
```

The `Ram_Ip_RamWriteSizeType` specifies the RAM section write size. The type shall be `uint8`, `uint16` or `uint32`, based on best performance.

Definition at line 161 of file `Ram_Ip_Types.h`.

6.4.4 Enum Reference

6.4.4.1 Ram_Ip_StatusType

enum `Ram_Ip_StatusType`

Ram ip status return codes.

This is the Ram State data type returned by the function `Mcu_GetRamState()` of the Mcu module.

Enumerator

<code>RAM_IP_STATUS_OK</code>	RAM_IP Ok status
<code>RAM_IP_STATUS_NOT_OK</code>	RAM_IP Not ok status
<code>RAM_IP_STATUS_UNDEFINED</code>	RAM_IP Status is unknown

Definition at line 115 of file `Ram_Ip_Types.h`.

6.4.5 Function Reference

6.4.5.1 Ram_Ip_InitRamSection()

```
Ram_Ip_StatusType Ram_Ip_InitRamSection (
    const Ram_Ip_RamConfigType * RamConfigPtr )
```

Initializes RAM section.

This function initializes RAM section.

Parameters

in	<i>RamConfigPtr</i>	Ram section configuration.
----	---------------------	----------------------------

Returns

`Ram_Ip_StatusType` Ram status

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